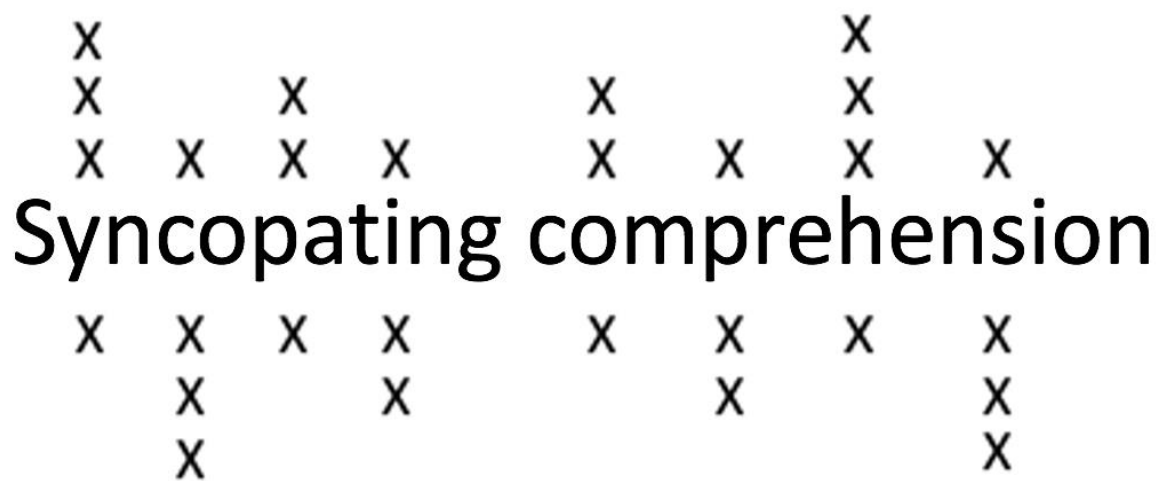




The Effect of Meter-Syntax Alignment on Sentence
Comprehension, Sensorimotor Synchronisation, and
Neural Entrainment

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Abstract

Why rhythm in language? And why is linguistic rhythm grasped by means of meter (a temporal grid structure) in the human mind? And specifically, why does the human mind prefer to align meter to language in certain ways and not others? This thesis examines the alignment of meter to syntactic phrase structure and its effect on language comprehension. This is investigated empirically in four experiments, whose results are situated within the relevant linguistic, musicological, cognitive, and neuroscientific literatures. The first two experiments show that meter-syntax alignment indeed affects sentence comprehension, and the second also shows an effect on sensorimotor synchronisation. The third experiment behaviourally replicates the comprehension result while also recording electroencephalography (EEG). This neural measurement shows how delta oscillations track the perceived meter rather than syntactic phrase structure: contradicting some recent theories. The final experiment applies meter-syntax alignment to an algebraic grouping task. By using simpler (better controlled) non-linguistic stimuli, the results in this experiment better constrain the mechanistic interpretation of the results so far. Specifically, I suggest that the effect of meter (and its alignment to syntax) on comprehension is mediated by an effect on short-term/working memory. The broader theoretical and practical implications of these experiments is finally discussed, especially with regard to theories of language processing, music-language parallels, and education.

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1. Introduction

It is easy to overlook how rhythm shapes the comprehension of language. In contrast to music, in which rhythm is often a prominent feature, rhythm in language mostly does its work without calling attention to itself. This thesis joins a growing many in turning the spotlight onto rhythm and its historically overlooked role in the cognition of language. An experimental exploration is presented of metrical rhythm and what happens when its alignment to syntactic phrase structure is varied. This alignment turns out to influence the ability to correctly comprehend the meaning of syntactically complex sentences (such as this one), with accompanying effects on sensorimotor coordination and neural entrainment. Making sense of this turns out to be theoretically informative and to have meaningful real-world implications.

The title of this thesis alludes to ‘syncopating comprehension’—what does this mean? In music, syncopation is a concept for describing a certain quality of metrical rhythm (i.e., music we might want to tap our foot along to) wherein a syncopated rhythm has conflicting cues to the alignment of a metrical grid (‘tripping up’ our attempts at foot tapping). By ‘syncopating comprehension’ I analogise between this alignment conflict in music and that between meter and syntax in language, while invoking consequences for comprehension. As will be seen, analogy between music and language figures prominently in my own thinking, and indeed has long historical roots which I hope this thesis can extend in some small way.

1.1. Personal motivation

Before providing a chapter summary, I will briefly motivate this research, not by reference to the scientific literature (which will come shortly) but with reference to my own personal experience. I have had the privilege of studying to be a performer on the classical guitar. Reflections on this experience provided the principal motivation for the following scientific inquiry and this experience in many ways continues to ground my thinking now. More than just an exercise in navel gazing, I hope this brief personal reflection provides context as to why the forthcoming research questions struck me as being important, and indeed, why they struck me at all in the first place.

The crucial reflection comes by way of my friend and former musical mentor Timothy Kain and the things I learned from him while studying musical performance at the Australian National University in Canberra. Tim is a wonderful musician and an inspiring teacher. One of the simple but revolutionary ideas he imparted was an active mindset with regards to rhythm. As a classical musician it is easy to sometimes ‘just play the notes’ on the page. That is, to try to reproduce the given sequence of pitches and rhythms that comprise the composed piece of music, and to do this by whatever means. But this is misguided. This mindset fails for the same reason that trying to speak a foreign language by reproducing a memorised sequence of syllables fails. The

failure in each case is a result of a mismatch between strategy and how the mind was designed to work.

The sense of ‘how the mind is supposed to work’ can be approximated by Karl Lashley’s prescient insight into serial order in behaviour (Lashley, 1951). In this classic paper, Lashley argued that complex syntactic sequences (like music and language) are controlled not by the chaining of associations (one note or syllable to the next) but rather by a central hierarchically organised plan. Accordingly, Tim’s simple seemingly obvious but profound insight into playing music was to be intensely mindful of rhythm and its hierarchical organisation into grouping and meter (and he had great tricks for helping you do this). If this structure (the central plan) was clear in your head, then everything fell into place, both in terms of motor execution as well as the ability to comprehend and communicate musical meaning. Too often, this structure was not clear in my head. But with years of practise, I became better at applying this mindset and it deeply transformed my musicality.

My original motivation in this thesis, therefore, was to understand this a bit better. Along the long and windy road from there to now, I, as I am very good at doing, got distracted a bit. I am very good at being distracted by interesting things. A productive fruit of this distraction was making a deep analogy between this experience I have just described and the problem of how we understand language given the sort of minds we have as humans. It turns out that rhythmic structure plays an analogous role in the linguistic story as it does in the musical one. Pursuing this insight led me to mistakenly become a cognitive scientist along the way to finishing this PhD thesis.

1.2. Acknowledgements

Many people deserve thanks for helping me get to the point of submitting. Firstly, my friends and close family. Their love and support have gotten me through many hard times. You know who you are and I could not have done it without you. In particular, I would like to dedicate this thesis to my mother Louise Schofield, who passed away 9 years ago to the day. Mum was a wonderful author of children’s literature and an advocate for the arts in her local community. The type of stories she told are very different to the one I will be telling here. Her stories are much more interesting.

Thanks are also owed to Tim Kain and the rest of the classical guitar community in Canberra. My time there deeply shaped who I am now, both as a musician and a person, and inspired this research that has occupied my mind for the last few years. Tim remains a role-model for what mentorship can be at its very best, and my then classmates continue to astound me with their musical accomplishments on the world stage, inspiring me to pick up the guitar again.

Many are then to blame for my inexplicable transformation from intellectually distractible guitarist to cognitive scientist. Michael Jacobson initially took me on as a PhD student and ultimately trusted me enough to let me go my own way once I found a clear path, even though it had diverged substantially from where we started and his own research area. Micah Goldwater

played a key role in helping me eventually find and traverse this path. Over the last few years he has been the primary support as I have slowly found my footing.

In the interest of saving ink/bits, here are just some of the many other wonderful academics that have helped in various ways, and to whom I am very grateful (in no particular order): Alex Holcombe, Peter Keller, Richard Cohn, Kirrie Ballard, Peter Reimann, Lina Markauskaite, Sylvie Nozaradan, Andy Milne, Matthew Coleshill, Gareth Roberts, Nick McNair, Mike X Cohen, Ling Wu, Ian Colley, and many others.

1.3. Chapter Outline

This thesis investigates how meter shapes our comprehension of language. Roughly speaking, this is to say that language has something like a musical beat. As a first approximation, the overarching research question being pursued is whether the misalignment of meter with syntactic phrase structure (what I call 'linguistic syncopation') affects the processing of linguistic meaning.

To this end, chapter 2 situates this question in an understanding of what language is generally and what sentence comprehension entails specifically. Of course, there is no simple and uncontested view of how we should think about language and how we as humans understand the meaning of sentences. The chapter, therefore, critically compares two prominent perspectives. This comparison serves two functions. The first is to give a sense of what it is that the alignment of meter could be influencing; which component processes could plausibly affect the comprehended outcome? The second function is to provide a backdrop for what will be one of the main theoretical contributions of the results of this thesis. This will be recapitulated in the final chapter, however, it is necessary to first plant the seed here.

Chapter 3 provides an overview of prosodic structure, which may be thought of as the rhythm of language. This includes an explanation of meter and its relation to prosodic constituent structure, and how both of these structures tend to align with syntactic structure. Attention is given to how current linguistic theory makes sense of how this alignment supports comprehension. Together with chapter 2, chapter 3 provides the necessary background in order to understand the significance of the research questions being pursued in this thesis.

After the theoretical build-up given in the previous two chapters, chapter 4 (Study 1) isolates the key research gap in the literature with regards to how meter-syntax alignment might affect language comprehension. Addressing this gap, two experiments are described that manipulate the alignment of meter to syntactic phrase structure and measure the effect of this on the comprehension of syntactically extended sentences (those with nonlocal dependencies). The second experiment also addresses the question of whether this alignment affects the coordination of the sensorimotor system as manifest in the ability to tap in time with metrically strong syllables during speech comprehension.

Reflecting on the positive results of the first study, chapter 5 introduces another relevant research literature that helps to make sense of them under a single coherent framework. The notions of dynamic attending and neural entrainment are introduced as specifically relating to, on the one hand, speech processing, and on the other, the processing of metrical rhythm.

Building on the previous chapter, chapter 6 (Study 2) articulates a contradiction in the literature concerning neural entrainment, meter, and syntax. An experiment is described that attempts to prise this apart and better understand the results of Study 1 in terms of underlying neural dynamics.

Reflecting on the results of studies 1 and 2, chapter 7 (Study 3) aims to further constrain the interpretation of why meter-syntax alignment affects sentence comprehension. Motivated by recent theoretical models, it is hypothesised that the effect is mediated by serial-order short-term memory. To address this, a final experiment is run in which the meter-syntax alignment paradigm is applied to a non-linguistic algebraic grouping task. The non-linguistic nature of this task is designed to more directly test the effect of meter on memory processes without confounds inherent in the meaningfulness of language (which affords predictive/semantic compensation for memory confusions). This experiment also manipulates the alignment of meter at two distinct hierarchical levels.

Chapter 8 concludes this thesis with a summary of the results and a discussion of their theoretical and practical implications.

2. From signal to meaning: Two tales to tell of how to get there

Language comprehension is a journey from signal to meaning. With speech we travel from vibrations in the ear to abstractions in the head and with other modalities (such as sign-language or reading) we start from similar humble beginnings and end at the same remarkable destination. This journey is taken for granted in our everyday lives, but it is truly a marvel of human cognition that it happens at all, let alone so rapidly and effortlessly. This thesis examines the role of metrical rhythm in this process. However, before this specific question can be investigated, this chapter sets the stage by laying out the broader context of what language comprehension entails more generally and how researchers have tended to think about it historically.

2.1. The problem of sentence comprehension

The shortest linguistic journey from signal to meaning is the one that takes place while comprehending a single word. *“Run!”*, *I shout*. Soundwaves reach your ear canal, are transduced into mechanical vibrations by the eardrum, are further transduced into electrical information in the cochlea, pass through various subcortical stages of early auditory processing, and eventually reach higher-order auditory areas that decode the sound and associate it with a meaning to be finally projected to your conscious awareness by mechanisms of attentional selection and conscious access. This all happens within a fraction of a second. The scientific reduction of this cognitive process fails to detract from its mystical wonder—“It’s still magic even if you know how it’s done” (Pratchett, 2004).

But word comprehension is only part of the journey of interest. This thesis, for example, is not merely a bag of words but a collection of sentences, paragraphs, sections, and chapters that cohere together in a certain way. Comprehension entails integrating the meaning of words into larger structures such as these, which together embody broader coherent arcs of meaning. A thesis consisting of just words and not these larger structures would not be a very good one.

The coalescence of words into a sentence is often characterised as a transformation from a sequence to a hierarchy. That is, words manifest themselves sequentially on a page or in speech, yet, when we understand them in the context of a sentence, their influence breaks free from the shackles of one word to the next. Instead, a linguistic tree is grown in the mind, whose branches words attach to. The connectivity of branch to branch and eventually to a unifying trunk bestows words with the ability to flexibly combine into different meaningful coalitions. This tree-like structure is what linguistics call ‘syntax’.

With regard to what sentences mean, syntax binds the component parts together in the right way. To understand the simple sentence “Carl loves Darcie”, for example, one must not only grapple with the ambiguity of words—what is *love*?—but also with the ambiguity of how they combine—*who* loves *whom*? This constitutes a ‘binding problem’ which is defined generally as

how we “integrate information across time, space, attributes, and ideas” into a coherent structure (Treisman, 1999; Treisman, 1996; Baggio, 2018, p. 47-96; Jackendoff, 2002, p. 197-230). As a specific type of binding, linguistic syntax is the basis for differentiating: Carl loving Darcie, Darcie loving Carl, both loving each other, or both being loving people. Without any binding: an amorphous mixture of Carl-ness, Darcie-ness, and love-ness; like mixing coloured paints into an undifferentiated grey; entropy. Sentence comprehension therefore requires putting words into structural relations (syntactic, semantic, discourse) that determine their role in the whole.

A more nuanced picture of syntax and its role in comprehension is however more complicated. Debates continue to rage in linguistics and neighbouring disciplines about how best to paint this picture, and a comprehensive review of the canvas so far would be beyond the present scope. This chapter will instead contrast two prominent perspectives.

The first perspective is synonymous with Noam Chomsky, the influential linguist and public intellectual. His ground-breaking theoretical work in the 1950s kickstarted modern linguistics and left a considerable influence on cognitive science more broadly. His perspective characterises language as an idealised formal system called a generative grammar. Syntax is at the very heart of generative grammar (Chomsky, 1995, 1965).

The second contrasting perspective takes various forms (Goldberg, 2013; Jackendoff, 2002, 2011; Jackendoff & Pinker, 2005; Pinker & Jackendoff, 2005; Croft & Cruse, 2004). They all, however, push back against the idea that syntactic recursion is what makes language special (Chomsky, 1995; Hauser, Chomsky, & Fitch, 2002). They also push back against the assertion that this linguistic capacity is the result of a chance genetic mutation, instead preferring to see it as incrementally evolved for social communication. Crucially, by de-throning syntax, this perspective also leaves room for phonology (and semantics) to play a more active role.

Comparing these two perspectives is aimed to provide the background necessary to understand how the research questions being explored in this thesis fit into a larger understanding of language and how it is processed in the human mind. The specific themes discussed in this chapter will also be recapitulated in chapter 8, wherein the results of this thesis will be discussed in light of these and other ideas.

2.2. View one: Chomsky’s infinity machine

2.2.1. Defining language

Chomsky’s work has largely been centred around answering the question famously posed by Wilhelm von Humboldt—how it is in language that we “make infinite use of finite means” (Humboldt, 1836/1971, p. 70). In other words, how do we so freely combine the limited set of words we know into an infinite number of (mostly novel) expressions that can be *made sense of* and *externalised* as speech, sign-language, or writing? Chomsky terms this the ‘Basic Principle of language’ (Chomsky, 2015/1995, p. ix) and goes about trying to explain it as the focus of his research program.

In *Syntactic Structures* (Chomsky, 1957) Chomsky makes two key advances. The first is a precise formal description of language that makes the Basic Principle mathematically explicit. The second is a methodology for evaluating evidence for or against a theory specified in such formally explicit terms.

A difficulty in explaining language is that there are an infinite number of sentences to explain. Chomsky's formal innovation for dealing with this was to apply earlier breakthroughs made in the mathematics of computation (e.g. Turing, 1936) to describing an infinity of sentences; describing language as an abstract machine (i.e. like a Turing Machine) composed of a system of rules (a grammar) operating over a lexicon (a dictionary). Just as $f(x) = 2x$ concisely enumerates the set of all (infinite) even numbers, Chomsky's *formal grammar* concisely enumerates the set of all (infinite) grammatical sentences consistent with its specified set of grammatical rules and words.

A key component of this grammatical rule set is *phrase-structure grammar*. This idea was operationalised in terms of rewrite rules (Post, 1943) of the form $Z \rightarrow X + Y$, which operate over syntactic constituents. For example, a sentence is defined as the rewrite rule $S \rightarrow NP + VP$ (read as 'a sentence can be rewritten as a noun-phrase plus a verb-phrase'). This formally specifies the hierarchical relations among syntactic constituents, and their *recursive* application down to terminal elements (parts of speech like: nouns, verbs, determiners...) specifies the full hierarchical structure of a sentence (figure 1). The hierarchical relations specified the iterative application of such rewrite rules can alternatively be represented as a tree structure. Fully specified structures of this sort are finally fleshed out as real sentences by *inserting* the appropriate category of word into terminal elements (e.g. rewriting 'N' as 'dog').

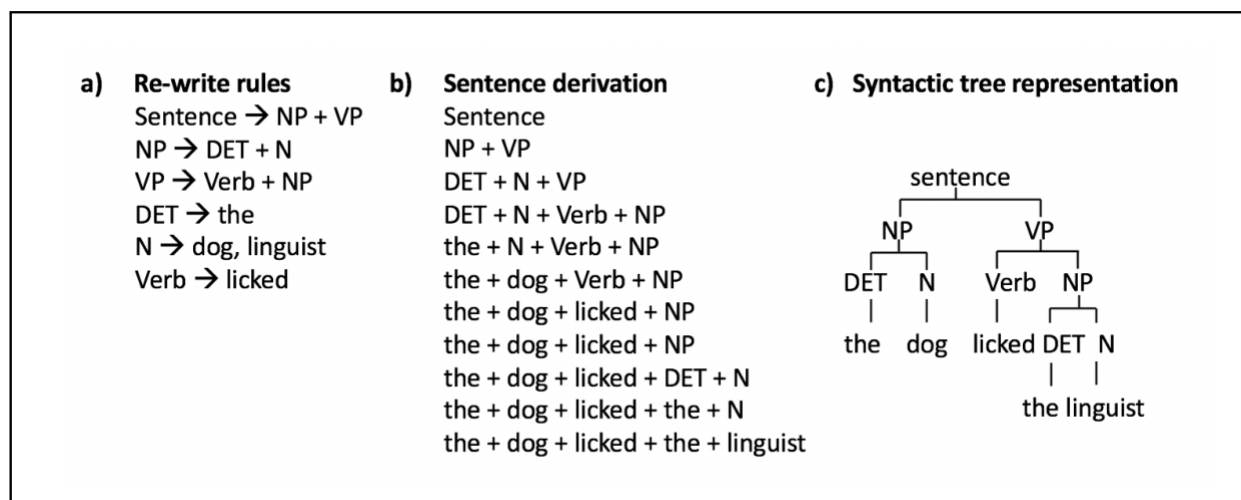


Figure 1: Example in the style of Chomsky (1957): a) phrase-structure grammar rewrite rules, b) a derivational sequence of how these rules can compose a sentence, c) alternate representation of how the derivational sequence underscores a syntactic hierarchy, represented using tree structure.

Then comes a *transformational grammar* stage. The basic idea is to take the output of the previous phrase-structure component and deform its structure by deleting or moving elements.

The motivation for this comes from the fact that sentences like ‘Chomsky likes formalisms’ and ‘formalisms are liked by Chomsky’ have the same meaning yet have contrasting surface structures. The theory of transformational grammar proposes that common meaning is structurally represented by common deep-structure and that different ‘surface structures’, which can be externalised as speech, are produced through transformations of the common deep structure.

There are many further nuances and reworkings in Chomsky’s later theories, however, a phrase-structure grammar complemented by some form of transformational component constitutes the essence of his formal system.

2.2.2. Falsifying linguistic infinities

How does one then determine whether a given formal grammar is a good theory? Are the formalisms of phrase-structure grammar and their transformation capturing deep linguistic truths? The difficulty is that, according to the Basic Principle, there are an infinite number of sentences to explain in a given language. It is therefore of limited value to judge a theory by how well it predicts a given sample. This is, of course, a fundamental problem faced in all empirical sciences and inductive reasoning more generally as there are always an infinite number of potential observations of relevance to a theory. This is why the primary epistemological directive in science is falsification rather than confirmation (Popper, 1959).

From this understanding, Chomsky sets about falsifying a competing theory of language through a famous carefully selected example sentence— “colorless green ideas sleep furiously” (Chomsky, 1957, p. 15-17). He first notes that it is intuited as being ‘grammatical’ by native speakers, thus a valid sentence among the infinite set in the language. He then notes that it lacks propositional meaning, as compared to a sentence like “big fluffy dogs smile gleefully”. And that its composition is not predicted by statistical association, in that neither the whole sentence nor any of its parts are likely to have ever been in the past experience of an English speaker.

If one reverses the order of the words in the sentence, however, it is no longer intuited as being grammatical despite being equally meaningless and statistically novel. The riddle is then what makes this so? Chomsky points out that statistical approaches such as those based upon finite-state Markov-chain models (Shannon & Weaver, 1949) are unable to account for this critical observation, thus suggesting a falsification of that approach (see for more extended review: Adger, 2018). Unsurprisingly, Chomsky’s answer to the riddle is that his model, equipped with phrase-structure grammar and transformations, can uniquely explain this and other critical examples. And specifically, the reverse ‘colourless-green sentence’ is ungrammatical in this model because the set of phrase-structure grammar rules in English requires a specific word order (and often functions word placement, and inflectional morphology; figure 2).

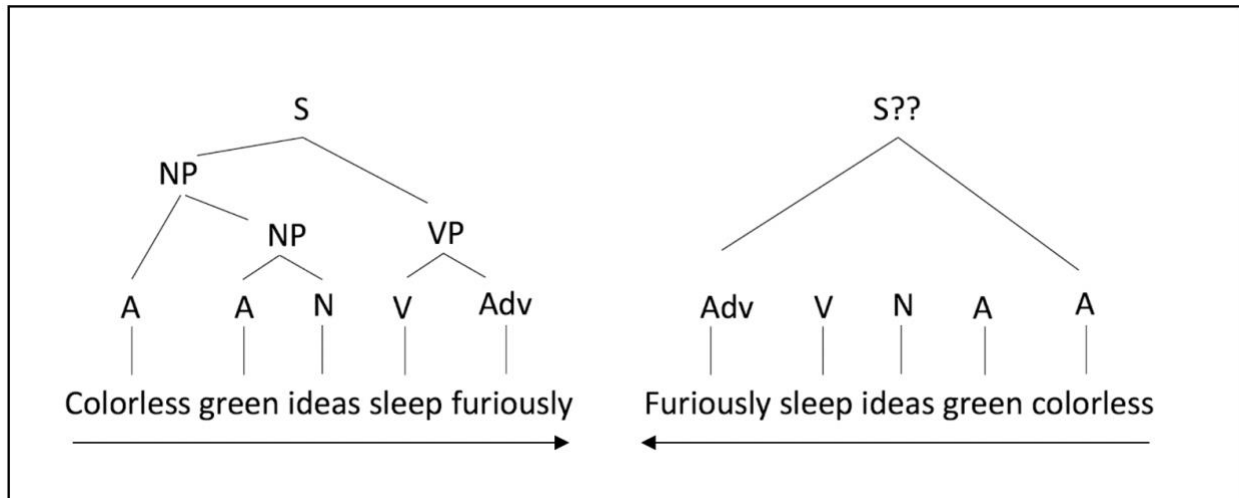


Figure 2: demonstration of how sequential order of words affects the formation of syntactic constituents.

Thus, to Chomsky, what is special about language and crucial to explaining the Basic Principle, is syntax. The *abstractness* of syntactic categories accounts for independence from meaning, as any word of the appropriate category can be slotted into terminal nodes at the end of derivation. For example, one could arbitrarily substitute ‘green’ in the ‘colorless green’ sentence for any other adjective (‘red’, ‘bouncy’, ‘colourful’...) and the sentence would still be grammatical. The *hierarchicality* of constituent relations then allow more expressive ways for sequences of words to be integrated into unified patterns (formalised in terms of the ‘Chomsky hierarchy’: Chomsky, 1959; Fitch, 2014; Fitch & Hauser, 2004). Such hierarchies remain opaque to statistical models that do not explicitly represent constituent structure, although considerable progress has been made in recent years in approximating this property of hierarchicality.¹

2.2.3. The competence-performance distinction

Moving beyond theorising about language in the general sense, Chomsky extends his claims to the mental architecture of language in individual minds. A key concept in doing this is the ‘competence-performance’ distinction. That is, one can logically separate the *tacit linguistic knowledge* somebody has—called competence—and the *processing strategies* by which this knowledge is actualised—called performance (Chomsky, 1965).

¹ It is worth noting that modern sequence-based models like Recurrent Neural Networks (RNNs) come close to capturing the dependency structures of syntax (RNNs; Elman, 1990; Lake & Baroni, 2017; Linzen, Dupoux, Goldberg, 2018; Futrell et al, 2018; Gulordava et al, 2018). Crucially however these approaches still fail on unusual edge-cases, thus fail strict systematicity (Fodor & Pylyshyn, 1988; Marcus, 2018; 2003; 1999).

This is roughly approximated:

$$\text{performance} = \text{competence} + \text{noise}$$

Where:

performance = linguistic forms found in real-world usage

competence = idealised linguistic knowledge of how to construct grammatical forms

noise = sensory + memory limitations, distractions,...

In these terms, it was argued that the object of linguistic inquiry should be the idealisation of competence rather than the messy real-world reality of performance. It was also argued that competence could be productively modelled as a formal grammar of the sort described in the previous section. Henceforth, a linguistic theory of this sort was called a *generative grammar* (Chomsky, 1965, p. 4-9) and has three major components (syntax, semantics, and phonology): the syntactic component generates sentence structures, and the phonological and semantic components interpret the generated structure as sound and meaning (see figure 3).

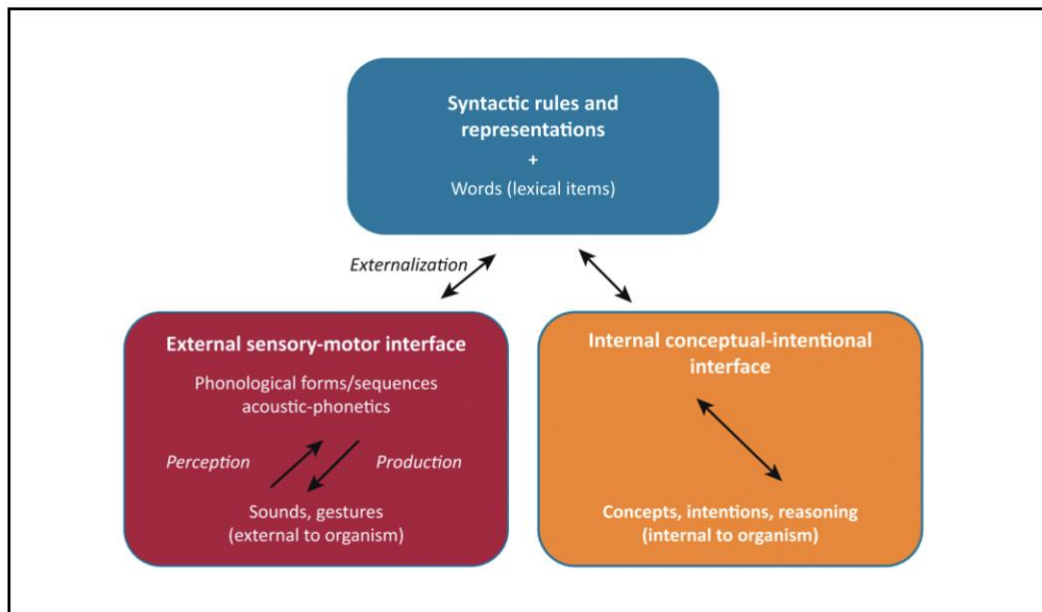


Figure 3: Schematic representation of Chomskyan architecture of language (taken from Everaert et al, 2015)

From this perspective, the question of determining sentence meaning, such as who-did-what-to-whom, is reduced “to the problem of explaining how [basic syntactic structures] are understood” (Chomsky, 1957, p. 92). This assumes classical compositionality where, given word meaning, syntactic structure fully determines what the whole sentence means—“the meaning of an expression is a function of the meanings of its parts and of the way they are syntactically combined” (Partee, 1984, p. 153). And the problem of externalising language as speech is then similarly solved by plugging the syntactically structured expressions into the phonological module to yield the appropriate sounds (Chomsky & Halle, 1968).

2.2.4. Minimalism

The most recent iteration of this thinking is articulated in the context of what is called the *Minimalist Program* (Chomsky, 1995). This represents an attempt to simplify previous theory guided by the assumption that language is a *perfect* system for interfacing sound and meaning. That is, this program expects the core generative system of language to conform to an optimal design that minimises computational redundancy and conforms to the Galilean principle “that nature is simple and that it is the scientist’s task to show that this is the case” (Chomsky, 1995, p. vii).

This notably led to a revision of phrase-structure grammar from deriving a sentence from the top-down according to a heterogeneous assortment of rewrite rules (starting from $S \rightarrow NP + VP$, and iterating down to words at the terminal nodes) to assembling sentences with just the operations of Select and Merge, which pairwise combine and label syntactic objects from the bottom-up (governed by θ -role assignment and general movement principles). These merged structures are then transformed by the ‘move’ operation before Spell-Out as sound (phonetic form; PF) at specific points in the derivation governed by phases, and finally make contact with meaning (logical form; LF).

2.2.5. Summary

Chomsky set out to explain the Basic Principle (how we can produce and make sense of an infinity of sentences) and found syntactic grammar to be the solution. But is this the right question to ask in order to reach an understanding of linguistic processing? This approach has led to the view that generating grammatical sentences is primary and that communicating with language is merely “peripheral to the core elements of language design” (Chomsky, 1995, p. x; Everaert et al., 2015). This has also led to the view that syntax is the core driver of language and that “externalization in one or another sensory modality (or none at all, as in thought) is an ancillary feature of language” (Chomsky, 2015/1995, p. x).

This downplaying of phonology leaves meter (the subject of this thesis) as precisely such an ancillary feature. That is, providing necessary support for language but no more, highlighting instead the wonders of syntax as the primary linguistic feature (see: Chomsky, 2015/1995, p. 13; p. 203).

2.3. Some criticisms

This mainstream generative-grammar perspective has been influential, but it has also been subject to criticism. In particular, its formal apparatus for grammatical derivation has sat awkwardly with psycholinguistic attempts to empirically study realtime language processing (Fodor et al., 1975; Clifton, 1981). And this tension has worsened in the current Minimalist Program instantiation of the theory (as discussed: Ferreira, 2005; Jackendoff, 1997, 2002, 2003, 2011). While a retort may be that generative grammar is a theory of competence rather than a theory of processing (performance), Culicover and Jackendoff note the following:

...an idealization always implies a promissory note: in principle, the theory of competence should be embedded in a theory of performance—including a theory of the neural realization of linguistic memory and processing. One of the criteria for an explanatory theory of competence is how gracefully it can be so embedded, to the extent that we can determine within our current understanding of processing and neural instantiation of any cognitive process.

— Culicover & Jackendoff, 2005, p. 10

2.3.1. Language from the top-down

How did Chomsky's perspective on language become disconnected from real-world processing? I argue here that this failure to gracefully integrate competence with performance arises from what Bradley Love calls a top-down approach to integrating across levels of analysis (Love, 2015). The levels of analysis Love refers to are those famously proposed by David Marr for analysing information processing systems: computational, algorithmic, and implementational levels of analysis (Marr, 1982).

Mainstream generative grammar is characterised by a top-down approach because it starts from the computational level description and unidirectionally works its way down. The Basic Principle (infinite sentences from finite means) is essentially a *computational problem* (an abstract description of required inputs and outputs). Chomsky then proposes a minimal *algorithmic architecture* to solve this problem: a lexicon of words and rules, and a process that assembles sentences with them through stepwise derivation. And recently neuroimaging studies have sought to confirm the *neural implementation* of this algorithmic architecture, with special interest on finding neural correlates of the core recursive computation *merge* (Friederici et al., 2017; Nelson et al., 2017; Zaccarella & Friederici, 2015).

The problem with this top-down approach, however, is that each level is underconstrained theoretically with respect to levels above it (Love, 2015; Fitch, 2014). In other words, many algorithms can satisfy the same computational specification (input-output requirements) and the same algorithm can be implemented in many possible physical substrates.

Chomsky makes two assumptions to deal with this. The first is that competence and performance can be cleanly separated, or minimally, that factors of performance are inconsequential to the design of language. This relates to the assumption (largely made without argument) that the language faculty did not evolve for the purpose of communication and is in fact badly designed for it. In other words, factors of performance (noise, memory limitations, robustness, etc) are deemed only relevant if the system was designed for communication (which is always noisy). The other assumption is that the architecture of the core language system is a *perfect* system for interfacing sound with meaning that minimizes computational redundancy. The problems with these assumptions will now be elaborated.

2.3.2. Evolutionary design for noisy pragmatic communication

In the real world, the journey from signal to meaning is beset with noise: from acoustic or visual noise in the signal, speaker or comprehender mistakes, confusions, and distractions. If the language system is designed for communication then it should take this inevitability into

account, like any well-engineered device does. Indeed, people have been shown to represent the certainty of linguistic inferences and adjust it dynamically to noise in an adaptive way (Gibson et al., 2017; Gibson, Bergen, et al., 2013; Levy et al., 2009; Ryskin et al., 2018, 2020). And the very linguistic forms we use (e.g. why we choose to use the word “cat” to denote cats, and so on...) also appear to be optimised for maximally robust communication under noise (Gibson et al., 2019, 2013; Hahn et al., 2020; Kanwal et al., 2017; Ramscar, 2020).

Comprehending noisy signals is also made difficult by time pressure. To account for this, comprehension is incremental and eager to make meaning from which to derive new inferences (Christiansen & Chater, 2015; Karimi & Ferreira, 2016; Tanenhaus et al., 1995). This contrasts sharply with the formal approach that waits around until a whole sentence has been uttered to carefully derive its syntactically licenced truth condition (as implied in the Chomskyan architecture; see Ferreira, 2005). Much like engineered systems that need adaptive realtime control, the language system also *anticipates* information before it arrives in order to enhance efficiency (Martin, 2016; 2018; Ferreira & Chantavarin, 2018; Levy, 2008; Levy et al., 2012; Poeppel & Monahan, 2011).

But, noise aside, are comprehenders even aiming for ‘truth’ in the first place (the ‘logical form’ in Chomskyan parlance)? As remarked by the philosopher Hannah Arendt —“Implicit in the urge to speak is the quest for meaning, not necessarily the quest for truth” (1977). Indeed, rather than a search for linguistic truth, modern psycholinguistic theory characterises comprehension as a “fast and frugal” estimation of meaning, strongly influenced by context and pragmatic goals. This is formalised in the so-called ‘good enough’ theory of processing (Christianson, 2016; Ferreira et al., 2002; Ferreira & Patson, 2007; Karimi & Ferreira, 2016; Sanford & Sturt, 2002), and is likened to the famous ‘two systems’ characterisation of human reasoning (Kahneman, 2011; Kahneman et al., 1982). Applied to language, this can be understood as there being a fast system that makes extensive use of contextually licensed heuristics and surface level semantic processing, and then a slower more algorithmic route that makes use of syntactic grammar (Ferreira, 2003; Ferreira & Çokal, 2015; Kuperberg, 2007; van der Lely & Pinker, 2014).

The Minimalist Program ignores these considerations: “Since [the core computational system of language] would have been subject to no selectional pressures, it would have assumed an optimal form in accord with natural law—specifically, [principles of Minimal Computation]—rather the way a snowflake forms” (Chomsky, 2015/1995, p. xi). Instead, Chomsky argues that the syntactic mechanism at the core of his theory of language (Universal Grammar generally, and the operation Merge especially; Berwick & Chomsky, 2016) was the result of a single random genetic mutation somewhere in our lineage, resulting in a discontinuous or ‘catastrophic’ emergence of full-syntax (and thereby full-language) all at once, and not gradually evolved for noisy pragmatic communication. The implication is then that the design of the language faculty would only be subject to so-called general laws of nature such as efficient computation (taken to mean nonredundant processing and storage). Empirical facts about real-world processing, from this perspective, could then be ignored: they only speak to the ‘performance system’ and not the core linguistic competence.

However, the assumptions that support this idealistic assertion—the ‘discontinuous’ syntax-first theory—are variously problematic. For instance, the plausibility of the ‘mutant-gene’ story has been called into question and argued to be highly unlikely on the grounds of basic evolutionary biology (Boeckx, 2017; de Boer et al., 2020; Everett, 2017; Fitch, 2010; Martins & Boeckx, 2019). And rather than there being strong constraints on grammatical structure from a genetically coded Universal Grammar, cross-linguistic research using computational phylogenetic methods shows that grammar is predominantly a product of cultural evolutionary processes (Dunn et al., 2005, 2011). Together, these growing bodies of evidence and theory together give support to an alternative evolutionary formulation specified in terms of good-ol’-fashioned Darwinian descent with modification, and in which cultural evolutionary (Dennett, 2017; Kirby, 2017) and social dynamics (Tomasello, 2008) play active roles.

This gradualist evolutionary perspective implies then that language had humble beginnings that were gradually tinkered with to yield more robust communication. In this early protolinguistic state, communicative utterances were likely holistic, and semantically and pragmatically grounded by gesture (Arbib, 2012; Everett, 2017) and affective prosody (Brown, 2017; Mithen, 2006). The communicative powers of such a protolinguistic system could have then been evolutionarily refined, both culturally and biologically. Syntax, in this context, is an abstract “scaffolding to help relate strings of linearly ordered words (one kind of discrete infinity) more reliably to the multidimensional and relational modality of thought (another kind of discrete infinity)” (Jackendoff, 1997, p. 18). Thus, can be conceived as an innovation to improve communication, whose underlying cultural and cognitive preconditions could gradually have evolved (Brown, 2017; Fitch, 2019; Jackendoff, 2002; Pinker, 2003; Pinker & Bloom, 1990; Pinker & Jackendoff, 2005; although see for critical perspective Christiansen & Chater, 2008 with pushback from Fitch, 2008).

So rather than there being an unmotivated and perfectly formed snowflake at the heart of the language capacity, it is suggested that language is no exception to the notion of biology as a *tinkerer* that gradually solves functionally motivated problems with whatever is at hand (Culicover & Jackendoff, 2005; Goldberg, 2003; Jacob, 1977). The problem that it solved was social communication, and the design it reached for the language system is framed as “a collection of innovations [acquired piecemeal], each of which improved the kind of communication system that could be learned through cultural transmission” (Culicover & Jackendoff, 2005, p. 543). Although Fitch reminds that we should not *just* focus on the communication but also on the evolution of broader aspects of cognition and conceptual reasoning which may incidentally support language (Fitch, 2010; Fitch, 2020).

2.3.3. Redundancy and robustness

What about the assumption of minimal computation applied to the algorithmic level?

Although perhaps formally not so elegant, a linguistic theory that incorporates redundancy may in fact prove empirically more adequate as a theory of competence, and it may also make better contact with a theory of processing. As a side benefit, the presence of redundancy may bring language more in line with other psychological systems, to my way of thinking a desideratum. (Jackendoff, 1997, p. 15)

Nature is replete with functional redundancy. It is an essential feature of most biological systems from ecosystems to genes (Nowak et al., 1997) and natural selection designs them in this way because redundancy imparts robustness (Edelman & Gally, 2001; Osterwalder et al., 2018; Simon, 1962). Cognitive systems are no exception and systematically represent and process information redundantly, such as how the visual system uses multiple redundant cues to infer percepts like colour or movement.

It is nonetheless true that nature also tends toward efficient computation (Friston, 2010; Friston et al., 2009). So how can redundancy and efficiency be reconciled? Chomsky's fatal mistake is applying the criterion of minimal computation to the computational problem of generating infinite use from finite means under the highly idealistic assumption that 'performance' constraints are irrelevant. That is, Chomsky assumes that language solves its computational problem in a vacuum.

As teasingly remarked by Daniel Dennett, Chomsky saw it as "somehow beneath the dignity of the mind" (Dennett, 1995, p. 387) to be in the business of solving a messy engineering problem with its language system, and would rather it instead to be like the simple lawful formation of a snowflake (that has only a few simple constraints that it can more-or-less perfectly satisfy). Agreeing with Dennett and others, I argue that language is better characterised as doing its best in a complex ill-defined task under numerous performance pressures. There seems to be no reasonable evidence to suggest that the design of the language architecture should be exempt from such pressures. Assuming otherwise is wishful thinking, leading to false conclusions: one system's redundancy is another's efficient solution.

2.4. View two: All roads lead to Rome (meaning)

As just summarised, the problem with the Chomskyan approach stems from assuming an *overly idealistic* bridge between what the system needs to do (the computational level problem) and how it does it (the algorithmic solution). Being able to make infinite externalisable and comprehensible use of finite linguistic means remains a useful abstraction of what the language system needs to accomplish at some level. However, language solves many problems and does so under many constraints. Ignoring or oversimplifying these constraints can be shaky ground on which to theorise: the language system has evolved to make use of what it's got in the circumstances it's given to achieve its various goals.

So instead of a problematic top-down approach, Love (2015) suggests progressing from the inside-out. That is, to posit algorithmic mechanisms that best satisfy behaviourally constrained descriptions of the computational level (above) and neurally constrained descriptions of the implementational level (below), and to iteratively refine in light of new evidence (see also: Krakauer et al., 2017; Poeppel & Embick, 2013).

Consistent with this approach, placing it in a larger theoretical framework, it may also be productive to consider language from the perspective of a complex adaptive system (Beckner et al., 2009; Kirby, 1999). Taking this view implies that the cognitive architecture of language emerges in evolutionary time from the complex interactions among the social agents that use it, which are in turn shaped by the cognitive constraints and affordances provided by their biology and by other factors in the environment. Together this dynamically co-determines the computational and implementation constraints that the algorithmic level emerges to solve.

2.4.1. Parallel Architecture

Ray Jackendoff's *Parallel Architecture* is a theory of linguistic competence that can be seen as having been constructed in this vein (Jackendoff, 1997, 2002, 2003, 2007, 2011). Its central point of difference with mainstream theory is its challenge to the 'syntactocentrism' inherent in assuming that "the fundamental generative component of the [linguistic] computational system is the syntactic component; the phonological and semantic components are 'interpretive'" (Jackendoff, 1997, p. 15). Instead, the central assumption is that syntax, phonology, and semantics are all equally generative systems in their own right, interacting through interface rules (figure 4).

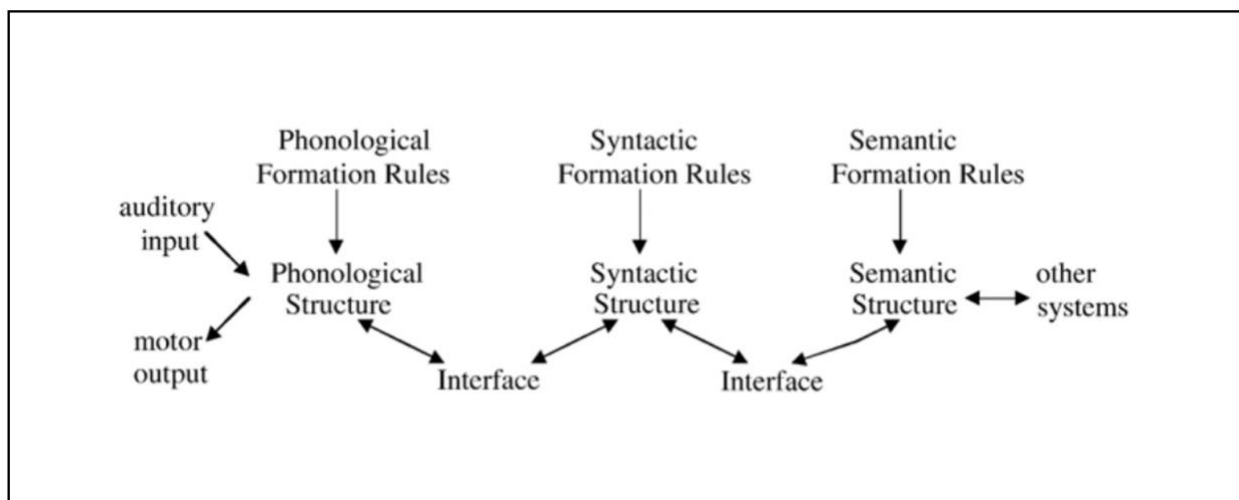


Figure 4: Schematic representation of Jackendoff's Parallel Architecture theory of language (taken from Jackendoff, 1997)

2.4.2. Syntax not necessary

While having multiple systems for generating structure (instead of just one) is redundant, Jackendoff shows that this allows a simpler theory of syntax. That is, syntax does not need to have complex invisible transformations of structure if semantics and phonology can also

contribute to meaning (Culicover & Jackendoff, 2005; see also theories of construction grammar: Goldberg, 2003; 2013).

Indeed, there are also a number of languages around the world with minimal or no recursive syntactic structure such as Riau Indonesian, the Amazonian language of Pirahã (Everett, 2005; Futrell et al., 2016), and pidgin or creole languages. This can be thought of as the use of phonology plus semantics (without syntax), sometimes referred to as a linear-grammar (Jackendoff & Wittenberg, 2017). Even in languages with recursive syntactic structure, the ability to use this structure can be disrupted in agrammatic aphasia (Gibson et al., 2016) and in children with Grammatical Specific Language Impairment (van der Lely & Pinker, 2014), resulting in their falling back onto linear grammar comprehension. And according to ‘good enough’ processing theories, those without impairment may still frequently make do with a frugal ‘linear interpretation’ (Ferreira et al., 2002; Ferreira & Patson, 2007; Christianson, 2016).

2.4.3. Its lexical all the way down

The Parallel Architecture also rejects the mainstream architectural division between the lexicon (*meaningful* words) and a separate syntax-based grammar component (*meaningless* rules), wherein the grammar assembles the *words* into sentences. Instead, *everything* is lexicalised, meaning that each item in the lexicon has its syntactic affordances ‘built-in’. This can be thought of as each item having connectors that constrain combination with others, like Lego blocks. Specifically, each lexical item is proposed to be a triplet of linked semantic, phonological, and syntactic structure (see theory of Relational Morphology: Jackendoff & Audring, 2016, 2018). Thus the process of comprehending a sentence in realtime is reconceptualised from being a stepwise derivational process driven by syntactic rules, to instead involve the parallel activation of lexical items in long-term memory, then tokenisation in working memory, and incrementally ‘clipping them together’ (in no necessary order; top-down or bottom-up) through a general unification operation (see for detailed elaboration: Jackendoff, 2002; 2007).

This general move toward lexicalisation, finer details aside, is shared with a number of other modern linguistic theories such as cognitive linguistics (Croft, 2001; Croft & Cruse, 2004; Langacker, 1987; Tomasello, 2008) and particularly construction grammar (Fillmore, 1988; Goldberg, 1995, 2003, 2013, 2019; Jackendoff, 2013). A key theoretical motivation for all these theories was the increasing realisation that sentence meaning could not be reduced to the ‘word meanings + syntax’ model of classical compositional theory (Partee, 1984). Archetypal examples that contradict the classical approach include idioms such as ‘kick the bucket’, whose meaning (to die) cannot be composed from syntactic combination of its constituent parts. Further analysis reveals that examples like this, and ones more subtle (such as coercion effects), are far more pervasive than previously acknowledged and typically ignored in mainstream generative grammar because they are hard to deal with.

Like with the redundant generation of structure, lexicalisation represents a rather extreme form of redundancy clearly described by Adele Goldberg’s claim that “patterns are stored [in the lexicon] even if they are fully predictable as long as they occur with sufficient frequency” (Goldberg, 2003; also adopted in the Parallel Architecture see: Jackendoff, 2013). This

contrasts sharply with Chomsky's assumption that the lexicon is an "optimal coding" of idiosyncrasies with no redundancy (Chomsky, 2015/1995, p. 216).

The advantage of storing more information (even if redundant) is that, for example, retrieving the meaning of a common phrase like "how is your day going?" as one whole piece of information is far quicker and less effortful than compositionally determining its meaning on the fly. And indeed, there is considerable evidence for frequency effects on processing for small binomial expressions and common phrases (Arnon & Snider, 2010; Morgan & Levy, 2016; Siyanova-Chanturia et al., 2011) as well as for more abstract pieces of linguistic structure (Casenhiser & Goldberg, 2005; Ferreira, 2003; Gibson, 1998; MacDonald et al., 1994; Goldwater & Markman, 2009). Indeed, storing large amounts of redundant information like this may well be the basis for how we learn to generalize and learn pieces of grammar like argument structure constructions (Goldberg, 2019; Goldwater, 2017).

Supporting this with neuroimaging data, there has been a failure to spatially dissociate syntactic from lexico-semantic processing (Fedorenko et al., 2016, 2018, 2020; Hagoort, 2019; Pylkkänen, 2019; Siegelman et al., 2019). In one particularly striking recent study, it was shown that scrambling the order of sentences (thus deforming syntactic well-formedness) had no significant effect on activation in the core language network as long as proximal words could be related by compositional semantics (Mollica et al., 2020), thus reinforcing the notion that the language network cares more about meaning than it does syntax.

2.4.4. Is phonology for externalisation?

It has been customary in the Chomskyan tradition to think of phonology as an ancillary consequence of 'externalisation'—"if humans could communicate by telepathy, there would be no need for a phonological component, at least for the purposes of communication" (Chomsky, 2015/1995, p. 221). However, as noted by Jackendoff, if humans could telepathically interface their conceptual systems, not only would there be no need for phonology, there would seemingly be no need for language either. Perhaps this statement makes more sense under the assumption that thought is just un-externalised language, and that language is the "syntax of human consciousness" (Bickerton, 1995). But this is itself an incoherent position with respect to modern theories of consciousness (see for detailed analysis: Pinker, 1992; Jackendoff, 1997, p. 183-186; Dehaene, 2014).

Contrary to this view, phonology may be a cognitively necessary component for enabling the expressive powers often attributed to syntax. Specifically, Jackendoff suggests that phonology is a 'conscious handle' for interfacing processes of consciousness with the unconscious systems responsible for syntactic and semantic processing—"somewhere between sensation and cognition lies a level of representation that is conscious" (Jackendoff, 1997, p. 192; 1987, see figure 5). In other words, when we comprehend a sentence, we are not aware of all the subroutines that yield apprehended meaning but we are consciously aware of and can mentally manipulate perceptual forms. And conscious access, more than just a cognitive convenience, significantly amplifies the cognitive potential of otherwise unconscious processes (Dehaene, 2014).

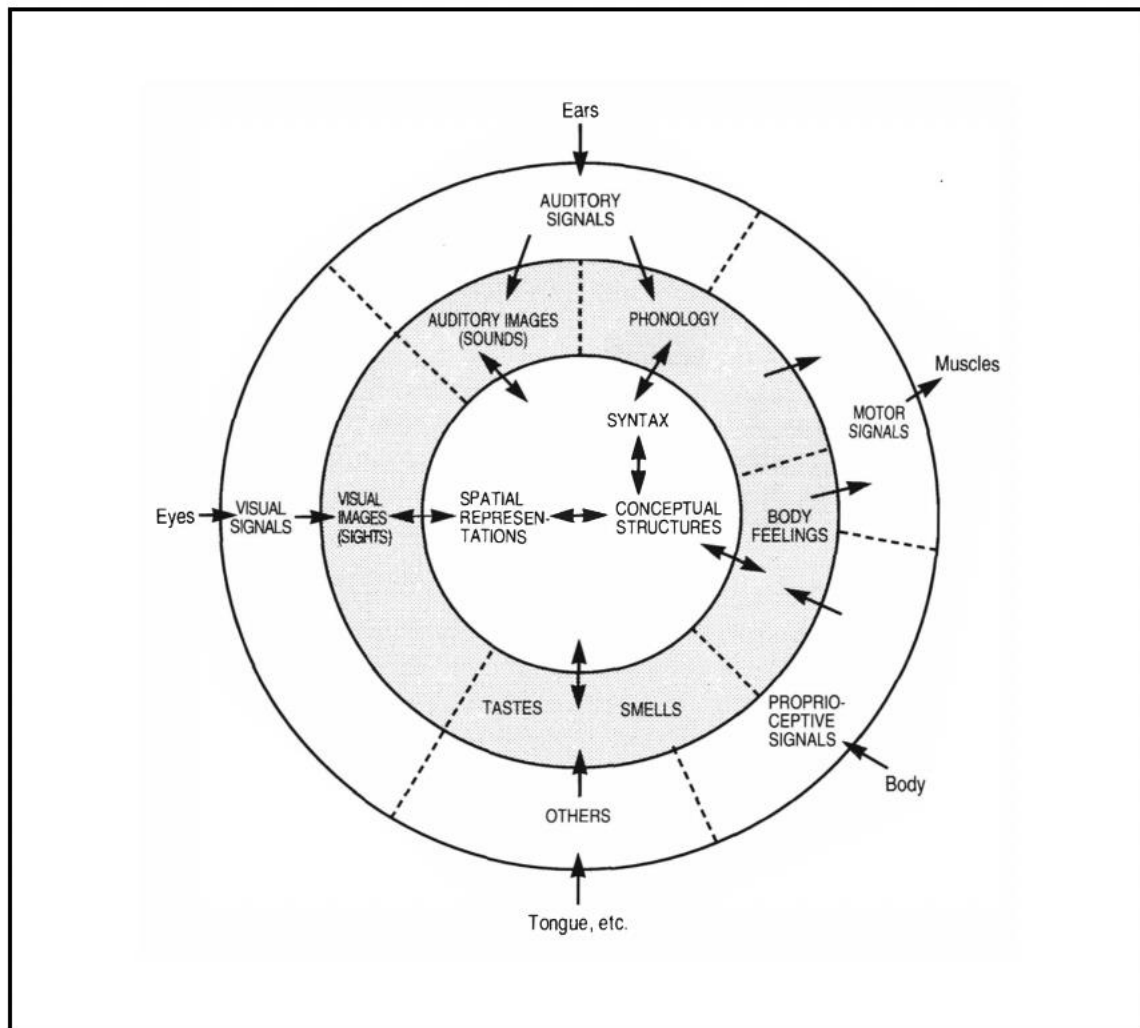


Figure 5: Schematic representation of Jackendoff's conception of the conscious accessibility of different mental representations. Specifically, only the intermediary (shaded) layer is thought to be consciously accessible (Jackendoff, 1997, p. 192)

While in principle one can imagine a cognitive system that syntactically computes linguistic meaning without phonology, the human mind is a distinct type of cognitive system whose constraints may preclude this clean separation. If this is true, phonology is not only functionally necessary for externalisation and social communication, but also, contra Chomsky and Bickerton, it would be necessary for cognitively enabling language to enhance internal thought (Jackendoff, 1997, ch. 8; Pinker & Jackendoff, 2005). Thus, rather than an ancillary imperfection—"the whole phonological system looks like a huge imperfection, it has every bad property you can think of" (Chomsky, 2000b, p. 118)—phonology is an equal partner in coalition with semantics and syntax.

2.5. Summary

I end this chapter with Herbert Simon's parable of two watchmakers (Simon, 1962):

There once were two watchmakers, named Hora and Tempus, who manufactured very fine watches. Both of them were highly regarded, and the phones in their workshops rang frequently—new customers were constantly calling them. However, Hora prospered, while Tempus became poorer and poorer and finally lost his shop. What was the reason?

The watches the men made consisted of about 1,000 parts each. Tempus had so constructed his that if he had one partly assembled and had to put it down—to answer the phone say—it immediately fell to pieces and had to be reassembled from the elements. The better the customers liked his watches, the more they phoned him, the more difficult it became for him to find enough uninterrupted time to finish a watch.

The watches that Hora made were no less complex than those of Tempus. But he had designed them so that he could put together subassemblies of about ten elements each. Ten of these subassemblies, again, could be put together into a larger subassembly; and a system of ten of the latter subassemblies constituted the whole watch. Hence, when Hora had to put down a partly assembled watch in order to answer the phone, he lost only a small part of his work, and he assembled his watches in only a fraction of the man-hours it took Tempus.

Suppose that Hora were Jackendoff and Tempus were Chomsky, and instead of making watches, they were our language systems trying to manufacture meaning out of an incoming signal. In an idealised world, the Chomskyan approach of stepwise syntactic derivation would be perfectly adequate, and perhaps more elegant than Jackendoff. But we do not live in an idealised world.

Thus, “the goal of language processing is to produce a *correlated* set of phonological, syntactic, and semantic structures that together match sound to meaning” (Jackendoff, 2007). And through such correlations (redundancy), the journey from signal to meaning is made more reliably. How prosody generally and metrical rhythm specifically contributes to this coalition will now be addressed in the next chapter.

3. Prosodic groups and grids and their alignment to syntax

Prosody is a universal characteristic of human linguistic communication. Whether the signal be auditory, visual, or even tactile, prosody imparts a rich further dimension surplus to a mere transmission of linguistic symbols. The focus of this chapter is prosodic *structure* as distinct from the affective or pragmatic aspects of prosody. This concerns the perceptual organisation of the signal into hierarchical constituent and metrical patterns and is "a complex grammatical structure that must be parsed in its own right" (Beckman, 1996). While syntax and prosody are independent from one another, they are systematically correlated in ways that support comprehension. A review of this literature will be now made, while also highlighting the significance of parallels between prosodic structure and rhythmic structure in music.

3.1. Introduction

Phonology describes the perceptual organisation of linguistic signals. The traditional focus on language specifically, however, is to some degree arbitrary. Just as the definition has historically shifted from concerning just auditory speech to also more recently acknowledging a phonology of sign language (Brentari et al., 2011), there is no principled reason to exclude further extensions to 'musical phonology' as the composer/conductor Leonard Bernstein once alluded (Bernstein, 1976) or a phonology of birdsong or ape-calls (Fitch, 2019). Thus, we can more generally take phonology, in this context at least, to refer to the perceptual organisation of communicative signals.

The focus in this chapter will especially be upon the *prosodic* (or suprasegmental²) component of phonology. Wagner and Watson (2010) define this as "a level of linguistic representation at which the acoustic-phonetic properties of an utterance vary independently of its *lexical items*," (my emphasis). And, in accounting for musical prosody, Palmer and Hutchins (2006), broaden the definition to the "acoustic properties of speech and music [that] can be manipulated to a certain extent without changing the *categorical information* (the words as they might be written or the musical pitches as they might be notated)" (my emphasis). These acoustic correlates typically comprise variation in timing, pitch, timbre, and intensity (language: Cutler et al., 1997; Shattuck-Hufnagel & Turk, 1996; Wagner & Watson, 2010; music: Palmer & Hutchins, 2006) and non-acoustic signals such as gesture can also play a part (language: Hubbard et al., 2009; Krahmer & Swerts, 2007; Swerts & Krahmer, 2008; music: Huberth & Fujioka, 2018).

Prosodic structure is a composite of two distinct structural dimensions—constituent structure and metrical structure—that, although often intertwined, have their own rules of formation and association with cues in the signal. These two perceptual structures will now be described.

² The parts of speech that are at a higher level of organisation than phonemic segments.

3.2. Constituent Structure

One of the important breakthroughs in 20th century phonology was to justify prosody as having its own independent hierarchical structure independent from syntax (Liberman & Prince, 1977; Nespor & Vogel, 1986; Selkirk, 1984). The so-called prosodic hierarchy that emerged from this describes an ordered set of prosodic categories, which assemble together into hierarchical relations (figure 6).

seven happy puppies are licking my face	Intonational phrase (ι)
seven happy puppies are licking my face	Phonological phrase (φ)
seven happy puppies are licking my face	Prosodic word (ω)
seven happy puppies are licking my face	Syllable

Figure 6: example of how the prosodic constituent hierarchy divides up an expression into hierarchically organised parts.

This hierarchy, however, was thought to differ from syntactic hierarchies in a way described by the strict-layering hypothesis (Nespor & Vogel, 1986; Selkirk, 2011). This hypothesis posited two prosody-specific hierarchical constraints. The first is *strict layering*, which requires that a given category of the prosodic hierarchy (say, an intonational phrase) only dominates categories of the level below it (i.e. phonological phrases, and not prosodic words, or syllables). This disallows *level-skipping* among the *ordered* set of prosodic categories. Secondly, prosodic hierarchies were thought to be *non-recursive*. This disallows prosodic categories from being hierarchically embedded in categories of the same type (e.g. prosodic words within prosodic words).

These constraints distinguish prosody from syntax, as syntax lacks categorical ordering (e.g. a NP can be embedded in a VP and vice versa) and can have recursive structure (e.g. embedded relative clauses):

Syntactic : [[This] [is [the dog [that likes [the cat [which has [a smile [that bit [the man]]]]]]]]]]]

Prosodic: [This is the dog] [that likes the cat] [which has a smile] [that bit the man]

3.2.1. Syntax-prosody correspondence

Prosody-syntax interface theories were then developed to describe how prosodic and syntactic hierarchies related to each other (Nespor & Vogel, 1986; Selkirk, 1984; Truckenbrodt, 1999, 2007). These theories are defined by “contributions from the theory of syntactic representation,

the theory of phonological representation, and the theory of the correspondence relation between the two” (Selkirk, 2011).

To implement this phonology-syntax correspondence, edge-based alignment frameworks proposed various ways by which either the right or left edge of prosodic and syntactic domains relate in systematic rule-like ways. Lisa Selkirk’s Match Theory is a recent extension of this theoretical tradition (Selkirk, 2011). This theory iterates on some of her own previous work in many ways, but one of its changes is to divide the traditional prosodic hierarchy into three zones, each of which is characterised by the sorts of cues that influence it (Îto & Mester, 2009, 2012; Selkirk, 2011):

- **Rhythm categories** (foot, syllable, and potentially mora): shaped by general phonetic and speech-rhythm factors.
- **Interface categories** (prosodic word, phonological phrase, and intonational phrase): shaped by the tendency to ‘match’ syntactic constituents (in addition to other prosodic cues).
- **Discourse categories** (utterance and above; in writing: paragraph, chapter, etc): this level is currently under-theorised but would likely be less shaped by phonetic or syntactic factors, and instead rely upon larger semantic and discourse structure for coherence (Auer et al, 1999).

The interface categories are therefore the specific domain for prosody-syntax correspondence where there is an expected “phonological mirroring of the syntactic constituents”: intonational phrases correspond with syntactic clauses, phonological phrases correspond with syntactic phrases, and prosodic words correspond with syntactic words (Selkirk 2011; figure 7). And rather than just aligning the left or right edge, Match Theory calls for a stronger ‘Match’ constraint that aligns *both* edges of prosodic and syntactic domains, while still retaining the possibility that either of these edges can be misaligned by other constraints that override it (such as phonological size constraints: Watson & Gibson, 2004).

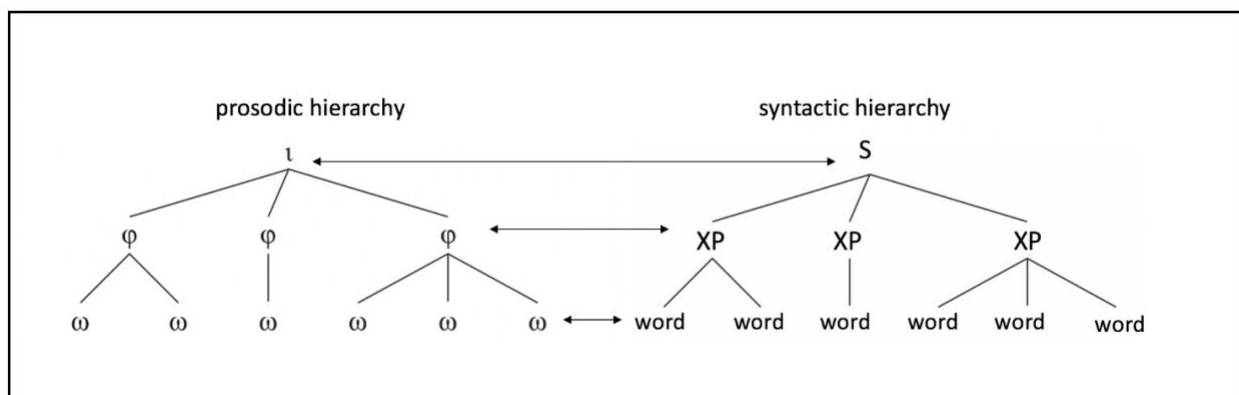


Figure 7: Schematic representation of how Match Theory calls for a mirroring of syntactic and prosodic constituents at the ‘interface level’.

This prosody-syntax mirroring affords the possibility of inducing syntactic structure from prosody. This is helpful to comprehension because prosody has a more direct relationship to properties of the signal than does syntax. Prosodic phrasing therefore allows syntactic comprehension to be more perceptually grounded (Carlson, 2009; Cutler et al., 1997; Frazier et al., 2006; Klatt, 1975; Lehiste, 1973; Schafer et al., 2000; Shattuck-Hufnagel & Turk, 1996; Wagner & Watson, 2010).

3.2.2. How are prosodic boundaries determined?

If prosodic phrasing can influence syntax, what sorts of cues allow prosodic boundaries to be determined from the signal? One such cue is *phrase-final lengthening* (Ferreira, 1993), which has two components. The first is *pre-boundary lengthening*, which is when the duration of the syllable(s) from the final (stressed) syllable to the boundary are lengthened and it has specifically been shown that the precise degree of lengthening correlates with relative boundary strength (Price et al., 1991; Shattuck-Hufnagel & Turk, 1996). The other component is *between-boundary pausing* (Hirsh-Pasek et al., 1987; Krivokapić, 2007; Martin, 1970), which are the temporal gaps placed *between* constituents. Both of these types of phrase-final lengthening are also found in sign-language (Nespor & Sandler, 1999; Wilbur, 2000). A recent theory from Kentner and Féry (2013) proposes to integrate these perspectives into a general one of prosodic grouping based on proximity cues and size constraints, and they note that this is shared with music (e.g. Lerdahl & Jackendoff, 1983).

The other main form of boundary cue is so-called *boundary tones* or *tonal scaling*, which relates to patterns of relatively high and low pitch culminating at prosodic constituent boundaries. Such pitch-based patterns constrain the boundaries of intonational phrases specifically (Ladd, 1986).

The link from such prosodic phrasing cues to syntactic parsing are also robustly supported in experiment and modelling work concerning speech production (Gee & Grosjean, 1983). Importantly, however, this modelling work makes clear that syntax does not *directly* affect these performance timing parameters, but rather makes certain mentally represented prosodic structures more likely (with their own formation constraints), which in turn shapes the acoustic realisation of speech rhythms that cue prosodic constituency (Ferreira, 1993; 2007; Watson & Gibson, 2004).

3.2.3. Current debates in syntax-prosody correspondence

In recent years, the extent to which prosodic and syntactic hierarchies truly differ has been called into question, leading to theories that posit a more direct syntax-prosody mapping than just aligning edges of each domain (Croft, 1995; Elfner, 2015; Schafer, 1997; Steedman, 1991, 2000; Wagner, 2010). A key motivation for this comes from a reassessment of the Strict Layering Hypothesis. Notably, tonal scaling was shown to reflect relative degrees of hierarchical embedding, thus suggesting that intonational phrases could exhibit recursivity (Féry & Schubö, 2010; Féry & Truckenbrodt, 2005; Kentner & Féry, 2013; Ladd, 1988). Similar arguments were also made for recursive prosodic words (Ito & Mester, 2009). Level-skipping was also shown in unfooted prosodic words (Selkirk 1996).

Other prosody-syntax boundary mismatches such as the following were reinterpreted:

Syntax: [Sesame Street is brought to you] [by The Children's Television workshop]

Prosody: [Sesame Street is brought to you by] [The Children's Television workshop]

Some argue that these apparent mismatches can be resolved by adopting a more flexible syntactic grammar capable of generating structures with either boundary (Steedman, 1991; 2000; Wagner, 2005; 2010). To resolve which boundary gets used, Steedman (1991) proposed a 'prosodic constituent condition', which restricts the formation of syntactic structures to precisely those already provided by prosodic structure. And by having "complete harmony" between hierarchical constituent boundaries for prosody, syntax, semantics and discourse structure, Steedman argues that these redundant cues can reinforce each other and make processing more robust.

3.2.4. Prosodic packaging

A subtly different account of prosody-syntax isomorphism is that prosody plays a more fundamental role as a basic memory-chunk in which subsequent processing takes place (Speer et al., 1996). In this 'prosodic packaging' approach, prosody constrains syntactic (and semantic) interpretation from the earliest stages of processing (as opposed to Steedman, where this constraint comes in later). This approach is also similar to the 'Intonational Unit Storage Hypothesis' by Croft (1995), where he suggests that grammatical constructions stored (or 'pre-compiled') in long-term memory generally correspond to intonational units in production due to working-memory constraints.

Taking this prosodic packing view, Speer and colleagues (1996) made three predictions regarding the alignment of prosodic and syntactic boundaries during comprehension:

1. When prosodic and syntactic boundaries coincide, syntactic processing should be facilitated.
2. When prosodic boundaries are placed at misleading points in syntactic structure, syntactic processing should show interference effects.
3. The processing difficulties that have been reliably demonstrated in reading experiments for syntactically complex sentences should disappear when those sentences are presented with a felicitous prosodic structure in listening experiments.

They went on to affirm these hypotheses in a series of experiments using sentences with temporary syntactic closure ambiguities and either congruent, incongruent, or neutral prosody-syntax alignment. The prosodic boundaries were induced through a combination of naturalistic phrase-final lengthening and tonal scaling cues, and participants listened to these sentences with the task of responding when they had understood the sentence's meaning. The results are clear (figure 8):

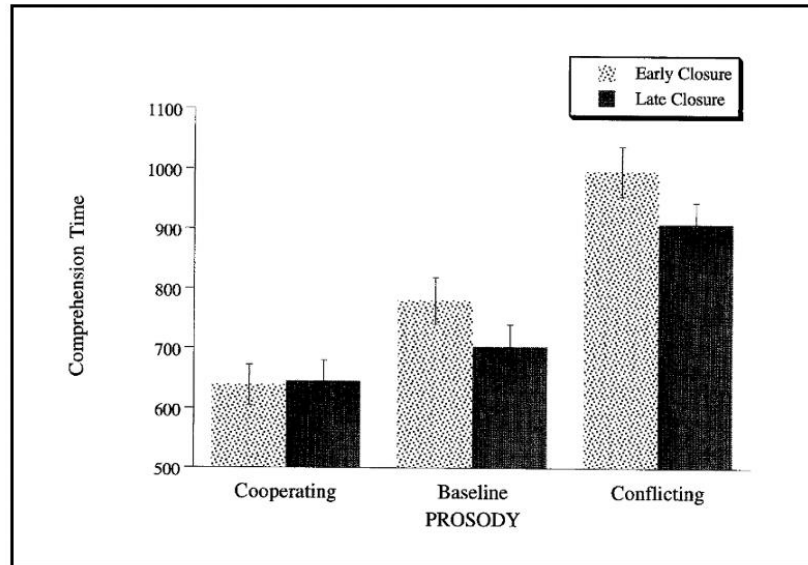


Figure 8: Response time results showing differences for either cooperating, baseline, or conflicting prosodies (Speer et al, 1996)

A follow-up study sought to replicate this effect while also more explicitly measuring comprehension accuracy (Kjelgaard & Speer, 1999). The results clearly replicate the original response times results (although with some differences) and crucially show a huge effect on comprehension accuracy (figure 9). The size of this effect was taken to suggest that prosody plays a foundational rather than just ancillary role in comprehension.

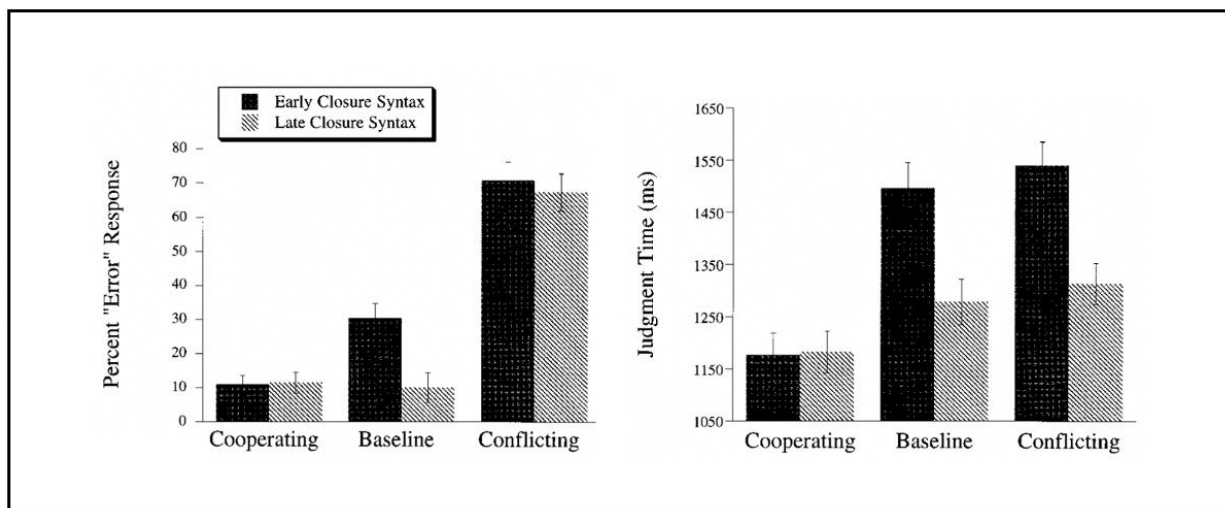


Figure 9: Left, shows comprehension accuracy and Right, response times (Kjelgaard & Speer, 1999)

Taken together, this prosodic packaging view suggests a more basic function than communicative disambiguation, as might be implied by syntax-prosody correspondence theories (Selkirk, 2011). Instead positing a more direct cognitive role, prosodic packing theories claim that prosody “serves to hold distinct linguistic representations together in memory” (Frazier et al, 2006).

Large effects of perceptual packaging on presumably abstract combinatorial thought are not unique to language. An analogous effect has also been observed for the processing of visually presented algebraic sequences with conflicting or supporting spatial grouping cues (Landy & Goldstone, 2007, 2010; Marghetis et al., 2016; chapter 7; figure 10). Thus, it appears to be a domain-general cognitive principle that perceptual representations affect working memory representations and thus constrain further combinatorial processing.

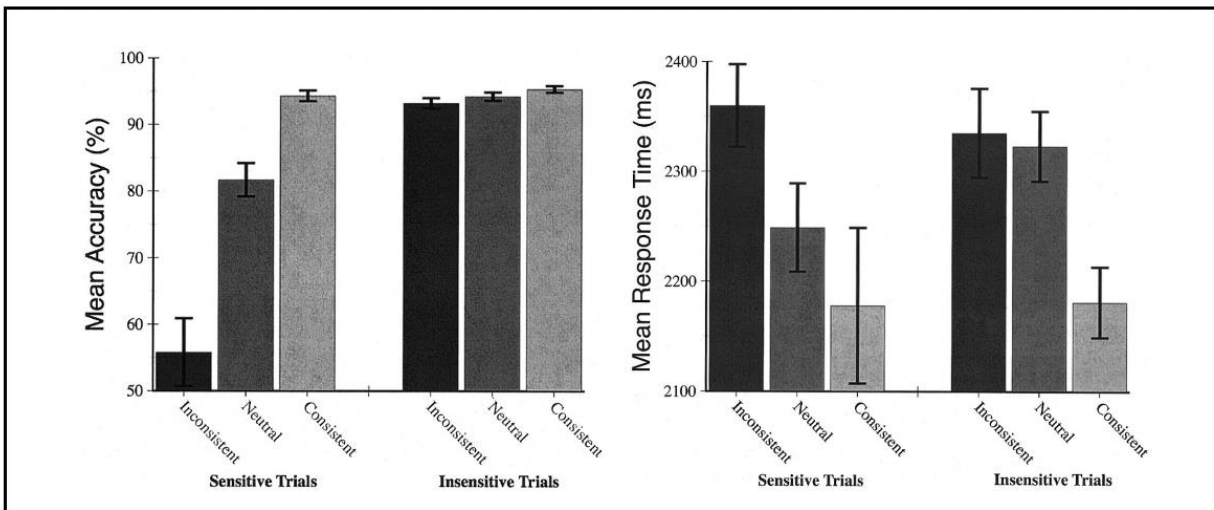


Figure 10: Results from Experiment 1 from Landy & Goldstone (2007a). Sensitive vs insensitive trials refers to whether the proximity manipulations cut across an algebraic syntactic boundary or not (see chapter 7 for more extensive explanation).

3.2.5. Summary of prosodic constituency

The prosodic hierarchy describes how language is hierarchically organised into perceptually distinct units. The alignment of prosodic and syntactic hierarchies has then been of theoretical interest and helps to explain how choices in prosodic phrasing of an utterance shape its syntactic interpretation. Notably, prosodic packaging theories posit prosodic constituents as playing a distinct cognitive role in holding information together in memory.

3.3. Metrical structure

Whereas prosodic constituent structure determines the perceptual units in a speech signal, metrical structure aligns these units with a hierarchical grid. This grid represents patterns of perceived prosodic prominence. This approach to representing prominence was developed in the tradition of metrical phonology (Halle & Idsardi, 1995; Halle & Vergnaud, 1987; Hayes, 1995; Liberman & Prince, 1977; Prince, 1983; Selkirk, 1984) and differs from the previous approach where prominence did not have its own independent structural representation (Chomsky & Halle, 1968).

As inherited from music theory, the metrical grid is depicted in terms of a series of points whose horizontal arrangement represents some sequence of notionally equal entities occurring over

time (e.g. beats or syllables), and whose vertical arrangement represents relative metrical strength among these entities. Metrical strength manifests within a given horizontal plane of the metrical grid as a regular alternation of relatively ‘strong’ and ‘weak’. This quality of metrical strength applies to *beats* in music (Lerdahl & Jackendoff, 1983) and to syllables (and their metrical projections) in language (Liberman & Prince, 1977; Prince, 1983; Halle & Idsardi, 1995; Hayes, 1995).

There are also important differences between how metrical phonology and music theory make use of this grid formalism. Particularly, the fact that music theory specifies beats as the fundamental grid element and phonology specifies syllables has consequences for how timing information is represented (Brown et al., 2017; Lerdahl, 2001; Patel, 2008). However, for present purposes, we will assume that their commonalities arise from a shared cognitive mechanism and that their differences arise from other factors that combine with this shared mechanism.

Importantly, prosodic prominence is a cognitively constructed percept not a description of speech acoustics. There is no universal phonetic correlate that guarantees its perception. Neither is the perception of prominence limited to the influence of local phonetic cues, but rather can be influenced by distal (nonlocal) context including the abstract syntactic, discourse, and informational structure of the unfolding linguistic message (Cole et al., 2019; Dahan, 2015; Turnbull et al., 2017). Prominence can also be influenced by non-auditory perceptual cues such as when speakers rhythmically gesture in time with their speech (Biau et al., 2016; Hubbard et al., 2009).

3.3.1. Meter and segmentation

One of the original aims of the theory of metrical phonology was to better account for how the patterns of stress observed in speech were part of a “hierarchical rhythmic structuring that organizes the syllables, words, and syntactic phrases of a sentence” (Liberman & Prince, 1977). Articulating an extreme version of this, Prince (1983) suggested that metrical stress could directly code prosodic grouping. More recent theory, however, takes an intermediary position that acknowledges a close but importantly non-determinative relation (Halle & Idsardi, 1995; Wagner, 2010; see also related discussion for music: Lerdahl & Jackendoff, 1983).

One way this is manifest is through metrical influence on speech segmentation. That is, from the raw acoustics of a speech signal, it is not straightforward to determine where one word ends and another begins. There is rarely an equivalent of ‘white space’ like there is in written text. Words tend to blur into each other without any consistent pausing to delineate boundaries.

How might meter help? The syllable structure of words has systematic tendencies in metrical organization (Hayes, 1995). For example, most English words have their primary stress on the first syllable (Cutler & Carter, 1987). Cutler and colleagues therefore proposed that competent English speakers interpret word-level speech stress as a segmentation cue for the left-edge of a content word (Cutler et al., 1997; Cutler & Butterfield, 1992; Cutler & Norris, 1988).

Building on this earlier work, Lee and Todd (2004) proposed a more general model for how duration, intensity, and pitch cues in the speech signal interact in determining cross-linguistic differences in rhythmic prominence perception and how this relates to ensuing segmentation strategies. Their model notably invites principled parallels to how rhythmic grouping and meter are perceived in music, and how segmentation strategies in both domains emerge from a common basic perceptual mechanism.

Metrical segmentation can also act at a distance. That is, the perception of metrical prominence is not solely determined by the local phonetic properties of a given isolated syllable but is rather richly constrained by the preceding context. This preceding context coagulates into metrical expectations, weighted by the strength of confirming evidence, which shape how future speech events are processed (Large & Jones, 1999; Liberman & Prince, 1977; Niebuhr, 2009; Pitt & Samuel, 1990; Quené & Port, 2005). With regard to segmentation, metrical cues in a preceding context (e.g. differences in pitch, intensity, speaking rate, and syllable duration) affect how lexically ambiguous sequences of syllables are perceived. For example, affecting whether someone hears the same speech input as “hand shake” or “handshake” (Brown et al., 2011, 2015; Dilley & McAuley, 2008; Kaufeld et al., 2019). Under certain circumstances, metrical context can even make words or syllables within a word perceptually disappear altogether (Baese-Berk et al., 2019; Dilley & Pitt, 2010; Morrill et al., 2014).

Meter is also implicitly constructed in silent reading (despite lack of overt sound) and shows similar distal segmentation effects. Breen and Clifton (2011, 2013) showed with eye-tracking that if a silently read word clashes with its preceding metrical context, then it is read more slowly. The electrophysiological signature of metrical violations in overt speech also mirrors that during silent reading (Breen et al., 2019). Further supporting the an active amodal role of meter, Kentner (2012; see also Kentner & Vasishth, 2016) shows that metrical constraints on speech rhythm, such as avoiding ‘stress-clashes’ (Kelly & Bock, 1988), affect the resolution of morphosyntactic ambiguities at an early stage of processing during both silent reading and when reading sentences aloud. This also implies an early role for meter in the planning of utterances during production.

Taken together, these results cohere with theories of meter and grouping in music, where complex distal interactions are also thought to shape segmentation (Lerdahl & Jackendoff, 1983). The distal nature of this also fits with more general perceptual theories such as *cue integration* (Kaufeld et al., 2019, 2020; Martin, 2016), which posit flexible integration of information over multiple timescales and sources in resolving the objects of perception and comprehension.

Formalising this segmentation function of meter, some theories of metrical phonology have explicitly included a bracketing notation on grid representations, whereby brackets denote grouping boundaries that are implied by the metrical accents (Fabb & Halle, 2008; Halle & Idsardi, 1995; Wagner, 2010; figure 11).

more complex/irregular metrical patterns there appears to be a threshold of ~160ms (Repp et al., 2005; see also ~300ms for anti-phase tapping in: Pressing, 1999). Importantly, these thresholds closely mirror the syllable-rate limits observed for speech intelligibility (Doelling et al., 2014; Ghitza & Greenberg, 2009), suggesting shared perceptual constraints.

However, a more substantial discrepancy concerns the representation of timing. The rows of a metrical grid in music theory formally represent periodicities/beats (Lerdahl & Jackendoff, 1983; London, 2004). Whereas, the rows in linguistic grids only represent sequential order between relative prominences in the syllable structure (where timing is left unspecified). As such, Lerdahl (2001, 2013) has argued that it is better to instead call them 'stress grids' so as to differentiate them from 'metrical grids' in music.

One possible justification for abstracting away temporal information in stress grids is that timing in language is simply more variable. In making this argument, Patel (2008) notes that the coefficient of variation in speech timing is ~33% (Dauer, 1983) compared to the typical ~5% as measured in sensorimotor synchronisation studies of musical rhythm (reviewed: Repp, 2005b; Repp & Su, 2013). This leads him to claim that "linguistic metrical grids are not abstract periodic mental patterns (like musical metrical grids) but are simply maps of heard prominences, full of temporal irregularities." (Patel, 2008, p. 141). He further notes that it is precisely the property of temporal regularity that allows musical rhythm to have additional properties such as 'silent beats' (London, 1993; Tal et al, 2017) and the related notion of rhythmic notion of syncopation (Patel, 2008, p. 141). Indeed, this conforms with those perspectives reviewed earlier which claim that speech is not isochronous (e.g. Nolan & Jeon, 2014).

However, there are problems with this specific comparison made by Patel. The ~5% estimate for music was based on experiments in the lab where participants are instructed to tap as accurately as possible in time with metronomically precise stimuli. This is starkly different from the ~33% estimate for speech, derived from the rhythms of naturalistic conversational speech. Lacrois, Palmer, and Peretz (2019) offer a more interpretable comparison by having participants tap in time with both naturalistically timed and isochronous speech stimuli and found that the coefficient of variation in the tapping was ~10% in both of those conditions (with the isochronous version being slightly less variable). Although this is higher than the 5% estimate, this is likely a result of the more complex and variable amplitude envelope inherent to speech sounds as compared to the pristinely controlled auditory tones typically used in tapping experiments (see: Schutz & Gillard, 2020). This may be formalised in terms of what has recently been called 'p-centre clarity' and how this may differ systematically between music and language (Villing et al., 2011; see also discussion in: Hawkins, 2014)

Timing variability is also not uniform across different styles of music. Some styles are rhythmically exact, like dance music, and this affords greater rhythmic complexity (Huron & Ommen, 2006; London, 2004, p. 86-87). Other styles are more flexible, where the elastic and more flexible pulse allows for additional expressivity and communicative potential (allowing prosodic phrasing to disambiguate syntactically ambiguous musical patterns). The music theorist David Temperley describes this in terms of a 'syncopation-rubato tradeoff' wherein

musical styles either optimise rhythmic complexity (with strict timing) or melodic/harmonic complexity (with flexible timing; Temperley, 2004). Consistent with this, Sam Mehr and colleagues recently proposed an evolutionary theory of music in terms of ‘credible signalling’ with two evolutionarily motivated modes of music that mirror this distinction (Mehr et al., 2020; see also my own commentary on this article: Hilton et al., 2021).

More generally, the advent of music recording technologies in the 20th century has influenced preferences around rhythmic precision (Katz, 2004). This may therefore mean that the prevalence of rhythmic precision in popular music today may be historically unrepresentative. Indeed, modern popular music differs dramatically from prevailing traditional folk varieties which tend to retain especially high degrees of (speech-like) timing variability while still maintaining a basic metrical structure (Johansson, 2017; see also for analysis of large corpus of traditional music: Mehr et al., 2019). Thus, the claim that musical rhythm is universally characterised by higher precision seems overstated.

If differences in timing precision between music and language are exaggerated, might there be isochrony after all? Amalia Arvaniti claims that there has been a conceptual confusion between *timing* and *rhythm* — “timing is concerned with the *durational characteristics* of events, while rhythm has to do with the *pattern of periodicities* that is [perceptually] extracted from these durations” (Arvaniti, 2009; p. 59, my emphasis). She also references the work of Dauer (1983), who was critical of strict typological differences between languages along stress/syllable/mora timing lines, however she argued that (quasi-)isochronous interstress timing was nonetheless a linguistic universal and functioned to support the regular grouping of speech sounds. While she broadly agrees with Dauer’s conclusion regarding isochronous timing, she disagrees with the methodological approach of measuring *surface-level durations* to assess this hypothesis.

There are two main reasons for this disagreement. The first is that perceived prominence can be contextual with no stable set of physical cues in the signal (Beckman & Edwards, 2010; Norcliffe & Jaeger, 2005; Turnbull et al., 2017), and indeed different languages/cultures often rely upon subtly different cues (Miller, 1984). The second reason is that rhythmic perception can be robust to a degree of variability through ‘quantising’ the perceived durations and by predictively continuing a perceived periodicity in the face of counter-evidence. Indeed, this is a fact of perception more generally (see: Goldstone & Hendrickson, 2010). Confirming this for speech perception, Lehiste (1977) observes that people are worse at discriminating durational differences between speech sounds than non-speech sounds, suggesting a perceptual compensation mechanism for expected timing variability in speech.

Recently, methods have been developed to quantify isochrony differently from duration-based metrics used in this previous research. Fourier (frequency-domain) analysis of the amplitude envelope of speech signals makes no assumptions about what prosodic units should be isochronous but rather, in a more theory neutral way, assesses directly the notion of periodicity in the vocalic amplitude envelope of the acoustic signal. Using such methods, Tilsen and Johnson (2008) analysed a speech corpus with this approach and found substantial periodicity

at especially the 1-2Hz and 4-5Hz range. They then quantitatively related differences in degree of periodicity to phonetic phenomena like consonant- and vowel-deletion.

One limitation of the Fourier transform is that it effectively assumes stationary periodicities. Using empirical mode decomposition, Tilsen & Arvaniti (2013) got around this limitation, replicating the presence of two primary spectral peaks cross-linguistically—~5-6Hz (syllables) and ~2Hz (accented syllables)— while also quantifying their stability and evolution over time. A similar cross-linguistic and cross-contextual result is shown by Ding and colleagues, who also derived the amplitude-envelope using a neurophysiologically plausible cochlea model (Ding et al., 2016b; see also review on related neurophysiology: Poeppel & Assaneo, 2020; figure 12³). These Fourier-based metrics have also proven informative in the clinical diagnosis of speech disorder, over and above the more commonly used nPVI metrics (Basilakos et al., 2017).

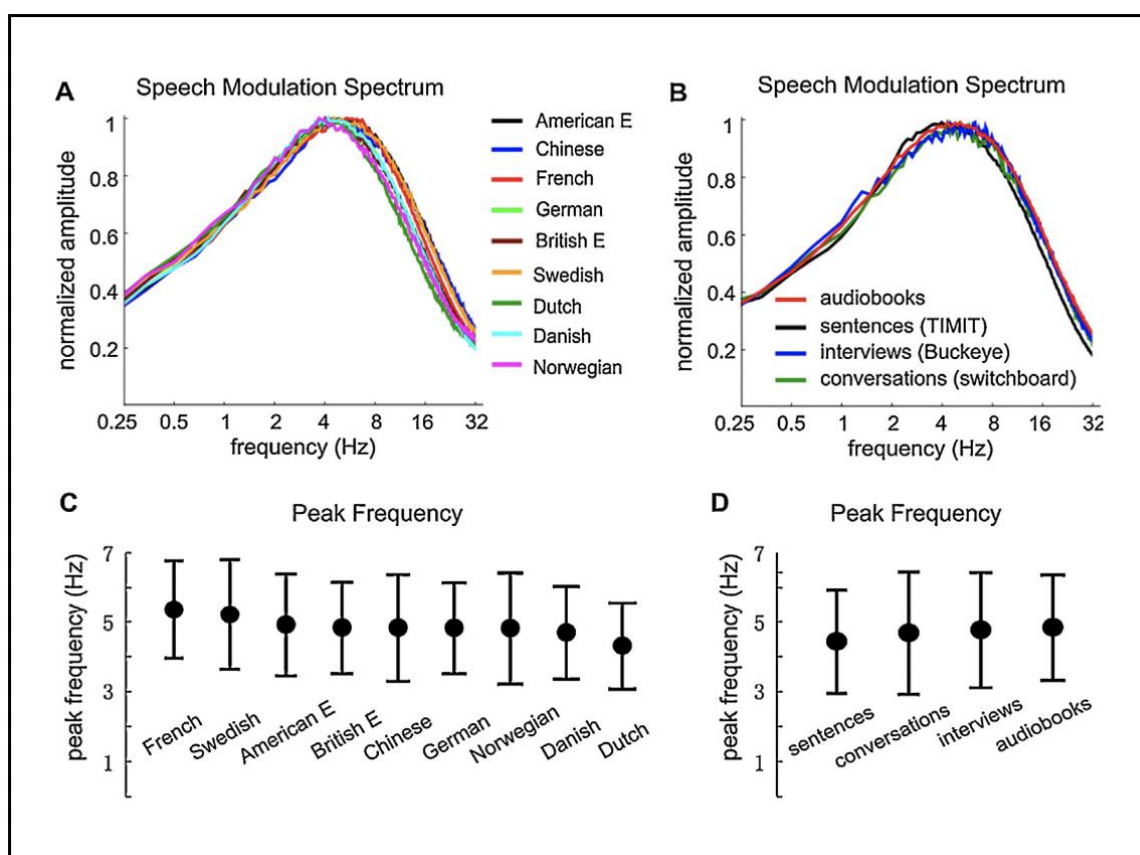


Figure 12: Results from Ding et al (2016). A & C: speech spectrum across languages, B & D: across linguistic contexts.

The periodic structure of speech can also be highlighted through certain experimental paradigms. Speech cycling experiments have participants cyclically repeat a phrase (e.g. “big

³ It is worth noting that the spread of this peak should not be interpreted as something akin to a quasi-normal distribution of frequency values. Interpreting such frequency-domain spectra is more complicated than this such that the spread of a peak can be influenced by a variety of factors including variabilities in frequency, amplitude, and most importantly, nonstationarities in these parameters.

for a duck”) while aligning a specific syllable (‘duck’) with an external isochronous beat (Cummins, 2009b; Cummins & Port, 1998; Tilsen, 2009). This task results in a *harmonic timing effect* whereby syllable timing converges to small-integer ratios of the main timing period, just like metrical timing in music. In other words, by enforcing a single periodicity, multiple phase-locked periodicities naturally emerge. However, it is possible that this increased metricality is a by-product of repetition, as these effects are weaker in non-repeated speech (Jacoby & McDermott, 2017⁴).

Periodic timing constraints in speech, similar to those in music, can also be found in more naturalistic contexts. Sam Tilsen showed that speech timing was more variable and prone to mistakes when the words being spoken imply metrical irregularities (2011).

Brown, Pfordresher, and Chow (2017) showed a similar result, although their theoretical framing of meter differed in an interesting way. Like Arvaniti (2009, 2011), they argued that many earlier attempts at quantifying speech meter were hampered by a confusion between rhythm and meter. To solve this, they motivate the adoption of a ‘musical model of speech rhythm’ wherein western musical notation is used to represent syllable timing. This allows a more precise representation of timing variability in syllable rhythm while simultaneously doing so in a way that highlights higher-level organisation into more regular periodicities. For example, this notational convention captures the intuition that changing the words in the nursery rhyme ‘twinkle twinkle’ results in changes to relative syllable duration that preserve higher level metrical regularity (figure 13).

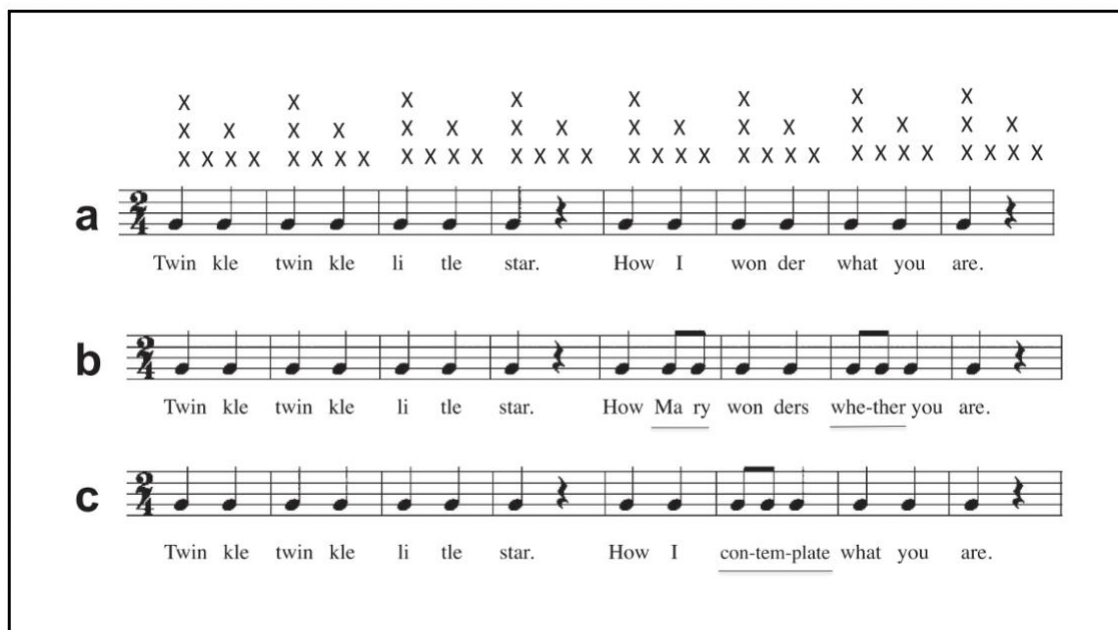


Figure 13: Example of how word substitution results in rhythmic intuitions that

⁴ Although, the result in this paper may have been a result of the particularly slow tempo chosen for the speech reproduction experiment, and the fact that the stimuli were non cyclically repeated, like previous experiments in that study.

maintain larger metrical structure (Brown, Pfordresher, Chow, 2017).

Assessing this model empirically, they asked participants to naturally read sentences while the implied meter was experimentally manipulated by shifting focus prominence such that sentences had metrical regular and irregular versions. The results showed a coefficient of variation of approximately 10% for metrically regular sentences and close to 30% for metrically irregular sentences, thus supporting this more musical approach to speech meter (which is invisible to the traditional 'stress grid' approach; figure 14). However, both the Tilsen (2011) and Brown and colleagues (2017) studies had small sample sizes, so further research is advised to increase confidence in this interpretation.

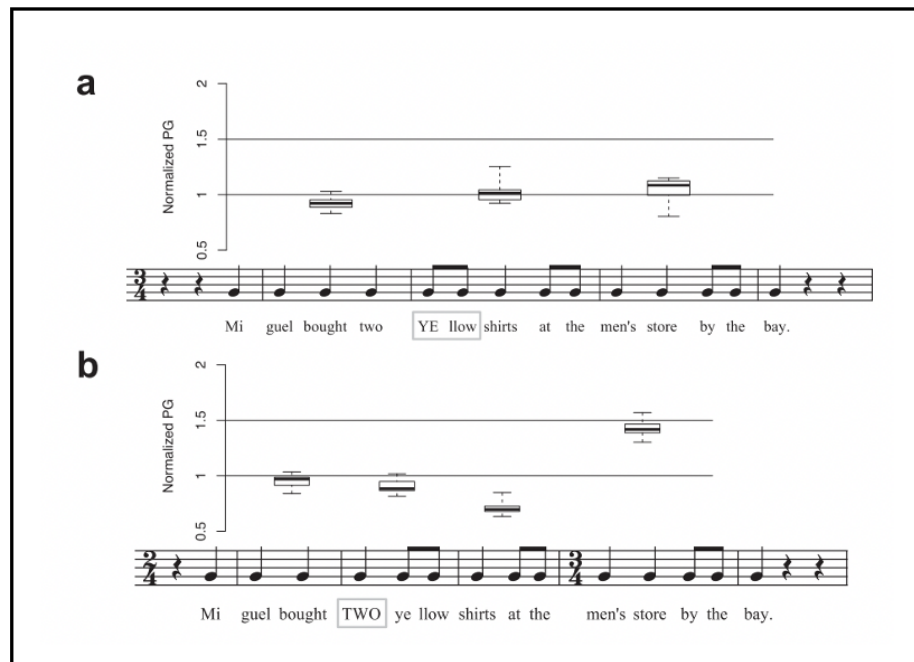


Figure 14: Results from Brown, Pfordresher, & Chow (2017): showing how differences in metrical regularity (top: regular, bottom: irregular) result in less stable metrical rhythm.

Speech is also not uniform in its rhythmicity. Some modes of speaking are more clearly metrical than others such as poetry, oratory/chanting, and joint speech (Brown, 2017; Cummins, 2003, 2020; Fabb & Halle, 2008; Lerdahl, 2001). For example, Mara Breen (2018) shows that meter accounts for variation in speech timing at up to five levels of metrical depth for people reading children's literature (such as Dr. Seuss's 'The Cat in the Hat'). Children's literature often has clear meter, and reinforces this structure using devices such as rhyming. Musical meter also combines with linguistic meter in systematic ways across cultures in the form of song (Gordon et al., 2011; Lerdahl, 2001; Palmer & Kelly, 1992). Joint-speech experiments show that people can synchronise their speaking relatively easily without rehearsal (Cummins, 2003). More generally, metrical coordination is also thought to support turn-taking in conversational speech (Hawkins, 2014; Levinson, 2016; Schultz et al., 2016).

Making sense of this picture, it seems that similarities and differences emerge from complex interaction between varying functional and structural properties of music and language and the different contexts they are realised in (Beier & Ferreira, 2018; Hawkins, 2014). Indeed, a significant portion of speech variability can be explained by prioritising referential communication, which involves rhythmic disruptions to plan what one will say (Ferreira, 2007) and additional variability contributed by the syllable structure of the words used to communicate (Hawkins, 2014). But a common drive for metrical coordination seems to be common to both (Port, 2003).

3.3.3. Summary of metrical structure

The metrical hierarchy describes how speech is aligned to a grid structure that shapes its prominence and timing characteristics. Although speech rhythm tends to be more variable in timing than most music, it was argued that the notion of a periodically structured metrical grid is still a useful construct in understanding language. Metrical alignment was shown to influence lexical access and speech segmentation at the level of syllables and prosodic words.

3.4. Summary

This chapter reviewed how language is prosodically organised into hierarchical groups and grids and how these structures tend to align with syntactic hierarchies. Prosodic structure was also shown to have similarities to rhythmic structure in music. Despite ongoing contention regarding whether speech rhythm meaningfully contains periodicities and whether it is metrical in the same sense as music, it was argued that the same basic processes and constraints underlie each domain.

4. Study 1: Linguistic syncopation

As discussed in the last chapter, speech rhythm is constrained by a hierarchical metrical structure that shapes its timing and accentuation characteristics. This metrical structure is also thought to confer cognitive affordances for perception, memory, and motor coordination, and to align with phrasal structure in systematic ways. This chapter shows that this alignment affects the robustness of syntactic comprehension and I discuss possible underlying mechanisms. In two experiments, meter-syntax alignment is manipulated while sentences with relative clause structures are either read as text (experiment 1, $n = 40$) or listened to as speech (experiment 2, $n = 40$). In experiment 2, the stability with which participants can tap in time with the metrical accents in the sentences they were comprehending was also measured. In addition to making more comprehension mistakes, sensorimotor synchronization was disrupted when syntactic cues clashed with the metrical context. I suggest that this reflects a tight coordination of top-down linguistic knowledge with the sensorimotor system to optimize comprehension.

4.1. Introduction

Speech has distinct rhythmic properties, described in the last chapter. To briefly reiterate: This is partly because it dances along to a (quasi) periodic and hierarchically organised timing and accentuation structure called meter. Speech acoustics reflect meter (Goswami & Leong, 2013; Tilsen & Arvaniti, 2013; Ding et al., 2016), but more than an acoustic pattern, meter is an internally constructed percept that resonates to neural and motor dynamics (Port, 2003; Hawkins, 2014; Beier & Ferreira, 2018; Tilsen, 2019).

The alignment of speech to metrical structure has cognitive consequences for perception, memory, and motor coordination. Optimal alignment may confer more robust perceptual discrimination; more coordinated speaking; more effective language learning; and more memorable cultural knowledge. In this study, I specifically ask whether metrical alignment affects the robustness of syntactic comprehension, and of a secondary interest, whether it also affects sensorimotor coordination.

4.1.1 Cognitive consequences of meter

Speaking with predictable metrical timing attracts attention and sharpens perceptual discrimination (Dooling, 1974; Pitt & Samuel, 1990; Quené & Port, 2005; Cason & Schön, 2012). And the stress patterns that punctuate speech rhythm affect perceptual grouping (Martin, 1972; Lee & Todd, 2004). For example, one can be pushed between hearing “crisis turnip” or “cry sister nip” depending on the prior rhythmic context and the metrical expectancies they induce (Dilley & McAuley, 2008; Brown et al., 2011, 2015; Kaufeld et al., 2019). Rhythm can even make syllables perceptually disappear altogether (Dilley & Pitt, 2010; Morrill, et al., 2014; Baese-Berk et al., 2019).

Speech spoken with regular stress patterns is easier to remember than when unstressed (Ryan, 1969; Robinson, 1977; Boucher, 2006) or when interstress timing exceeds a range of 0.5-2 seconds (Ghitza, 2017; Rimmele et al., 2020). This is paralleled in music (Mathias et al., 2015; Gorin et al., 2018). Recent models posit mechanisms to explain this based upon entrained neural oscillations (Hartley et al., 2016; Ghitza, 2017; Plancher et al., 2018). Indeed, children who tap in time with rhythms more accurately show stronger neural entrainment to speech as well as more robust auditory short-term memory (Woodruff Carr et al., 2014).

Metrically regular sentences are easier to speak than irregular ones, resulting in fewer speech errors (Tilsen, 2011) and more stable timing (Brown et al., 2017). Metrical perception is also impaired in populations with disordered speech fluency such as stuttering (Falk et al., 2015; Wieland et al., 2015) and aphasia (Stefaniak et al., 2020). External metrical pacing has proven helpful in remediating these motor coordination difficulties (Toyomura et al., 2015), even for improving gait in Parkinson's Disease (Benoit et al., 2014). Neural entrainment is also implicated as a key mechanism underlying these relations to motor coordination (Morillon et al., 2019; see also: Cummins, 2009; chapter 5)

4.1.2 Meter-syntax alignment

Does linguistic structure make use of these cognitive affordances? Meter (and prosody more generally) has strong relationships to linguistic structure and the broader pragmatics of linguistic communication (Ladd, 2008; Calhoun, 2010; Cole, 2015). As reviewed in the last chapter, many theories predict systematic alignments of prosodic structure with syntactic phrase structure. One aspect of this is the systematic tendency for syntax to influence the placement of the strongest metrical stress in the phrase (referred to as 'nuclear-stress' or 'phrasal stress'). In English, the rightmost content word typically receives this phrasal stress (e.g., "seven happy **P**uppies"). More generally (and cross-linguistically), stress follows the most deeply embedded syntactic constituent, which happens to be the rightmost for head-initial languages like English (Cinque, 1993). Exceptions (e.g., Bolinger, 1972; Ladd, 2008) drive ongoing theoretical debate (Wagner, 2010; Selkirk, 2011; Zubizarreta, 2014). For example, there are some open questions about whether prominence serving an information focus function (Breen et al., 2010) is represented metrically and whether the cues for this interact with those serving a more structural function (Kentner & Vasishth, 2016; Wagner & McAuliffe, 2019). Despite these nuances, the various different proposals all acknowledge a systematic relation of meter to syntax in some form.

In this chapter I consider the possibility that this relation (hereafter 'meter-syntax alignment') arises to optimize the robustness of comprehension through the cognitive affordances of rhythm. That is, the cognitive demands of syntactic comprehension may themselves be rhythmically distributed. Meter-syntax alignment may therefore, at least partly, serve to adaptively coordinate meter (with the affordances it brings) to comprehending and producing syntactic sequences (with the demands it has).

4.1.3 Evidence

A few key points of evidence help to motivate this hypothesis. First, syntactic structure cues the perception of phrasal stress over and above acoustic cues (Kentner & Vasishth, 2016; Cole et al., 2017, 2019; Wagner & McAuliffe, 2019; Bishop et al., 2020). This underscores not only the cognitively constructed nature of metrical perception—it is *not* acoustic—but crucially suggests that it actively seeks out meter-syntax alignment. In light of my hypothesis, this may be seen as a form of perceptual expertise (see: Goldstone et al., 2015) that biases metrical perception towards interpretations learned to be optimal in some way (more on this in chapter 8). This top-down influence dovetails other work showing that the perception of prosodic boundaries is also influenced by syntactic expectations independent from acoustic cues (Cole et al., 2010; Buxó-Lugo & Watson, 2016).

Second, if meter serves a cognitive function then one would expect its presence even when language is not explicitly externalised, as in silent reading or inner speech. Building on the notion of implicit prosody (Bader, 1998; Fodor, 2002; Frazier & Gibson, 2015), a growing number of studies show that meter is both generated during silent reading and that it actively influences syntactic parsing (Breen, 2014; Breen et al., 2019; Breen & Clifton, 2011, 2013; Kentner, 2012; Kentner & Vasishth, 2016). Reading comprehension is also more robust when people imagine a vivid inner-voice (auditory perceptual simulation), and this is suspected to relate to a stronger activation of syntax-aligned prosodic representations (Zhou et al., 2019; Zhou & Christianson, 2016).

Finally, if meter-syntax alignment optimizes processing then misalignment should have measurable consequences. Indeed, disruptions to meter affect the discrimination of morphosyntactic violations (Schmidt-Kassow & Kotz, 2008; Canette et al., 2020; Chern et al., 2018; Fiveash et al., 2020; Kotz & Schmidt-Kassow, 2015); lexico-semantic integration (Rothermich et al., 2012; Rothermich & Kotz, 2013); lexical priming (Gordon et al., 2011); and the comprehension of syntactically extended sentences with relative-clause structures (Roncaglia-Denissen et al., 2013).

However, a limitation of these studies is that they manipulate metrical regularity rather than alignment. If the benefits of meter come by way of entrained neural oscillations (see chapter 5), then irregular meter may simply provide a less predictable target to entrain to and thereby a less stable neuro-oscillatory substrate. This is distinct from the question of whether certain alignments (independent from regularity) optimize syntactic comprehension. The only study above that did manipulate alignment directly (Gordon et al., 2011) did so for simple sentence structures and only measured priming in a lexical decision task (in addition to neural measures).

Meter-syntax alignment also has measurable consequences for how infants and children learn language (Langus, Mehler, & Nespor, 2017; Morgan & Demuth, 1996/2014). Infants develop sensitivity to the patterns of metrical prominence in a language as early as 5 weeks of age (Nazzi et al., 1998, 2000), and perhaps even prenatally (Abboub et al., 2016). Once sensitivity to this rhythmic structure is developed there seems to be an innate bias toward ‘prosodically bootstrapping’ aspects of grammar learning like syntactic head-direction from features such as

nuclear-stress (Benavides-Varela & Gervain, 2017; Bernard & Gervain, 2012; Christophe et al., 2003; Christophe et al., 1997; Massicotte-Laforge & Shi, 2015). That is, word order generalizations such as ‘determiner goes before the noun’ in English can be bootstrapped by assuming that the latter is the one that is metrically prominent within a phrase group, thus simplifying the probabilistic learning task faced by infants in learning a language where they have to assign new words to different syntactic categories (and perhaps also learn those categories in the first place).

This effect of meter-syntax alignment on language acquisition is likely complementary to other aspects of prosodic phrasing. For example, more general prosodic phrasing cues for grouping such as between-boundary-pauses, final-lengthening, and edge-tones have been shown to perceptually ground the apprehension of syntactic constituency in acquisition (Hawthorne & Gerken, 2014; also see: Männel & Friederici, 2009).

4.1.4 Metrical timing and grammar

Although not a direct form of evidence for the meter-syntax alignment question, transfer effects have also been observed between musical rhythm and linguistic grammar. Specifically, individual differences in the ability to perceive and produce metrical rhythms (often in the context of music) are associated with relatively enhanced phonological processing (Hausen et al., 2013; Kraus & Chandrasekaran, 2010; Tierney et al., 2017; Woodruff Carr et al., 2016), reading abilities (Bonacina et al., 2018; Gordon et al., 2015; Ozernov-Palchik & Patel, 2018; Woodruff Carr et al., 2014), and competence in producing syntactically grammatical constructions (Gordon, et al., 2015a; Gordon, et al., 2015b). Gordon and colleagues (2015a) specifically showed that the performance of typically developing children on a rhythm discrimination task accounted for 48% of the variance in morpho-syntactic competence after controlling for IQ, socioeconomic status, and prior musical experiences. A follow-up analysis then further showed that this transfer effect of rhythm on grammar manifest primarily in complex syntactic constructions such as relative-clause structures and structures involving passivization movement (Gordon, et al., 2015b).

In the opposite direction, a number of speech and language disorders show deficits to metrical timing. The ability of dyslexic people to process metrical rhythms, for example, is predictive of later reading attainment (Goswami et al., 2013; Huss et al., 2011). And metrically priming can be used to restore impaired morphosyntactic processing in dyslexic children (Canette et al., 2020; Przybylski et al., 2013), and even boost morphosyntactic processing in normally developing children (Chern et al., 2018). Children with developmental language disorders are also less sensitive to meter-syntax alignment than typically developing children (Richards & Goswami, 2019), likely related to perceptual deficits relating to the perception of speech stress (Goswami, 2015). Synthesizing these and many other related findings, an *Atypical Rhythm Risk Hypothesis* has recently been proposed (Ladányi et al., 2020), which sees atypical rhythm abilities as a risk factor for a broad range of speech and language disorders.

The grammatical sub-type of Specific Language Impairment (G-SLI) is also characterised by difficulty with both extended syntactic constructions (i.e. those involving non-local

dependencies) as well as extended phrasal prosody (reviewed: van der Lely & Pinker, 2014). This population is also characterised by abnormalities in the caudate nucleus of the basal ganglia as well as certain areas of a prefrontal loop involving the inferior frontal gyrus and its white matter connectivity to the superior temporal lobe along the arcuate fasciculus. As van der Lely & Pinker note (2014), this corresponds with the dorsal stream of Hickok and Poeppel's influential neural model of speech processing (2007). This dorsal stream crucially involves both the sensorimotor interface and the articulatory network, and has strong connectivity to posterior superior temporal areas implicated in syntactic processing (Matchin & Hickok, 2019). It is notable then that these areas also overlap with those for rhythm and beat-based processing, which also rely upon a sensorimotor interface between motor and auditory areas (Patel & Iversen, 2014; Chen, Penhune, & Zatorre, 2009; Schubotz, 2007). Highlighting the relevance of such sensorimotor interactions, differences in white-matter connectivity between these areas correlates with a robust behavioural measure of speech rhythm as well as abilities for language learning (Assaneo et al., 2019).

This association between extended syntactic constructions and extended prosodic structure in grammatical specific language impairment (Marshall et al., 2009; Marshall & van der Lely, 2009; van der Lely & Pinker, 2014) notably dovetails the results from Gordon and colleagues (2015b), which showed rhythm abilities only correlating with grammatical performance for extended syntactic constructions. It also dovetails the results of Brod and Opitz (2012), who found that musical experience was associated with the ability to learn non-local dependency structures in an artificial language (and not the local dependency structures). In discussing findings such as these, Patel and Morgan (2016) propose two possible explanations: 1) enhanced verbal short-term/working memory in musicians, 2) musical engagement may differentially reward predictions based on hierarchical structure.

4.1.5 The present study

As we have seen, meter can have a number of effects on language processing. There is also precedent for the hypothesis that the systematic alignment of meter to phrase structure optimizes syntactic comprehension. The present study aims to provide a critical test of this hypothesis that, in addressing previous limitations in the literature, both measures syntactic comprehension directly and manipulates meter-syntax alignment independently from metrical regularity.

Experiment 1 does this for reading. Experiment 2 extends this to speech, while also measuring the effect of meter-syntax alignment on online processing through its effect on sensorimotor synchronisation.

4.2 Experiment 1

4.3.1 Experimental design

4.2.1.1 Manipulating syntactic complexity

Complex sentences with relative clauses were used for the primary materials. This means that there are two agents in the sentence that must be relationally bound to their grammatically licenced predicates. For example, in the sentence “the boy that helped the girl got an ‘A’ on the test” one has to determine 1) who helped whom, and 2) who got the ‘A’. These bindings can then be probed using forced-choice prompts as a measure of syntactic comprehension (e.g., asking “did the girl help the boy?”, to which a participant would respond “yes” or “no”).

Complexity can be further manipulated through the extraction of the relative clause (figure 15). Object-extracted sentences are more difficult to process than subject-extracted sentences (in English) because of an increased memory load in tracking syntactic dependencies (Gibson, 1998; Gibson et al., 2005) and as a result of being less frequent in usage (Levy et al., 2012). By having these two levels of syntactic complexity, I am able to assess whether the effect of meter-syntax alignment interacts with syntactically induced memory load. If it does, this would be consistent with meter mediating the robustness of the sentence representation in short-term memory, and thus help to narrow down the interpretation of any congruity effect.

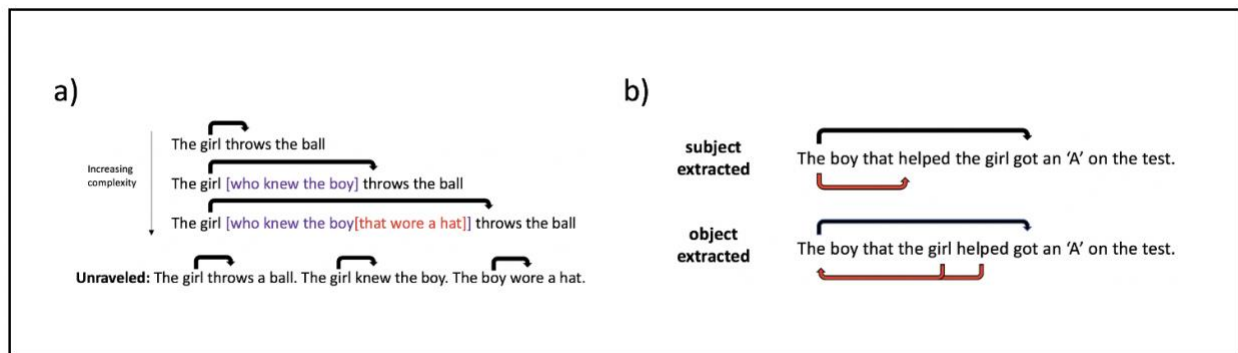


Figure 15: Two ways of manipulating syntactic complexity: a) interposing relative-clauses, b) modifying the extraction of those relative clauses.

4.2.1.2 Sentence materials

Thus, 48 English sentences were composed, each from 12 monosyllabic words (these sentence materials were adapted and extended from those used in Fedorenko et al., 2009; some sentence-final words had two syllables). Each sentence had subject- and object-extracted versions. A further 25 filler-sentences of assorted structure and word length were also used to avoid sentence structure from becoming too predictable (see Appendix A for full sentence list).

4.2.1.3 Syncopating meter-syntax alignment

Congruent and incongruent meter-syntax alignments were defined for both sentence types (figure 16a). The congruent condition aligns primary stress on the rightmost content-word of small phrase groups (clitic group/phonological phrase) to yield regularly spaced metrical accents. The incongruent condition shifted the congruent alignment ‘to the left’ by one position.

For the subject-extracted sentences, this alignment formed a regular binary meter (stress every second syllable) for the noun phrase constituent and a ternary meter (stress every third syllable) for the verb phrase constituent. Enforcing metrical regularity, I rhythmically reduced the function words in the verb phrase (e.g., having “on the” have the same duration as “test”) so that the whole sentence formed a regularly timed binary meter throughout (such reductions are common in natural speech, see: Martin, 1972; Brown, Pfordresher, Chow, 2017). The same procedure was applied for the object-extracted sentences, except that due to differences in sentence structure, a ternary meter formed the most parsimonious meter-syntax alignment throughout the sentence.

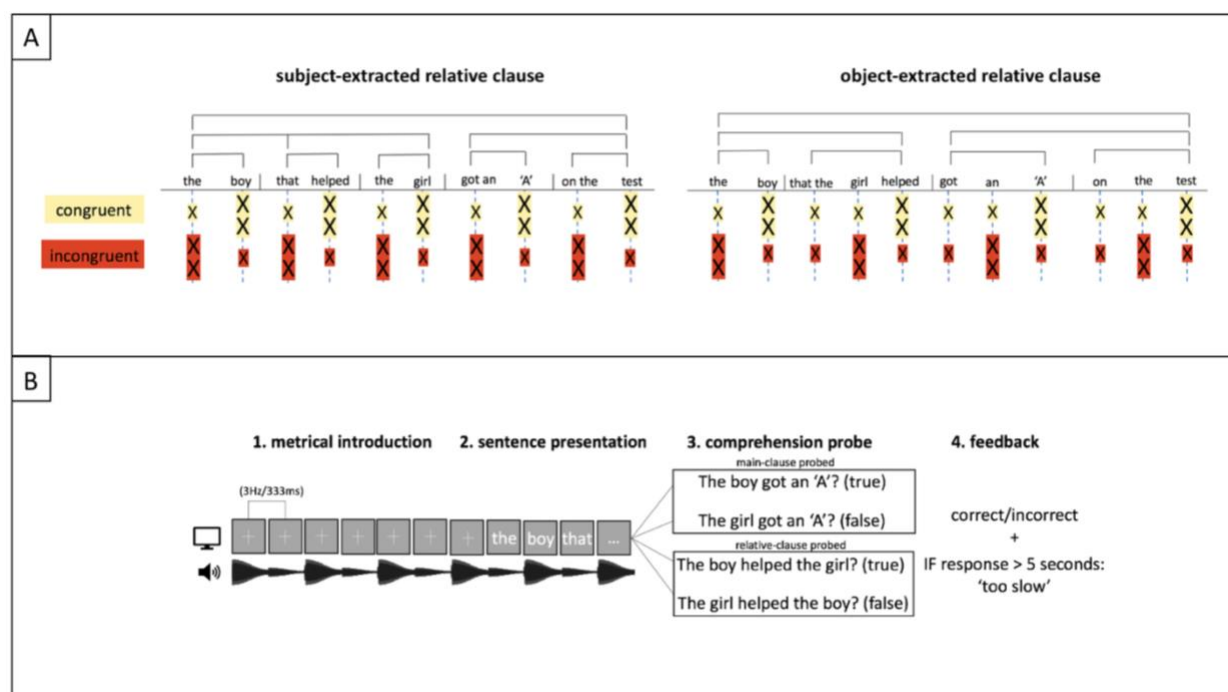


Figure 16: Top (A): Definitions of meter-syntax alignment for both subject-extracted (binary meter) and object-extracted (ternary meter) sentences. Bottom (B): Trial schematic.

4.2.1.5 Auditory materials

In other studies, meter is often manipulated by using sequences of words with different lexical stress patterns. For example, “**S**ally is **H**oping to **T**Ravel to **C**anada” forms a regular ternary pattern, versus “**S**ally is a**V**oiding a **T**RIP to **C**Anada” forms an irregular meter (Tilsen, 2011). This approach cannot manipulate meter-syntax alignment independently from metrical regularity. My approach was to hold the sentence materials constant across the different metrical conditions and instead manipulate meter with rhythmic auditory tones that accompany the sentence presentation (similar approach taken in: Cason et al., 2015; Cason & Schön, 2012; Falk et al., 2017; Falk & Dalla Bella, 2016).

These auditory stimuli were generated using a custom Python script and consisted of a 333Hz pure tone in which a 3Hz beat was induced by amplitude-modulating the signal with an asymmetric Hanning window with 80% depth and a 1:19 ratio of rise-to-fall time. Every second

(binary meter) or every third (ternary meter) tone then received a further 50% amplitude boost to cue strong beats.

4.2.1.5 Experimental procedure

Participants were seated at a chair in front of a computer in a quiet room and supplied with headphones adjusted to a comfortable listening volume. The experiment was run by a custom program written in Python. After an introduction to the task led by the experimenter, participants completed a series of practice trials and were encouraged to ask any clarifying questions. They were told that the experiment was about how rhythm affects reading and that they were to imagine their inner-voice speaking the sentences in the rhythmic pattern provided in each trial. They were also told that this rhythm might make comprehension easier or harder, and that this was part of the experiment: they should not ignore the rhythm. They then completed the main experiment as a single block interspersed with short breaks.

Participants began each trial by pressing the space key. A series of auditory tones then provided a metrical context that either aligned or misaligned with the following sentence, during which time participants attended to a fixation-cross centre screen (figure 16b). Next, the words began visually appearing in the place of the fixation cross as white text on a grey background synchronized to the continuing auditory tones until the sentence was finished. At the end of the sentence, the probe question appeared centre screen (e.g. “the boy got an A?”) and the participant responds as quickly as possible with either “y” (yes) or “n” (no) keys on a keyboard. Additionally, they were given the option of “d” (don’t know) and encouraged to use this if they got distracted and did not register the sentence properly and would otherwise randomly guess; these were quite rare (across all participants: 27 during congruent trials, 37 during incongruent trials) and were discarded from the final analysis. If participants took longer than 5 seconds to respond, they were prompted to speed up on the next trial. Corrective feedback was given after each trial and participants were encouraged to balance speed with accuracy.

The participant saw one of four possible probe questions for each sentence, probing either the main or relative clause, and framing the question such that the correct answer could be either “yes” or “no”. The sentence materials, and their assignment to syntactic complexity, meter-syntax alignment, and probe-clause factors was balanced and randomized in each participant, as well as their order of presentation. Probe question type was also balanced and randomized in this way.

4.2.1.6 Participants

40 native English-speaking volunteers (20 female, 20 male) from the Sydney area took part in this study and were naive to its purpose. They were between 22 and 52 years of age ($M = 28.3$, $SD = 6.2$). They had normal hearing, and normal or corrected vision, and no prior history of speech or language disorders. They provided written consent before commencing the experiment, and the protocol was approved by the University of Sydney ethics committee.

4.2.1.7 Predictions

The design manipulated factors of syntactic complexity (subject- or object-extracted constructions), meter-syntax alignment (congruent or incongruent alignments), and probed clause (main or relative clause): a 2 x 2 x 2 factorial design. I expected meter-syntax alignment to affect both the number of comprehension mistakes and length of response times, in line with other studies of prosody-syntax congruity (e.g., Kjølgaard & Speer, 1999). Thus, I predicted main effects for all three factors: more comprehension mistakes and longer response-times for object-extracted sentences, relative-clause probes, and most importantly, incongruent meter-syntax alignments. Additionally, I predicted an interaction between meter-syntax alignment and syntactic complexity: a stronger cost of misalignment for the more demanding sentence type (e.g., motivated by studies like: Fedorenko et al., 2009).

I preregistered my predictions and analysis plan (<https://osf.io/42msj/>), however, deviate subtly in a way that allows a fuller description of the data. Specifically, unlike the study I based the general design and stimuli on (Fedorenko et al., 2009), I later decided to analyse trials that probe the relative-clause in addition to the main clause. To account for this additional complexity, I used mixed-effects modelling rather than the planned repeated-measures analysis of variance. This change does not affect the significance of the results.

4.2.2 Results

Model comparison Comprehension data (whether correct or incorrect on each trial) were analysed using mixed-effects logistic regression with fixed-effects for congruency, relative-clause extraction, probed clause, and random intercepts for participants. Model comparison using a likelihood ratio test showed no additional benefit for including random intercepts for sentence items ($\chi^2(1) = 0.017$, $p = 0.896$), or for trial number (accounting for practise effects; $\chi^2(1) = 1.242$, $p = 0.265$); these were not included in the final model on grounds of parsimony. While there was also no significant improvement for a trial-number by congruency interaction ($\chi^2(1) = 2.915$, $p = 0.088$), I include it in the final model to facilitate comparison to the response-time model in which this interaction was significant. Additionally, there was a significant improvement for including the probe framing (whether the correct answer is 'yes' or 'no') in the model ($\chi^2(1) = 34.984$, $p = <.001$). There was no significant improvement for including the interaction between syntactic complexity and meter-syntax alignment ($\chi^2(1) = 0.522$, $p = 0.470$), but include it in the model as it was part of our original predictions.

Accuracy All three hypothesized main effects were confirmed (figure 17; table 1), with more mistakes being made for incongruent meter-syntax alignment, when the relative clause was probed, and for object-extracted relative-clause constructions. The syntactic complexity by metrical alignment interaction was, however, not significant.

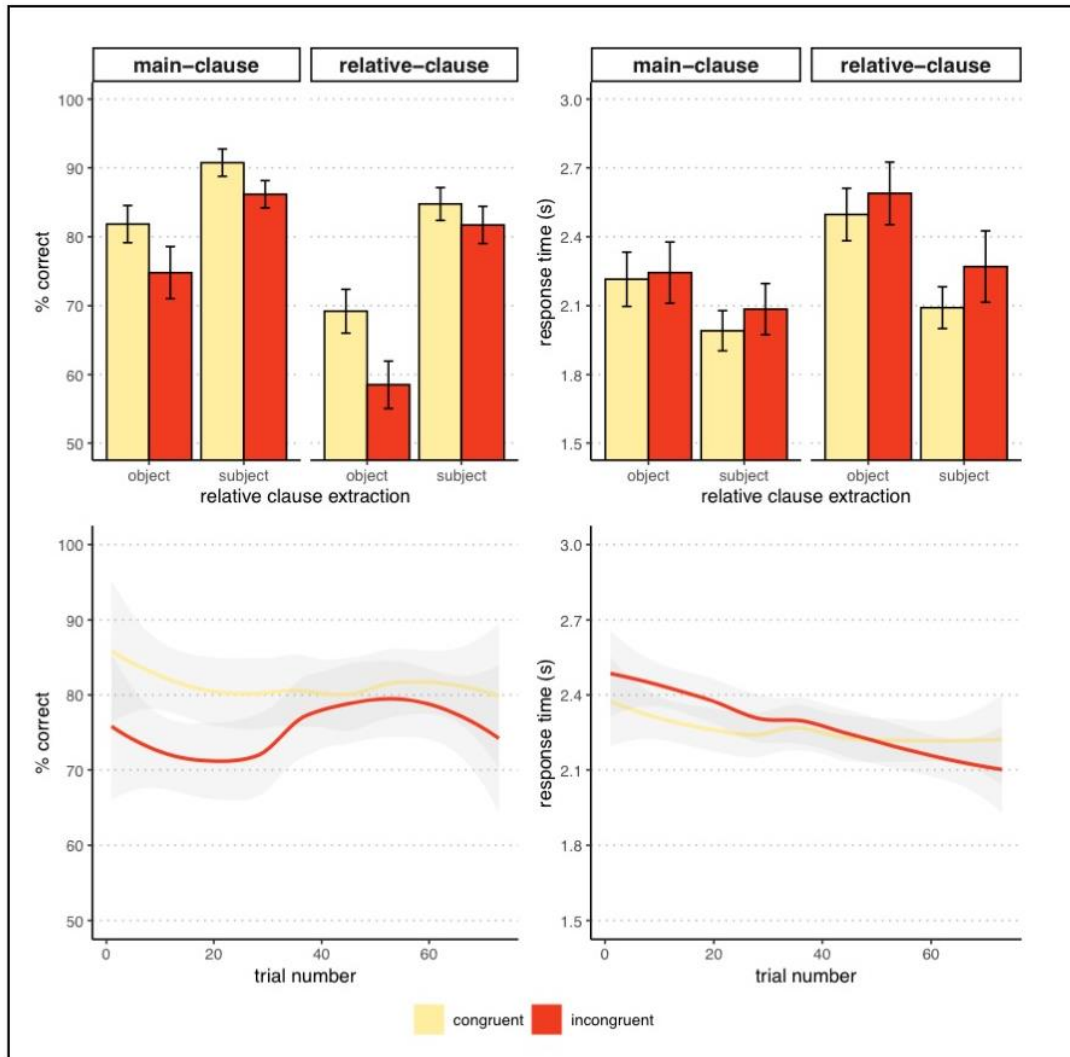


Figure 17: Results from experiment 1. Left: comprehension accuracy. Right: response times. Bottom: how these results distribute over the trials in the experiment (averaging over conditions). Error bars represent SEM.

Response times

Model selection Response times (RTs) were defined as the duration from the presentation of the probe question on the screen to the point at which the participant pressed a key to indicate their response. This was measured in seconds, then log-transformed for further analysis. Responses that were more than 3 standard deviations from the mean were excluded as outliers (these trials were also excluded from the accuracy analysis above). These data were analyzed using a linear mixed-effects model of the same basic structure as the comprehension model. Model fit was incrementally improved by including probe framing ($\chi^2(1) = 37.852$, $p = <.001$), an interaction between meter-syntax congruency and trial number ($\chi^2(2) = 22.964$, $p = <.001$), and whether the participant indicated a “yes” or a “no” for their response ($\chi^2(1) = 11.714$, $p = <.001$). The syntactic complexity by congruency interaction once again was not favoured ($\chi^2(1) = 0.451$, $p = 0.502$).

Table 1: Estimate of fixed effects for accuracy model

	β	Std. Error	Z value	p value
(Intercept)	1.282	0.229	5.600	<0.001 ***
Congruency (incongruent)	-0.866	0.267	-3.243	0.001 **
RC extraction (subject)	0.907	0.183	4.968	<0.001 ***
Probed clause (relative-clause)	-0.672	0.122	-5.487	<0.001 ***
Probe framing (positive)	0.719	0.123	5.850	<0.001 ***
Congruency (congruent) x Trial number	-0.002	0.004	-0.483	0.629
Congruency (incongruent) x Trial number	0.009	0.004	2.262	0.024 *
Congruency (incongruent) x RC extraction (subject)	0.181	0.249	0.726	0.468

Results All three main hypothesized effects are confirmed: slower response times for incongruent meter-syntax alignment, object-extracted sentences, and relative-clause probes (figure 17; table 2). Participants were also significantly faster at accepting rather than rejecting probes, and were even faster when accepting was the correct answer. For transparency I note that the crucial main-effect of meter-syntax alignment is only significant when the interaction between congruency and trial number is included in the model, which although provides a significant improvement to model fit, was not part of my original analysis plan (the effect of meter-syntax alignment on comprehension accuracy in table 1, however, is robust with or without this interaction term).

Table 2: Estimate of fixed effects for response times model (log-transformed units)

	β	Std. Error	t value	p value
(Intercept)	0.793	0.050	15.926	<0.001 ***
Congruency (incongruent)	0.093	0.036	2.413	0.016 *
RC extraction (subject)	-0.109	0.019	-5.711	<0.001 ***
Probed clause (relative-clause)	0.101	0.022	5.295	<0.001 ***
Probe framing (positive)	-0.056	0.023	-2.405	0.016 *
Response (yes)	-0.081	0.024	-3.422	<0.001 ***
Congruency (congruent) x Trial number	-0.001	<.001	-1.733	0.083
Congruency (incongruent) x Trial number	-0.003	<.001	-4.842	<0.001 ***

4.2.3 Discussion

Participants read sentences rhythmically while being probed for syntactic comprehension. I found that they both made more mistakes and took longer to respond to comprehension probes when meter did not align with phrase structure in the typical way. This supports my hypothesis that meter-syntax alignment mediates the robustness of syntactic comprehension and that the typical alignment is optimal. This effect, however, did not interact with syntactic complexity (memory load) as I originally predicted.

Prior studies on meter and reading have investigated how prior text can induce metrical expectations that may clash with the processing of a target section (Breen & Clifton, 2011, 2013; Kentner, 2012). The present study is unique in that it holds linguistic materials constant across conditions and instead manipulates meter through alignment with a rhythmic tone sequence. This allowed us to investigate the effect of metrical alignment independently from metrical regularity (meter was regular across all conditions). Additionally, this highlights how meter during reading is dynamically enacted and more interesting than simply a concatenation of lexical stress patterns (consistent with: Zhou & Christianson, 2016; Zhou et al., 2019).

A limitation of my design is that I assumed that words aligning with amplitude accented tones would be perceived as metrically strong. Metrical strength does not necessarily correspond to acoustic intensity in either music (Lerdahl & Jackendoff, 1983) or language (especially the 'nuclear stress': Wagner & McAuliffe, 2019; although see Breen et al., 2010). Although I think it unlikely, it is possible that some participants were perceiving metrical accent in different positions than assumed in the experiment, thus contributing additional variability in the results.

4.3. Experiment 2

Experiment 2 sought to replicate these findings and address some of its limitations. Rather than *visually* presenting sentences as a sequence of words to read on a screen, sentence materials were instead presented as *auditory* speech. To ensure that participants were perceiving the meter as intended, I instructed them to tap their finger in time with the strong beat both during the metrical introduction to each trial and to continue such tapping during the subsequent speech stimuli. Further extensions to the design were also employed, which are now described.

4.3.1 Experimental design

4.3.1.1 Meter-syntax alignment

Experiment 2 expanded the basic logic of the meter-syntax alignment manipulation to allow for three levels of alignment for each sentence type. To achieve this, both sentence types were set to a ternary meter in one of three possible alignments determining its congruency. The congruent condition, as before, had the metrically strong beats align with the right-edge of the syntactic phrases. The other two incongruent conditions then aligned this strong beat with either a phrase-medial position (incongruent-1; this was the same as the incongruent condition from Experiment 1) or with the left-edge of the phrase (incongruent-2; see figure 18a).

A limitation of the adopted design is that in order to fit the subject-extracted structures into a ternary meter (and thus to allow three possible alignments) certain rhythmic compromises were required that confound comparisons regarding syntactic complexity. Specifically, the bolded words in, for example, “the **boy** ____ that **helped** ____...” were followed by a silence (with the duration of a syllable) to fit the ternary meter (see figure 18a). This confounds the interpretation of the syntactic complexity effect and congruity by complexity interaction; however, this was not the focus of this study and it does not confound the central congruity manipulation. The subject-extracted sentences here serve to show the robustness of the effect to different structures and to ensure structure was not predictable between trials.

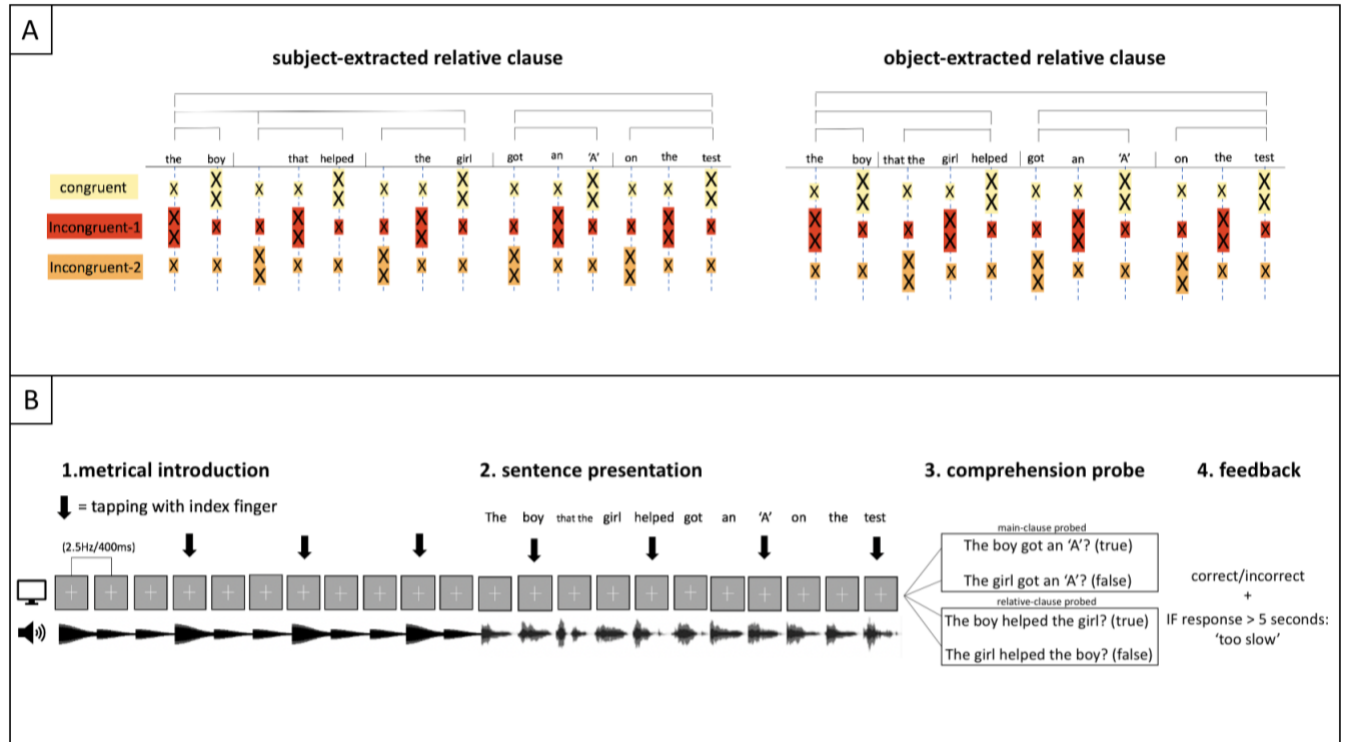


Figure 18: Top (A): Meter-syntax alignment definitions. Bottom (B): trial schematic.

4.3.1.2 Sentence materials and speech synthesis

To accommodate the additional meter-syntax alignment conditions, ensuring enough trials per condition, the original 48 sentences and probes from Experiment 1 were extended to a new total of 72 sentences with the same basic properties (see Appendix A).

Speech stimuli were synthesized and preprocessed from these sentence materials with a custom python script, using Google's text-to-speech API (i.e. the voice from Google Translate). This script firstly generated audio-files for each word in the sentence individually. These individual speech sounds were then preprocessed to standardize duration, amplitude, and pitch qualities in the signal. Finally, they were strung together into a whole sentence with a constant syllable rate of 2.5Hz (this slower presentation-rate was arrived at through piloting). This synthesis procedure allowed us to dismiss the possibility that there were any systematic prosodic cues in the signal which could possibly influence the perception of meter or prosodic phrasing.

Auditory beat stimuli were the same as Experiment 1 except for now being at the slower 2.5Hz tempo.

4.3.1.3 Sensorimotor synchronisation

To assess the effect of meter-syntax alignment on sensorimotor synchronization, I measured how accurately participants tapped in time with the speech they were trying to comprehend (figure 18b). Their tapping was recorded on a MIDI drum-pad (Korg Nanopad) that registered

timing and pressure of each tap, and they used their dominant index-finger. Participants received no auditory feedback from their tapping and volume was calibrated such that they could not hear the physical sound of the tapping.

To estimate the effect of meter-syntax alignment on sensorimotor synchronization, I compared tapping variability to the speech stimuli to a rhythmically matched non-linguistic control task. Replacing the speech sounds of the object-extracted sentences with auditory tones yields a simple undifferentiated triplet rhythm that is the same regardless of alignment, yielding a single non-linguistic rhythm to compare against the three object-extracted alignments (rhythm 4 in figure 19a). However, due to the alternating long-short rhythm of the subject-extracted sentences, each alignment condition produces a cognitively distinct rhythmic pattern, yielding a separate rhythm for each of the three alignment conditions (rhythms 1, 2, 3 figure 19a). Participants were presented six trials for each of these four different rhythm types. The start of each of these trials had an additional accented bass tone which served to indicate where the participant should tap. This dropped out after four repetitions, leaving the participant to continue the pattern.

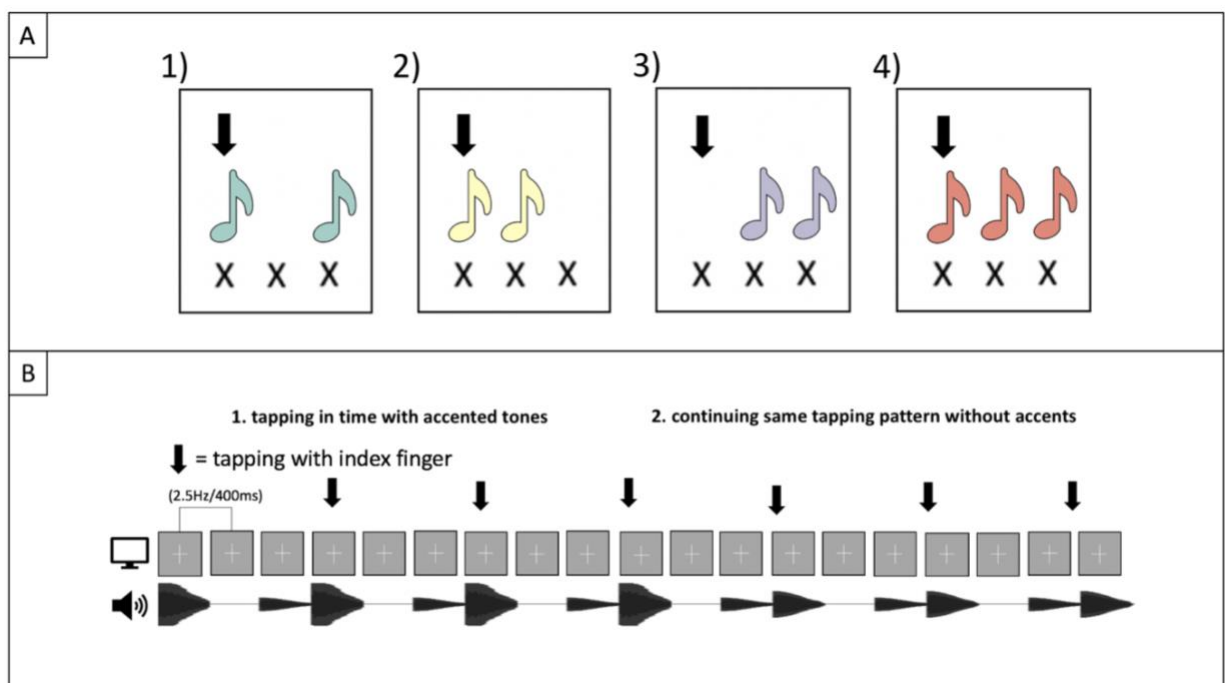


Figure 19: Top (A): The four rhythmic patterns used for sensorimotor synchronization (no language) part of the experiment. Bottom (B): trial schematic of the non-linguistic tapping task (using the first of these rhythms as an example).

4.3.1.4 Experimental procedure

The same basic experimental procedure as Experiment 1 was used, except that participants were now required to tap on a drum-pad in time with metrically accented beats. They were told that the experiment was investigating how this rhythm affects their comprehension, so were to not ignore the rhythm. Participants were given instructions on tapping and a series of practise

tasks. They then completed the non-linguistic rhythm control task in a single block (lasting ~15mins), each participant seeing a unique randomised order of trial types.

After the non-linguistic metrical rhythm section, participants completed the 72 sentences of the main language task in a single block (interspersed with short breaks). They were encouraged to balance attention equally between the language aspect of the task and tapping as accurately as they could with the beat. The trial structure during the main experimental block was similar to that of Experiment 1, except that the metrical tones from the introduction to each trial stopped playing while the auditory speech played (see figure 18b). Importantly, this meant that there were no physical cues to meter during sentence presentation. Thus, it was presumed that the metrical percept during the speech would be induced by preceding distal metrical context, and reinforced and maintained by the participant's finger tapping.

4.3.1.5 Participants

40 new English-speaking volunteers from the Sydney area (17 female, 23 male) were recruited, who met the same basic requirements as in experiment 1. They were between 21 and 65 years of age ($M = 29.3$, $SD = 8.6$).

4.3.1.6 Predictions

Meter-syntax alignment I predicted that the congruent alignment would result in fewest comprehension mistakes and fastest response times, as compared to either incongruent-1 or incongruent-2 alignments. In adjudicating between the two incongruent alignments, we note that incongruent-1 places the metrical stress in a medial position in each group ("on **THE** test") and incongruent-2 a left-aligned position ("**ON** the test"). Prior theory notes a tendency for aligning prominences to edges of metrical constituents (see: Prince, 1983), thus I weakly predict incongruent-1 to have a stronger cost on comprehension over incongruent-2.

Sensorimotor synchronization I also predicted that meter-syntax alignment would affect the variability of sensorimotor synchronization, such that congruent alignment would show the least variable tapping and incongruent-1 the most variable.

4.3.2 Results

Model selection The same analysis approach is used here as in experiment 1, with model selection procedures leading to the same basic structure. Additionally, including a predictor based on sensorimotor synchronization (tapping ratio; see section 3.2.3) did not improve model fit ($\chi^2(1) = 1.376$, $p = 0.241$), however, including an interaction between this synchronization metric and congruency did improve model fit ($\chi^2(1) = 6.422$, $p = 0.011$).

Accuracy The results replicate those of experiment 1: more mistakes were made for (both) the incongruent conditions, object-extracted sentences, and when the relative-clause was probed (figure 20; table 3). Like before, there was also an interaction between congruency and trial number: indicating that participants were able to adapt to the incongruent meter-syntax alignment over time.

Interestingly, there was also an interaction such that those with more variable tapping scored higher in comprehension for the object-extracted sentences. This seems to suggest that participants gave less attention to the tapping aspect of the task for the more difficult sentences. Additionally, while the negative effect on comprehension was numerically larger for the incongruent-1 alignment over incongruent-2, this difference was not itself statistically significant ($\chi^2(1) = 2.408$, $p = 0.121$).

Table 3: Estimate of fixed effects for accuracy model

	β	Std. Error	Z value	p value
(Intercept)	0.573	0.256	2.235	0.025 *
Congruency (incongruent-1)	-0.817	0.224	-3.647	<0.001 ***
Congruency (incongruent-2)	-0.498	0.221	-2.533	0.024 *
RC extraction (subject)	0.802	0.255	3.140	0.001 **
Probed clause (relative clause)	-0.480	0.091	-5.303	<0.001 ***
Probe framing (positive)	0.651	0.091	7.133	<0.001 ***
RC extraction (object) x tapping ratio	0.295	0.130	2.265	0.023 *
RC extraction (subject) x tapping ratio	-0.200	0.168	-1.190	0.234
Congruency (congruent) x Trial number	0.005	0.004	1.146	0.252
Congruency (incongruent-1) x Trial number	0.013	0.004	3.561	<0.001 ***
Congruency (incongruent-2) x Trial number	0.008	0.004	2.168	0.030 *

Response times Again, I adopt the same basic model structure as for experiment 1 for analysing log-transformed response times (outliers 3 SDs from mean also excluded). However, unlike with the comprehension accuracy results, adding tapping variability as a predictor to the model did not significantly improve model fit ($\chi^2(1) = 0.833$, $p = 0.362$) nor did its interaction with congruency ($\chi^2(3) = 4.387$, $p = 0.223$). Both probe framing and response further improved fit ($\chi^2(1) = 14.002$, $p = <0.001$; $\chi^2(1) = 32.216$, $p = <0.001$). Trial number also improved model fit ($\chi^2(1) = 41.072$, $p = <0.001$) and including an interaction between this and congruency improved the model further ($\chi^2(2) = 6.310$, $p = 0.043$). However, including this interaction explained away the variance of the effects of congruency and probe framing, and likelihood tests preferred the simpler models that then dropped these from the model. I adopt this latter model, but the alternative without the congruency by trial number interaction is included in the appendix (which show significant fixed effects for both incongruent alignments).

Table 4: Estimate of fixed effects for response time model (log normalized units)

	β	Std. Error	t value	p value
(Intercept)	0.941	0.039	24.227	<.001 ***
RC extraction (subject)	-0.092	0.014	-6.460	<.001 ***
Probed clause (relative clause)	0.097	0.014	6.878	<.001 ***
Response (yes)	-0.102	0.014	-7.049	<.001 ***
Congruency (congruent) x Trial number	-0.003	<.001	-7.810	<.001 ***
Congruency (incongruent-1) x Trial number	-0.002	<.001	-3.820	<.001 ***
Congruency (incongruent-2) x Trial number	-0.002	<.001	-4.110	<.001 ***

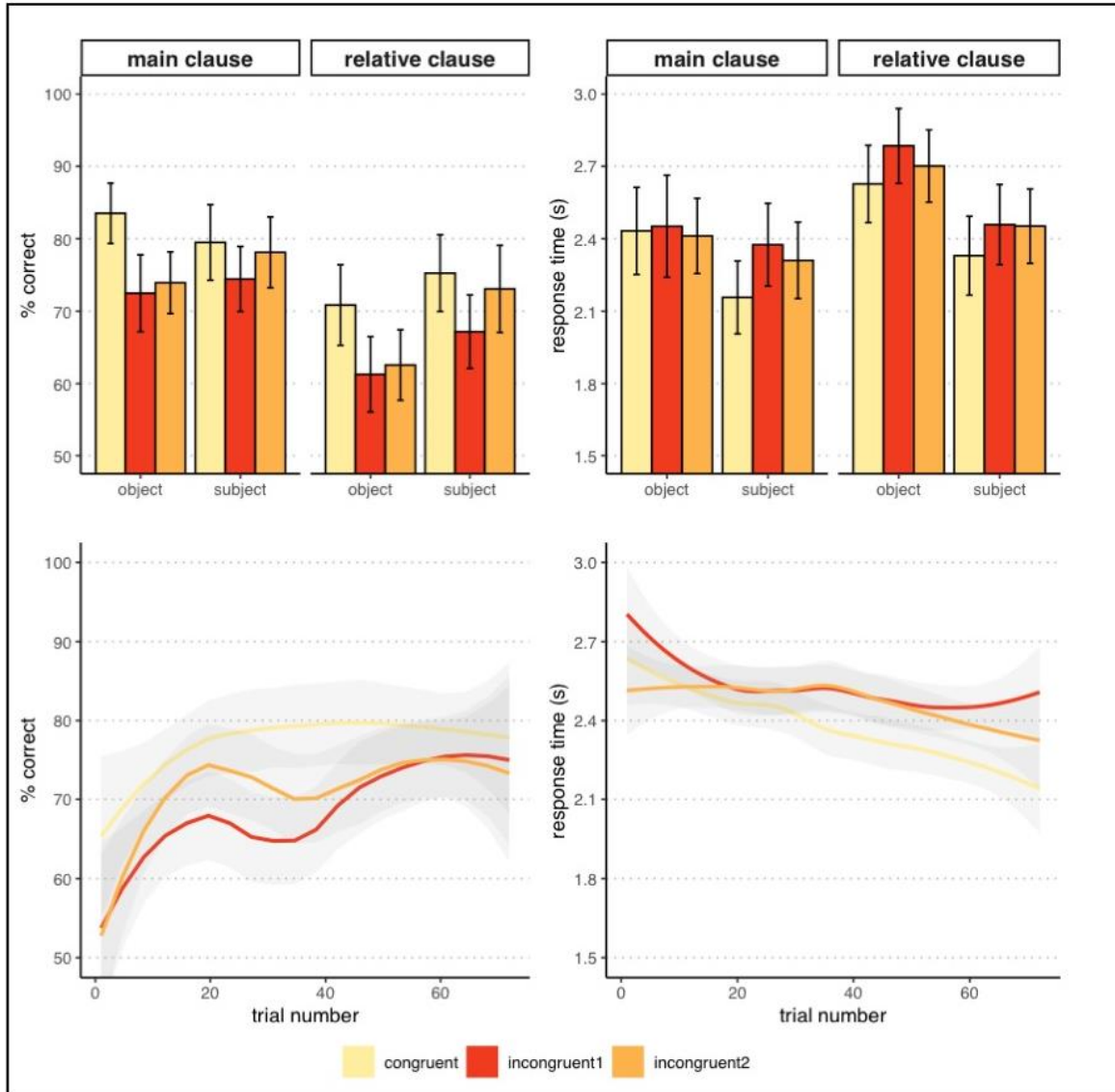


Figure 20: Results from experiment 2. Left: accuracy results, Right: response time results. Bottom: how these results distribute over the trials in the experiment (averaged over all conditions). Error bars represent SEM.

Sensorimotor synchronisation Finally, to assess the effect of meter-syntax alignment on sensorimotor synchronization I analysed the ratio of tapping to speech to tapping to rhythmically matched non-linguistic tones: scores greater than 1 indicate more variable tapping over and above that contributed by basic rhythmic properties. Starting with the non-linguistic rhythms, I first computed the asynchronies between each tap and the target beat. Standard deviations were then computed for each trial and then averaged within each participant and for each of the four rhythm types (figure 19). The same procedure was then applied to the tapping in time with speech during the main task, except prior to averaging, the trial-level variability scores were divided by the condition-averaged non-linguistic rhythm scores, yielding the tapping ratio score. The final participant-level averages are shown in figure 21.

Model selection These trial-level ratio scores were analysed using a linear mixed effects model with fixed effects of meter-syntax congruency and relative-clause extraction, and random intercepts for participants. Including trial number did not improve model fit ($\chi^2(1) = 1.556$, $p = 0.212$), nor did the interaction between congruency and syntactic complexity ($\chi^2(1) = 1.012$, $p = 0.603$).

Results Confirming my predictions, there was significantly greater tapping variability for both incongruent alignments over the congruent alignment (figure 21; table 5). Incongruent-1 alignment was also significantly more variable than the incongruent-2 alignment ($\chi^2(1) = 16.465$, $p = <.001$). Additionally, there was a significant effect of syntactic complexity, such that tapping in time with object-extracted sentences contributed additional variability (presumably because these sentences more challenging to understand and pulled more attention away from synchronization).

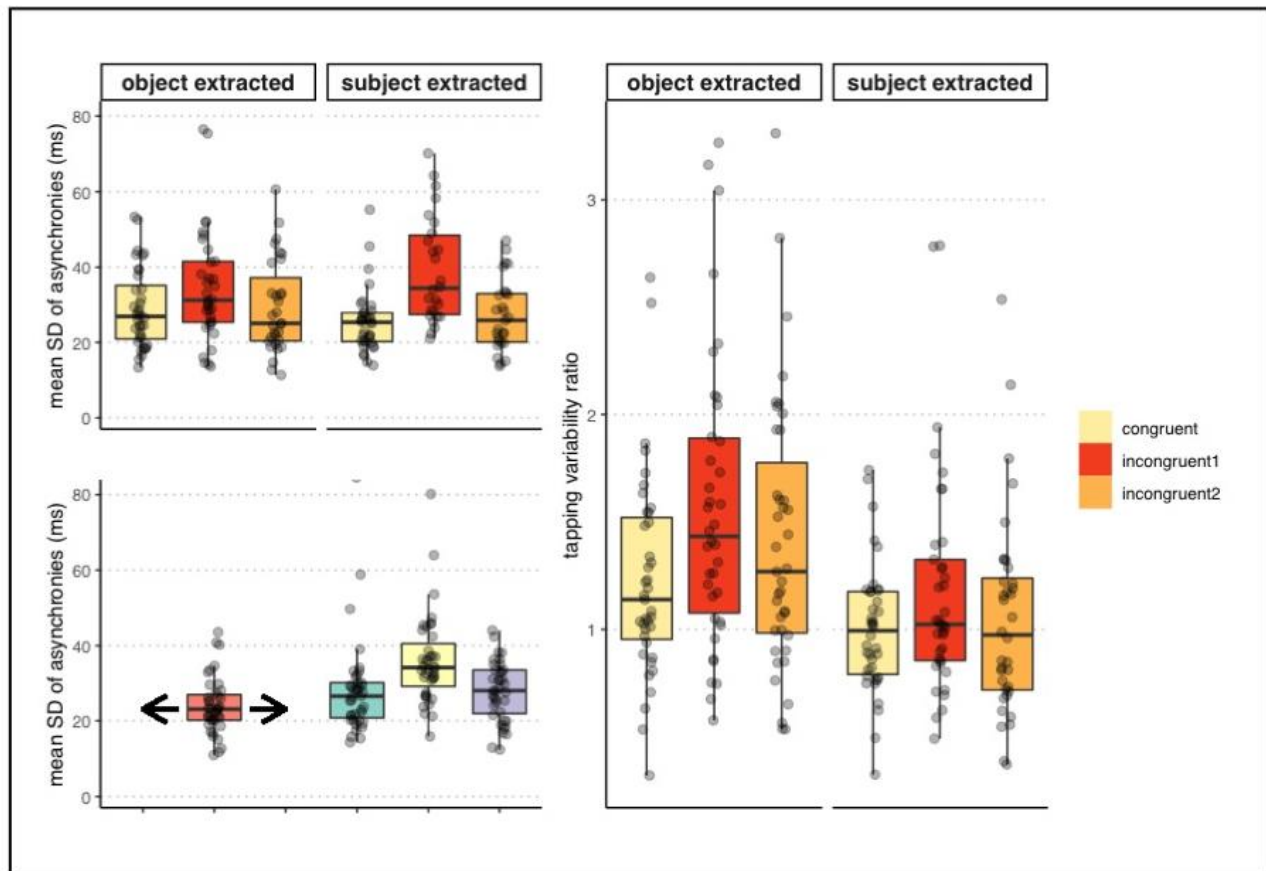


Figure 21: Top left: mean SD (i.e., 'stability of tapping') of sensorimotor synchronisation during the speech part of the task. Bottom left: mean SD of tapping along to non-linguistic rhythms with matched metrical properties. Right: The result of dividing the scores of top-left by the bottom-left. This can be interpreted as showing the degree of tapping stability while controlling for basic metrical properties.

Table 5: Estimate of fixed effects for tapping ratio score

	β	Std. Error	t value	p value
(Intercept)	1.285	0.066	19.473	<0.001 ***
Congruency (incongruent-1)	0.249	0.035	7.048	<0.001 ***
Congruency (incongruent-2)	0.098	0.036	2.738	0.006 **
RC extraction (subject)	-0.295	0.029	-10.018	<0.001 ***

4.3.3 Discussion

Experiment 2 replicated and extended experiment 1. Applying the same paradigm to auditory speech (rather than visual text), I showed that congruent meter-syntax alignment resulted in better performance than alignment shifts ‘to the left’ (incongruent-2) or ‘to the right’ (incongruent-1). The difference in the size of the effect between the two incongruent alignments was only statistically significant for the sensorimotor synchronization results, however, the trend was fairly consistent across the different measurements (congruent > incongruent-2 > incongruent-1). Should these differences turn out to be robust, they would provide a useful empirical target to constrain theory on why metrical alignment influences syntactic comprehension. Regardless, the difference between the congruent alignment and both incongruent alignments was robust for all measurements in the experiment, consistent with my primary hypothesis.

The effect of whether the comprehension probe targeted the main or relative clause was also strongly replicated in this study. While not theoretically relevant to meter-syntax alignment, this effect appears to be strong and robust at least within the current paradigm and from what I can gather, surprisingly few studies have addressed this topic (somewhat related results in: Newman et al., 2009; Lowder & Gordon, 2014). This effect could be especially of interest to noisy-channel theories of sentence processing (e.g., Gibson, Piantadosi, et al., 2013).

Unlike experiment 1, I additionally had participants tap their finger in time with the metrical beat. This was partly to ensure that participants were perceiving the intended metrical alignment by having them externalise their perception of the strong beats. But more interestingly, this also allowed me to test whether meter-syntax alignment affected sensorimotor synchronization: it did.

Why might the stability of tapping be affected by the alignment of meter and syntax? I suspect this is caused by conflicting metrical cues. One metrical cue comes from the metrical introduction of each trial induced by timing and accentuation patterns (reinforced by finger tapping). The conflicting metrical cue arises from syntactic predictions (Levy, 2008), which bias the perception of metrical prominence toward the syntactically expected position (Bishop et al.,

2020; Cole et al., 2019). Additional top-down inhibition may therefore be required to resolve the conflict and maintain sensorimotor coordination. Like with other congruity effects, the difficulty in inhibiting this automatic syntax-driven metrical reflex likely then explains the effect observed here.

While tapping in time with speech may seem like an artificial activity that could distract from comprehension, prior research shows that it can boost the perception of speech (Falk et al., 2017; Falk & Dalla Bella, 2016) and audition generally (Morillon & Baillet, 2017). And while people do not normally tap along with speech, they do frequently use ‘beat gestures’ implemented by rhythmic hand movements. Beat gesture is known to affect metrical prominence perception (Krahmer & Swerts, 2007) and syntactic parsing (Biau et al., 2018; Holle et al., 2012).

It is evident, however, that some of the variance in tapping stability comes from balancing attention between tapping and comprehending. More challenging sentence structures predicted more variable tapping (table 5), and more variable tapping predicted better comprehension performance for these more challenging sentences (table 3). This is consistent with prior work suggesting that top-down aspects of syntax and meter are controlled by a shared attentional resource (Schmidt-Kassow & Kotz, 2008). Despite this, this is orthogonal to the main meter-syntax alignment effects of interest here.

Concerns may also be had about the strictly isochronous speech stimuli and whether the results here would generalize to more rhythmically variable and naturalistic speech. First, I note that many naturalistic modes of speaking do have stricter timing, especially joint-speech (Cummins, 2020). More pertinently, I agree with others who have argued that the rhythmic idiosyncrasies of conversational speech do not represent a departure from metrical structure but rather a shift in the balance of top-down and bottom-up cues that support it (Beier & Ferreira, 2018; see also Hawkins, 2014). Thus, although induced differently in naturalistic speech, I see no reason to think that the effects of meter would be any different.

4.4 General discussion

Across both experiments, supporting the main hypothesis, I found that the alignment of meter with phrase structure affected syntactic comprehension. This effect was robust for whether sentences were rhythmically read (experiment 1) or listened to as speech (experiment 2). Comprehension was optimal when meter aligned with syntax in the typical way for English phrases (stress on the rightmost content word); There were more comprehension mistakes and lengthier response times when the alignment deviated from this linguistic norm. Metrical regularity was held constant in all cases.

The second experiment also showed that misaligning meter and syntax resulted in more variable sensorimotor synchronization. This provided an *online* measure of meter-syntax congruity during sentence processing, complimenting the *offline* comprehension measures (accuracy and response times) after sentence presentation. Interpreting this difference in

variability, I suggest an interference between an ongoing metrical representation and automatic, hard to inhibit, metrical reflexes tethered to syntactic predictions: a ‘metrical Stroop effect’.

I did not find an interaction between meter-syntax congruity and syntactic complexity. I had originally envisaged this as revealing whether congruity affects structural integration and its underlying memory-based processes. Although my data fail to support this hypothesis, it does not rule it out as the effect could be quite subtle. Roncaglia-Denissen and colleagues (2013), for instance, only reported such a meter-by-syntax interaction (with their metrical regularity manipulation) for differences in the P600 (an electrophysiological marker of structural integration). They did not report any such interactions in their behavioural results.

Additionally, the relative difficulty of the two sentence types may not relate as strongly to memory processes as I assumed. Specifically, object-extracted relative clause structures may be more difficult to process (for English speakers) because of their rarity in usage and therefore their low structural plausibility (Levy et al., 2012; Zhou et al., 2018), rather than being more difficult because of intrinsic structural properties (Gibson, 1998; Gibson et al., 2005). If this is true, the interaction is not so clearly expected: consistent with the result. Further research is needed.

I now turn to possible explanations for the main effect of congruity: Why are certain alignments better than others? As discussed in the introduction, meter is thought to have cognitive affordances for perception, memory, and motor coordination. Dynamic attending theory (Large & Jones, 1999; Pitt & Samuel, 1990) and related theories of neural oscillations and entrainment (Lakatos et al., 2008; Henry & Herrmann, 2014; Hartley et al., 2016; Rimmele et al., 2018; Morillon et al., 2019) have emerged as viable frameworks for making sense of how and when these affordances arise.

In the present study, it seems unlikely that the observed differences in syntactic comprehension were the result of metric modulations of low-level perception (such that words aligned to strong beats were perceived more acutely). The sentence materials were designed to minimize lexical ambiguities, to be morphologically as simple as possible, and to contain no additional prosodic information that could bias interpretation.

A more likely possibility is that meter affected comprehension by modulating lexical access and accessibility in memory rather than (just) the acuity of perceptual details. Metrically accented syllables make contact with lexical representations quicker in online processing, and are more accessible in short-term memory in offline processing (Gow & Gordon, 1993). These effects are also reflected in neural markers of lexical integration (Rothermich et al., 2012; Rothermich & Kotz, 2013; Gordon et al., 2011; Roncaglia-Denissen et al., 2013; also stronger activation of the language network more generally: den Ouden et al., 2016). Similar effects are also found for how metrical rhythm modulates memory for non-linguistic items (Hickey et al., 2020; Jones & Ward, 2019; Plancher et al., 2018; Thavabalasingam et al., 2016).

How might this explain the differences in syntactic comprehension observed in the study? Across both experiments, the congruent conditions align metrical accent to only content words (e.g., “the **BOY** that_the girl **HELPED** got an ‘**A**’ on the **TEST**”), whereas the incongruent conditions align mainly with function words (e.g., “**THE** boy that_the **GIRL** helped got **AN** ‘**A**’ on **THE** test”). Function words are relatively predictable given content words, but the opposite is not true: ‘boy girl helped’ is comprehensible but ‘the that the’ is not. Therefore, a stronger lexical activation and memory trace for more informative words in a sentence representation may support more robust comprehension and contribute to the observed pattern of results (consistent with: Aylett & Turk, 2004).

An alternative (potentially complimentary) explanation stems from how meter is implicated in serial-order short-term memory. Serial-order is thought to be cognitively represented by associating memory items with a temporal context signal arising from an entrained neural oscillator during encoding (Ng & Maybery, 2002). A recent model extends this to a hierarchy of such oscillators (Hartley et al., 2016) in a way that resembles the hierarchy of oscillators thought to underly meter (Port, 2003; Hawkins, 2014).

How might this explain the results? Metrical rhythm not only boosts serial-order recall generally, but it also changes the structure of memory confusions such that transpositions of order are more likely between rhythmic groups for items in the same metrical position than adjacent items in the same group (Gorin et al., 2018a; Mathias et al., 2015). And within groups, memory is most robust at the beginning and end and least robust in the middle (Hartley et al., 2016; Hurlstone, 2019). Thus, the tendency to align metrical stress to the most deeply embedded syntactic constituent (Cinque, 1993) may function to distribute the pattern of serial-order robustness over a phrase in a way that is optimal for syntactic comprehension: ensuring that memory confusions are least likely to result in corrupted interpretations.

Finally, a related idea is that there is a temporal window for integrating sequence information into memory chunks that may also be driven by entrained neural oscillations (Ghitza, 2017; Rimmele et al., 2020). Such temporal constraints appear to shape syntactic parsing (Schremm et al., 2015, 2016; Meyer et al., 2017) and sensorimotor synchronization (Mates et al., 1994). It is possible, therefore, that meter-syntax alignment affects syntactic comprehension by pushing around this window such that non-local dependencies are more-or-less likely to be integrated into the same memory chunk.

It remains for future research to disentangle and adjudicate between these possible mechanisms. Regardless of the mechanism(s) involved, the result highlights the importance of meter for linguistic processing beyond the more established effects on low-level perception. This has implications for a number of theoretical and applied areas.

Children with developmental language disorders (such as dyslexia) are thought to have impaired metrical entrainment and sensorimotor synchronization abilities, in both linguistic and musical contexts (Huss et al., 2011; Cumming et al., 2015; Colling et al., 2017; Fiveash et al., 2020). They are also less sensitive to the alignment between meter and syntax (Richards &

Goswami, 2019). I speculate that interventions could have efficacy in targeting more directly this alignment between rhythm and (top-down) sentence structure.

In language learning more generally, a recent study by Blanco-Elorrieta and colleagues (2020) compared the ability to discriminate speech in noise for both native and second-language speakers. They showed that linguistic ability not only makes comprehension more robust, but that it tracks with the ability to use top-down linguistic knowledge to drive neural entrainment to the speech signal. This mirrors my claim that syntactic predictions drive metrical predictions to enhance processing.

This importance of rhythm also underscores parallels with music. Gordon and colleagues (2015), for example, found that for typically developing children, the ability to discriminate metrical rhythms in a musical context was a strong predictor of grammatical development, even after controlling for non-verbal IQ, socioeconomic status, and prior musical abilities. These data are consistent with the claim that music and language play a mutually supportive role during development (Brandt et al., 2012; François et al., 2013), and that embedding speech in music benefits language learning (Schön et al., 2008; Vanden et al., 2020). Child-directed speech, song, and literature also has particularly emphasised rhythmic and musical qualities (Falk & Kello, 2017; Moser et al., 2020) and often consistently aligns this with syntactic phrase structure (Breen, 2018).

In conclusion, I showed that the alignment of meter to syntax affects the robustness of syntactic comprehension. This is important because meter (and its underlying neuro-oscillatory substrate) is often thought of as simply modulating low-level perceptual discrimination (Pitt & Samuel, 1990; Quené & Port, 2005; Niebuhr, 2009). But as this result highlights, it also has relevance to deeper aspects of cognitive processing. The synchronization results further highlight how top-down syntactic knowledge drives adaptive metrical reflexes. This likely relates to our spontaneous tendency for ‘beat gesture’ while we speak. More generally, I suggest that this reflects how our sensorimotor system is tuned to optimize syntactic comprehension.

5. Tuning the inside to the outside: The neural dynamics of music and language

“My experience is what I agree to attend to.”
- William James (1890)

“Clocks tick, bridges and skyscrapers vibrate, neuronal networks oscillate.”
- Buzsáki & Draguhn (2004)

The study presented in the last chapter showed that the alignment of metrical rhythm with syntactic phrase structure influenced sentence comprehension, and that it also influenced sensorimotor synchronisation. This chapter now situates these findings, and the literature discussed so far, in a broader literature in cognitive neuroscience that has addressed the question of how rhythmic neural dynamics can constrain and provide affordances to cognition. It will be argued that dynamic attending and the mechanisms of neural entrainment provide an informative theoretical framework that helps to make sense of why metrical alignment might support sentence processing and why this might relate to coordination of the sensorimotor system. Evidence from the study of speech and language disorders are also discussed in relation to these ideas.

5.1. Dynamic Attending Theory

Dynamic attending theory asserts that our internal biology (inside) and our external environment (outside) are rhythmically structured. Perception, attention, and memory (and their coordination) are then thought to be optimised by tuning the inside to the outside (Jones & Boltz, 1989; Jones, 1976). The precise nature of these rhythms and precisely what is meant by ‘tuning’ was later formalised in terms of nonlinear oscillators and processes of entrainment (Large & Jones, 1999). A nonlinear oscillator mathematically describes a periodic self-sustaining process with a limit cycle (a central tendency). A cascade of such oscillators with different temporal limit cycles can then become coupled together into a hierarchy of internal rhythms. These can then become *tuned* to multiscale rhythms in the environment as described by general physical processes of entrainment (Strogatz & Stewart, 1993).

One of the goals of dynamic attending theory was to explain why certain rhythmic events are cognitively more difficult to follow than others. This difficulty was formalised in terms of how well the fixed set of internal rhythms matched those of the external environment being attended. Regular rhythms are those that corresponded with either a single biologically relevant timescale or multiple (phase-aligned) nested timescales. Such rhythmic regularity affords what Jones and Boltz (1989) call *future-oriented attending*. This allows reliable predictions about future external events due to the rhythmic relation between the inside and the outside. These predictions allow the allocation of attentional energy to those anticipated points in time and this is thought to optimise their processing.

Irregular rhythms are those that fall outside of the relevant biological timescales or required non-phase locked nested rhythms. Such irregular rhythms do not afford temporal predictions and are not energetically sustainable. If such irregularity persists, the system defaults into an *analytic mode of attending*, where attentional energy is evenly distributed.

Future oriented attending and *analytic attending* are the extreme states of the system. The key to dynamic attending theory is that it formally describes how a whole spectrum of intermediary states are dynamically negotiated in realtime by this dance of entrainment between internal and external rhythms (see figure 22).

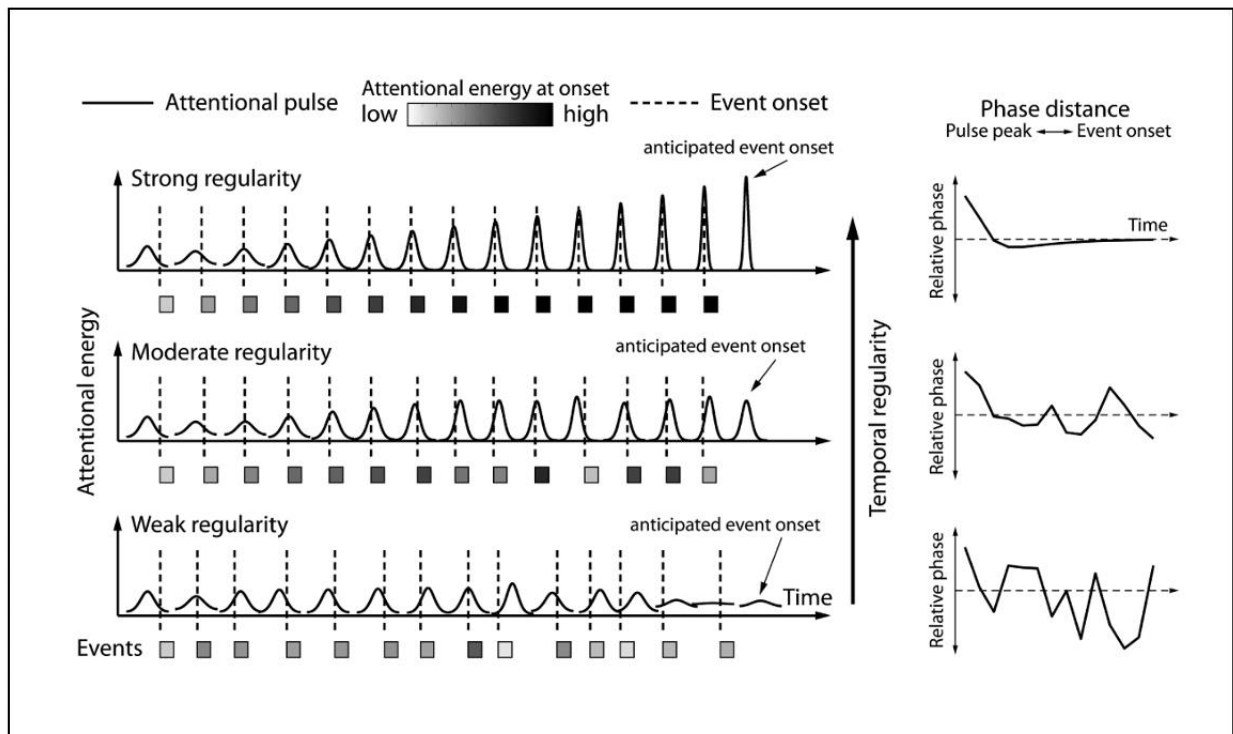


Figure 22: Example of how the mechanisms of Dynamic Attending Theory dynamically allocate rhythmic attention in a way that is sensitive to temporal regularity (taken from: Henry & Herrmann, 2014).

5.1.1. Dynamic attending and speech

In Jones' original paper (1976), she used both music and language as examples of regular environmental rhythms that afford future-oriented attending. Indeed, given that these are human generated signals, it is unsurprising that they have culturally evolved to fit our biologically determined internal rhythms.

One particular application of dynamic attending theory to language is the 'attentional bounce' theory (Pitt & Samuel, 1990). It proposes that the rhythms of prosodic prominence in speech attract attentional energy due to their rhythmic regularity, afford future-oriented attending, and thereby enhance phonemic discrimination. This has been demonstrated behaviourally (Pitt &

Samuel, 1990), and shown to also enhance the detection of subtle prosodic cues such as lengthening (Zheng & Pierrehumbert, 2010). Importantly, in line with the predictions of DAT, not only is a regular alternation of stresses important, but so is the regularity of their timing (Quené & Port, 2005).

Beyond language, effects involving a modulation of perceptual acuity by temporal attending have been observed for both general auditory (Hickok et al., 2015; Jones et al., 2002), and visual (Bolger et al., 2013) stimuli. Although, some more recent studies have failed to replicate some of these effects (Bauer et al., 2015), suggesting that such attending rhythms may be more limited to perceptual-motor interactions (Kunert & Jongman, 2017) and timing-based discriminations (Prince & Sopp, 2019).

5.1.2. Dynamic attending in the brain

Consistent with the basic premises of dynamic attending theory, modern cognitive neuroscience has embraced rhythmic oscillations of brain activity as an important aspect of brain function (Buzsáki, 2009; Buzsáki & Draguhn, 2004; Lakatos et al., 2019; Singer, 2013; Wang, 2010), shared across species (Buzsáki et al., 2013). And research confirms that tuning the neurally-instantiated inside to the outside is beneficial for various aspects of processing (Henry & Herrmann, 2014; Schroeder & Lakatos, 2009).

Neural oscillations are an emergent macro-scale phenomenon arising from periodic fluctuations of the membrane potential of spatially clustered populations of neurons. These fluctuations effectively modulate the firing rate of action potentials. Functionally, this excitatory state of a neural population is thought to allow better encoding of information and better communication with neural populations that are excited in the same temporal window (see: Fries, 2005, 2015). These neural oscillations have been behaviourally shown to implement attentional selection, as predicted by dynamic attending theory, by amplifying or suppressing perceptual gain depending on the phase of the relevant oscillation (Calderone et al., 2014; Lakatos et al., 2016; Schroeder & Lakatos, 2009).

The auditory cortex specifically has been shown to have preferred resting oscillations in delta (.5-3Hz), theta (4-Hz) and gamma (25-35Hz) bands (Lakatos et al., 2005; figure 23). These neurally preferred rates correspond closely with timescales of phrasal, syllabic, and phonemic rates in speech (Giraud et al., 2007; Meyer, 2017). The fact that nonhuman primate brains entrain in a similar way to speech at these timescales to humans (Zoefel et al., 2017) suggests that these physiological properties of the auditory cortex were likely prior constraints that shaped the cultural emergence of language, rather than the cultural emergence of language determining the nature of our internal biological rhythms. And indeed, showing that DAT is relevant to auditory processing, acoustic modulations at these timescales co-modulates both brain and auditory perception in systematic ways (Henry et al., 2014, 2015; Henry & Obleser, 2012; Obleser & Kayser, 2019).

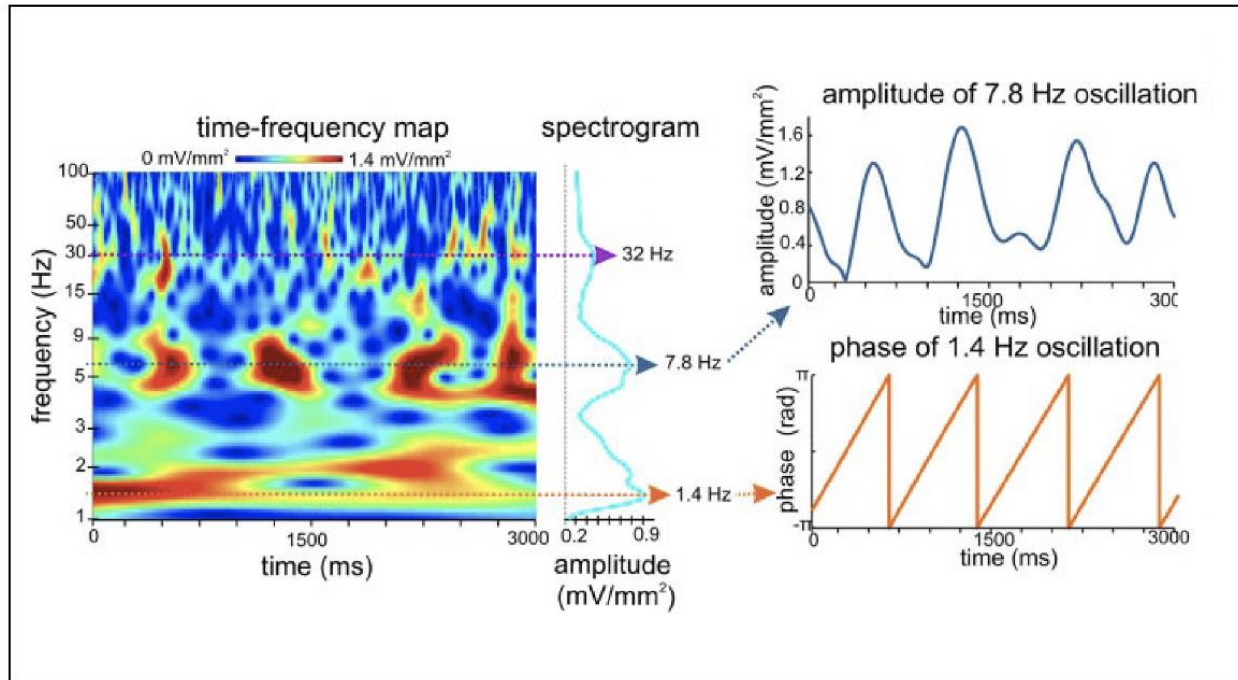


Figure 23: intracortical recordings from macaque monkey auditory cortex, showing systematic cross-frequency neural coupling at timescales relevant to speech and music: delta (1.4Hz), theta (7.8Hz), and gamma (32Hz; taken from: Lakatos et al., 2005)

5.2. Neural entrainment and speech

This hierarchy of neural rhythms is then clearly implicated in speech processing. Specifically, it has been observed that there is *phase locking* between neural theta rhythms (~4-8Hz) and the amplitude envelope fluctuations of syllables (Ding & Simon, 2012; Luo et al., 2010; Luo & Poeppel, 2007). It has then been shown through various means that interfering with this phase locking interferes with speech intelligibility. This can be done by presenting syllables outside of the theta-range (Ghitza, 2012; Ghitza & Greenberg, 2009); by removing amplitude envelope 'edges' (Doelling et al., 2014); and by artificially disrupting neural theta rhythms through brain-stimulation (Wilsch et al., 2018; Zoefel et al., 2018). All of these methods show such neural entrainment to be causally relevant to speech processing.

Making sense of this theoretically, it has been proposed that phase locking of neural and speech rhythms serves a *segmentation function*. Theories initially proposed two timescales of segmentation corresponding to gamma and theta rhythms (Giraud & Poeppel, 2012; Luo & Poeppel, 2012; Peelle & Davis, 2012; Poeppel, 2003, 2014). The basic logic is that phase-amplitude coupling between a faster and slower oscillator (e.g. theta and gamma) supports *discretisation* at the slower rate. Specifically, the oscillations in neural excitability of the slower rate implement selective attention for the encoding at the faster rate, effectively adding in gaps that serve to break the continuous stream into discrete units that can be cleanly decoded at the higher level (Giraud & Poeppel, 2012; figure 24).

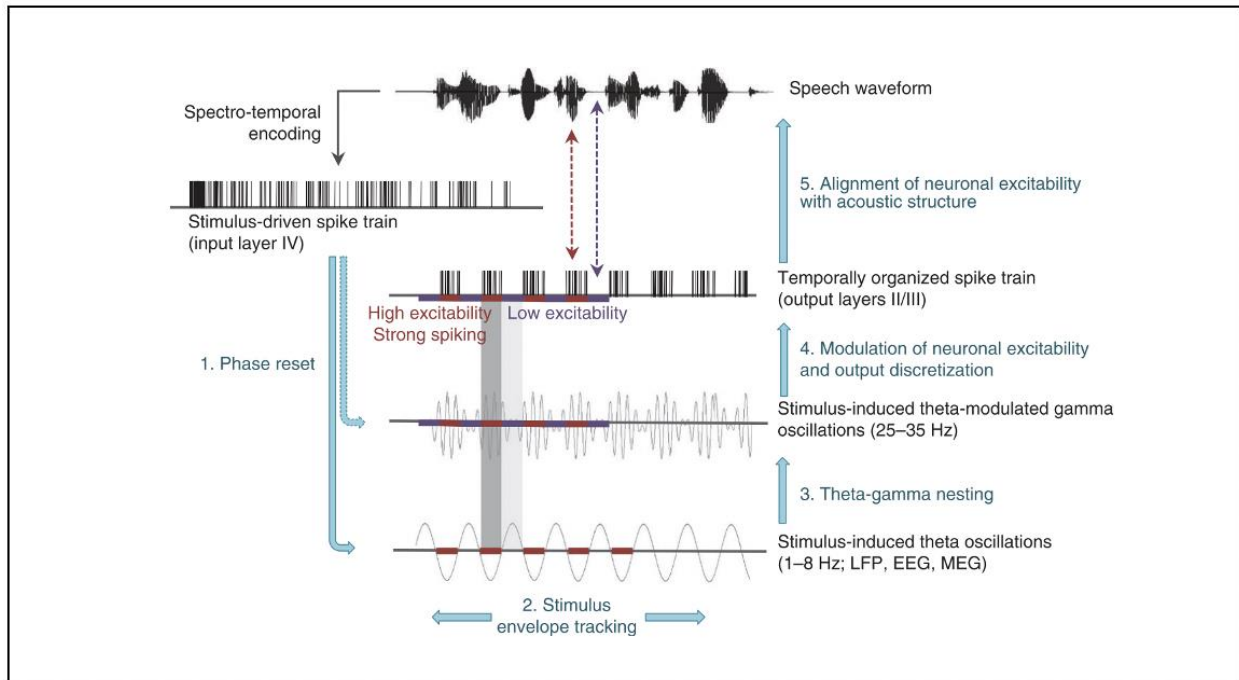


Figure 24: proposed mechanism for how nesting of theta and gamma oscillations can support prosodic segmentation through phase-amplitude coupling (taken from: Giraud & Poeppel, 2012).

This oscillatory mechanism for discretising a continuous input has been explicitly modelled computationally (Ghitza, 2011; Hyafil et al., 2015) and shown to beat previous state-of-the-art speech segmentation algorithms (Hyafil et al, 2015). Unlike these other algorithms, the oscillation-based models capture the characteristic drop in intelligibility when the input exceeds the 'packaging rate' defined by the theta oscillator (Ghitza, 2011). These recent experiments and models are also remarkably consistent with earlier psycholinguistic work showing how meter influences the segmentation of speech (Cutler, Dahan, & Donselaar, 1997; Dilley & McAuley, 2008).

5.2.1. Phrasal segmentation with delta rhythms?

Recent work has speculated whether the slower delta rhythms may also have a similar segmentation function for producing phrase-sized prosodic units (Meyer, 2017; Ghitza, 2017; Meyer et al, 2019; figure 25). Indeed, delta oscillations systematically entrain to speech signals, often in a way that seems to be driven by phrasal prosody (Bourguignon et al., 2013; Ghitza, 2017; Gross et al., 2013; Keitel et al., 2018).

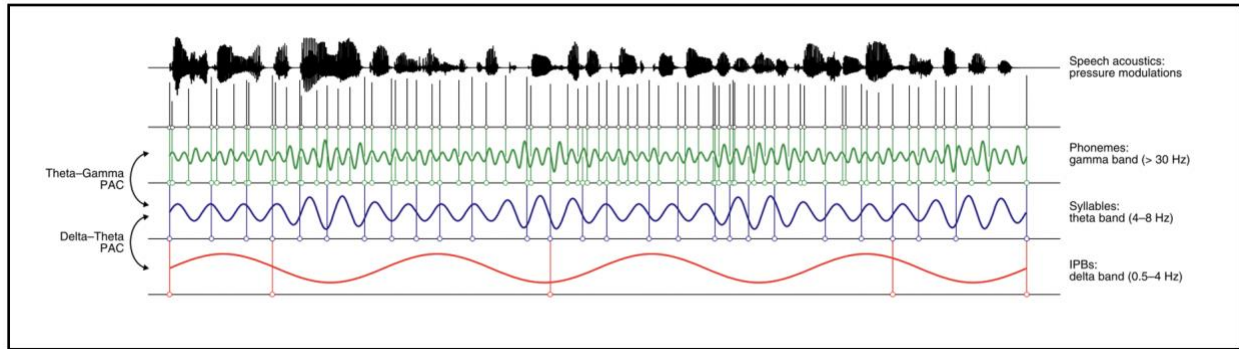


Figure 25: Schematic representation of how hierarchical neuronal oscillations are proposed align with the prosodic hierarchy in language (Meyer, 2017).

Unlike the clear and consistent amplitude modulations of syllables, phrasal units do not always have an acoustic correlate. This, however, does not stop delta oscillations from tracking them. An influential study by Nai Ding and colleagues (2016) showed this by presenting isochronously timed speech at a constant syllabic/word rate while recording brain activity using magnetoencephalography (MEG). The key manipulation was whether those words also formed syntactic phrases and clauses. Despite the speech stimuli lacking prosodic cues, delta oscillations tracked the rates of syntactic phrases (2Hz) and clauses (1Hz; see figure 26). Crucially, when these (Chinese) sentences were played to participants who could not understand Chinese, only the acoustically indexed syllable rate (4Hz) was tracked by neural oscillations (see also: Sheng et al., 2018; Zhang & Ding, 2017). Follow-up studies further confirmed that this neural tracking of supra-syllabic units is dependent on conscious attention. For example, it is absent when distracted with another task (Ding et al., 2017) or when sleeping (Makov et al., 2017), whereas theta tracking of bottom-up speech cues remains in both cases.

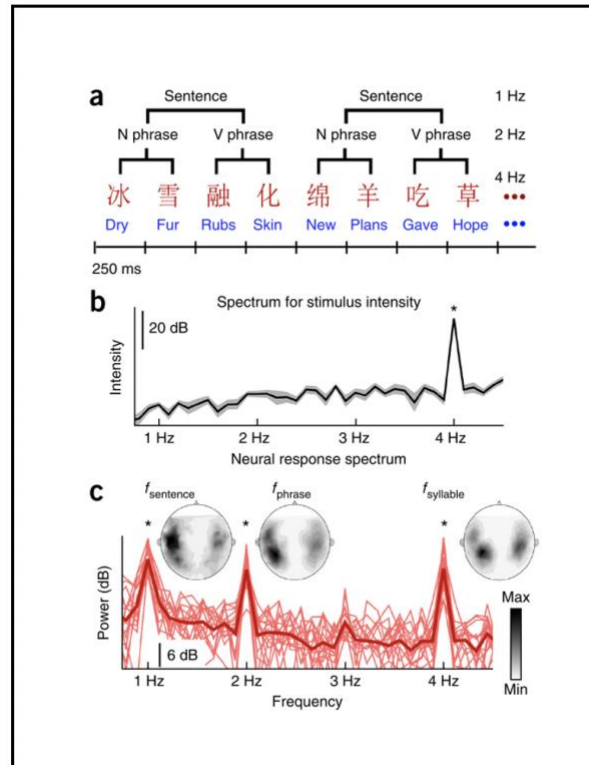


Figure 26: a) example of language stimuli, b) spectrum of speech stimuli, c) spectrum of neural response demonstrating tracking of abstract syntactic levels of structure (Taken from Ding et al., 2016).

5.3. Neural entrainment and rhythmic timing

Dynamic attending theory and neural entrainment have also proved to be useful theoretical constructs for understanding rhythm and meter in music. ‘Neural resonance theory’ uses the principles of dynamic attending to model metrical timing in music (Large, 2008; Large & Kolen, 1994; Large & Snyder, 2009; figure 27). Its main advantage over other models of timing is that it provides a principled means by which to account for flexible metrical induction from naturalistic stimuli. Naturalistic rhythms are often quasi-irregular and can sometimes have ‘silent beats’ where a strong beat has no physical counterpart in the signal. Neural resonance theory provides a clear way to account for flexible sensorimotor coordination to such rhythms (Large & Palmer, 2002; Loehr et al., 2011).

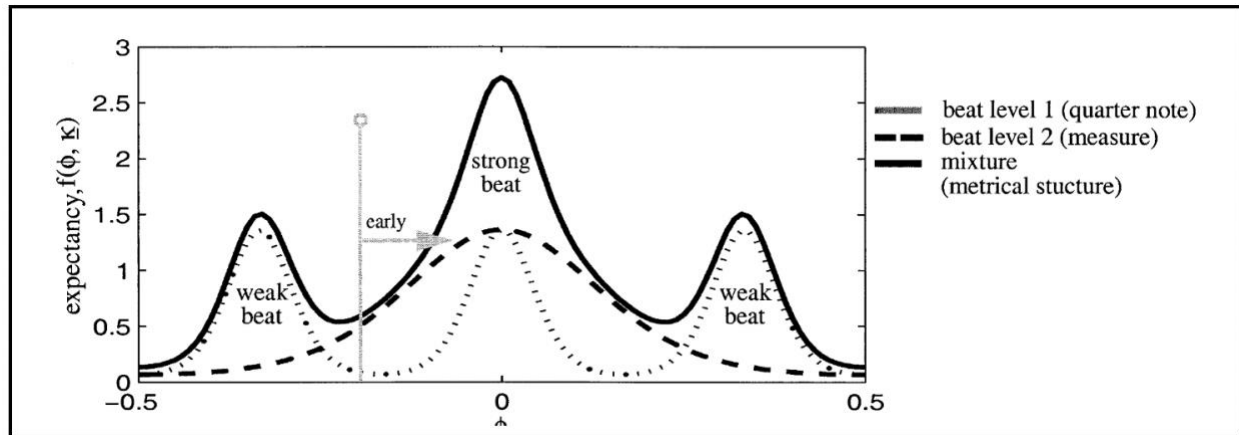


Figure 27: Schematic representation of how the multiple hierarchical levels of meter are proposed to be represented within Neural Resonance Theory (Large & Kolen, 1994).

The predictions of the model are supported by a number of experiments that test how people induce a meter from rhythmic stimuli (Snyder & Krumhansl, 2001; Snyder et al., 2006; Toivainen & Snyder, 2003). The model has also been supported by studies that measure the timing variability of how people tap in time with rhythms (van Noorden & Moelants, 1999; although also see Repp & Su, 2013 for broader review encompassing some limitations). And although most of this metrical timing work has focused on music, it is also consistent with the timing of speech production in speech-cycling experiments (Cummins, 2009b; Cummins & Port, 1998; Tilsen, 2009, 2011).

In addition to this experimental support for the theory, as a more general theoretical desideratum, neural resonance theory is also grounded in basic biophysical principles from which these higher level cognitive properties of temporal attention and motor coordination can be seen as emerging from (Kim & Large, 2019; Large et al., 2015; Large & Snyder, 2009).

5.3.1. Musical meter and neural resonance

In an influential confirmation of the theory's neural underpinnings, Nozaradan and colleagues (2011) showed that neural oscillations entrained to metrical structure. In their experiment, EEG was recorded while participants were presented with an isochronous auditory beat (at 2.4Hz) and where asked in different conditions to either listen carefully for glitches (control condition) or to imagine a binary (1.2Hz) or ternary (0.8Hz) metrical pattern (metrical imagery conditions; i.e. imaging a strong beat either every 2nd or 3rd beat). The results were clear, showing that the brain not only entrained to the beat physically present in the signal but also to the imagined metrical periodicities (figure 28).

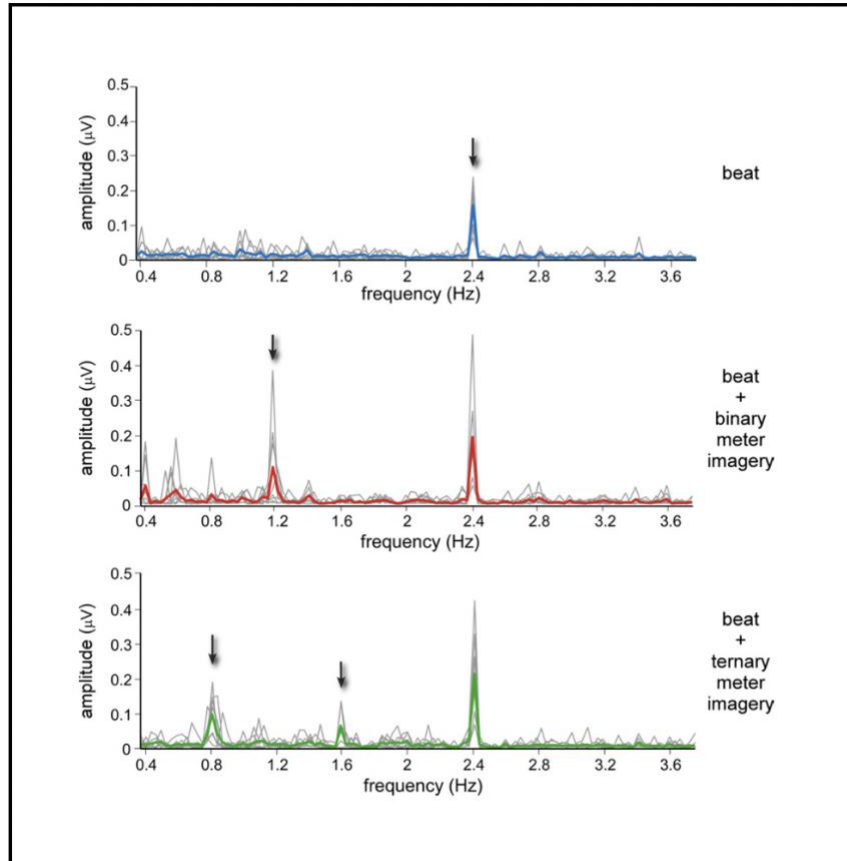


Figure 28: neural spectra while listening to an isochronous beat in either control or metrical imagery experimental conditions (taken from Nozaradan et al, 2011).

The neural tracking of meter has been broadly replicated (Nozaradan et al., 2012, 2016a, 2017; Tierney & Kraus, 2014), and shown to be especially sensitive to bass sounds (Lenc et al., 2018), body-movement (Chemin et al., 2014), and to be predictive of behavioural abilities in sensorimotor synchronisation (Nozaradan et al., 2015, 2016b). Crucially, a recent study by Doelling and colleagues supported the interpretation that such findings are indeed driven by the entrainment of an internal oscillator rather than just being an ‘illusion of entrainment’ driven by the frequency-domain representation of a sequence of event-related potentials (Doelling et al., 2019).

This meter-induced neural resonance can even occur in the complete absence of sensory stimuli all together, entirely produced by mental imagery (Okawa et al., 2017) or when these oscillations entrain to syncopated rhythms in which acoustic energy does not align with beats (Tal et al., 2017). This approach can also be used to study subjective perceptions of metrically ambiguous polyrhythms (Stupacher et al., 2017). And Cirelli and colleagues have shown such metrical entrainment to take place in infants as young as 6 months of age, and for the amplitude of this entrainment to be sensitive to prior musical experiences (Cirelli et al., 2016).

5.3.2. Periodic and nonperiodic timing

In a recent review and theory paper, Johanna Rimmele and colleagues outline some limitations of the classical dynamic attending model of timing and its recent neural instantiations (Rimmele et al., 2018). They point out that although it accounts for the processing of *periodic* stimuli, it fails to account for how the brain entrains to *aperiodic* stimuli.

Our day-to-day environment is full of biologically relevant aperiodic events. For example, in speech, just hearing the word “the” out of context makes it likely that one will hear a word following it shortly after. Or in music, certain motifs can elicit temporal expectations outside of a metrical context. For example, the opening four notes of Beethoven’s iconic fifth symphony set up a temporal expectation such that upon hearing the first three notes one expects to hear the fourth at a very particular time (the briefness of this motif is far too short to set up temporal expectations in the classical dynamic attending sense).

Supporting the notion that aperiodic timing had similar effects on attention to periodic timing, a recent study showed that participants were equally adept at using periodic and aperiodic cues to facilitate the detection of a near-threshold auditory tone (Morillon et al., 2016). A double dissociation has also been shown for periodic and aperiodic predictive mechanisms in the brain by looking at the respective deficits of patients with lesions to either the cerebellum or the basal ganglia (Breska & Ivry, 2018; see also dissociations in electrophysiological dynamics: Breska & Deouell, 2017).

Accounting for these findings, Rimmele and colleagues proposed an integrated theory of how periodic and aperiodic events drive entrainment. In their view, the ongoing oscillatory activity described by dynamic attending represents a *pre-existing neural constraint* on processing that only adapts in a slow bottom-up way to temporal patterns in the stimulus (periodic prediction). This basic foundation is then supplemented and made more adaptive to aperiodic dynamics by top-down memory-based predictions that intervene on the ongoing oscillatory cycle through *phase resetting* mechanisms. Combined, they provide a flexible way to anticipate predictable events and to align internal processing resources accordingly (see figure 29).

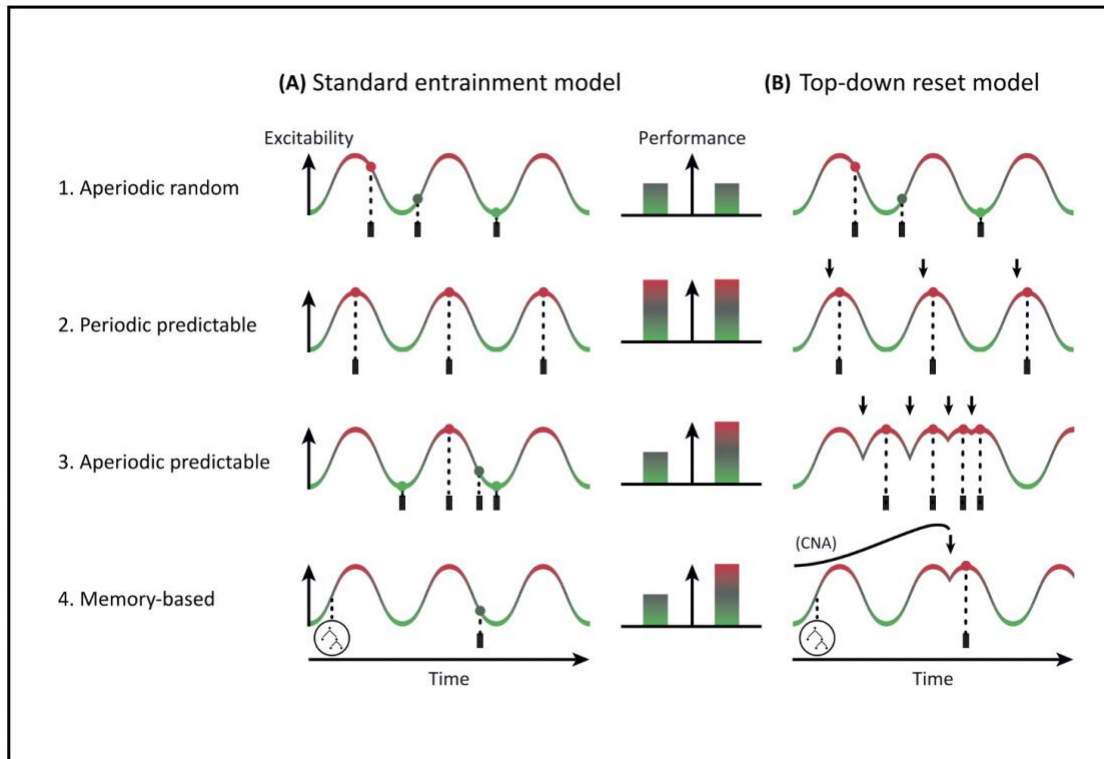


Figure 29: Schematic representation comparing how the classical DAT model (A) compares against one that is also supplemented by top-down phase resetting mechanisms (B; Rimmele et al, 2018)

5.4. Motor contributions to flexible aperiodic timing

One of the primary sources of such aperiodic temporal prediction within this framework is the motor system (Patel & Iversen, 2014; Schubotz, 2007). This relates to the notion of *active sensing*, which is a theory that characterises perception as an active process relying upon motor sampling routines (Morillon et al., 2015; Schroeder et al., 2010). For example, we actively shape our visual experience of the world through coordinated movement of our body, head, and eyeballs to select the most important things for us to see.

Our motor system can also influence perception in a less explicit way through corollary discharge signals, which are copies of motor commands sent to sensory areas (Crapse & Sommer, 2008). This is thought to be reflected in beta and delta oscillations and to modulate sensory gain by phase-resetting ongoing neuronal oscillations in the target sensory areas (Arnal et al., 2015; Lakatos et al., 2019). Indeed, Schneider and colleagues (2014) have provided detailed anatomical evidence of the required neural circuits in mice. And in humans, a growing number of experiments strongly suggest that self-initiated action influences auditory perception by these mechanisms (Morillon et al., 2014; Morillon & Baillet, 2017).

However, explicit movement is not necessary for motor-to-sensory predictions to take place. Indeed, a number of theories posit motor simulation as a general timing mechanism (Patel & Iversen, 2014), and have even suggested that the motor system might be better thought of as a

general purpose timing area (Schubotz, 2007). But although movement is not necessary for the motor system to contribute timing predictions, recent studies show that congruent movement often enhances and coordinates these motor-based predictions (Manning & Schutz, 2015, 2016; Morillon & Baillet, 2017).

This close relation between temporal predictions and the motor system is thought to explain our tendency for impulsive movement in response to rhythmic music: “we may feel the drive to move our bodies to the metrical beat to establish a metric model that generates the right sort of auditory predictions” (Koelsch, Vuust, & Friston, 2018). The processing of musical rhythm more generally is associated with activity in motor networks in the brain (Bengtsson et al., 2009; Chen et al., 2008). Movement also has been shown to influence the interpretation of metrically ambiguous rhythms (Chemin et al., 2014; Phillips-Silver & Trainor, 2005). And the beta and delta oscillations implicated in corollary discharge signals are routinely found in the perception of meter (Fujioka et al., 2015, 2009; Snyder & Large, 2005). Indeed, such neural tracking of musical rhythms, associated with beta and delta, has been shown to be modulated by musical expertise (Doelling & Poeppel, 2015).

There also seems to be a similar coordination between motor and perceptual networks in speech processing. For instance, frontal and motor areas modulate oscillatory activity in auditory areas via beta and delta oscillations during speech processing (Keitel et al., 2018; Park et al., 2015), and specific synchronisation between speech-motor and auditory networks seems to be rate-limited to the syllable rate of speech (Assaneo & Poeppel, 2018). Even ocular muscles have been shown to become entrained to higher-level delta oscillations during sentence processing (Jin et al., 2018). One notable recent study showed that a tendency for spontaneous speech synchronisation was predictive of abilities in speech processing and language learning, and was associated with structural connectivity differences in the brain’s dorsal white-matter connections (Assaneo et al., 2019).

Demonstrating the functional relevance of all this, moving in time with speech stimuli has been shown to enhance phonological processing, as shown by studies that have participants tap their finger in time with speech (Falk et al., 2017; Falk & Dalla Bella, 2016). Even just observing a speaker’s hand gestures has a similar enhancement effect, modulating activity in speech related brain areas and facilitating processing (Biau et al., 2018; Dargue et al., 2019; Holle et al., 2012; Hubbard et al., 2009).

5.4.1. Impairments to motor control impair syntax?

The contribution from the motor system to flexible entrainment also becomes clear in certain clinical populations, such as the so-called KE family, in which the FOXP2 gene was famously identified as being important for language (Enard et al., 2002; Fisher et al., 1998). While there was originally a focus on the syntactic deficits associated with this genetic mutation (Gopnik, 1990), it was later observed that the affected family members also had deficits in rhythmic timing but otherwise normal hearing (Alcock et al., 2000). Indeed, the authors noted that “neither [the affected KE members’] linguistic nor oral praxic deficits can be at the root of their impairment in timing; rather, the reverse may be true”.

The FOXP2 gene is expressed in motor-circuits such as the striatum, the basal ganglia and in inferior frontal areas (see for review: Fisher & Scharff, 2009), which are also implicated in beat-based processing and speech timing (Breska & Ivry, 2018; Grahn & Rowe, 2009; Kotz & Schwartz, 2010b). These areas are also similar to those identified in a recent meta-analysis of brain-imaging studies for rhythm and syntax processing, particularly finding overlap in left inferior frontal areas (Heard & Lee, 2020; Herdener et al., 2014). And these same brain areas are also implicated in grammatical-specific language impairment (van der Lely & Pinker, 2014), in which there is also a co-morbidity of extended syntax processing as well as extended aspects of prosodic rhythm.

This also accords with the observation that Parkinson's Disease patients, whose neuropathology is characterised by atrophy to these regions, also have a comorbidity of syntactic processing (Friederici et al., 2003; Lieberman et al., 1990) and rhythm processing (Kotz & Schmidt-Kassow, 2015; Breska & Ivry, 2018). Rhythmic cueing has then been shown to transiently restore some function relating to their syntactic deficit (Kotz et al., 2005; Kotz & Gunter, 2015).

5.4.2. Developmental language disorders and oscillations

Usha Goswami specifically draws these relations between neural entrainment, motor coordination, metrical timing, and language processing in the context of developmental language disorders and dyslexia specifically. She proposes in her oscillatory 'temporal sampling' framework that the impairments arise from atypical entrainment at the lower theta and delta frequencies, which results in impaired phonological processing (Goswami, 2011, 2015, 2018).

One component of this impairment may be a reduced sensitivity to amplitude-envelope rise time (AERT) information in speech signals (Corriveau et al., 2007; Cumming et al., 2015; Goswami et al., 2002; Richards & Goswami, 2015). AERT is an important cue to metrical accent in speech signals that is plausibly used to bottom-up entrain neural oscillators. Thus, children that are less sensitive to it may be impaired in other aspects of speech processing such as segmentation and coordination of working memory, as well as being less sensitive to alignment of metrical and syntactic cues more generally (Richards & Goswami, 2019).

In speech-disorder populations, there also seems to be an association between their language impairments and sensorimotor synchronisation for pathologies such as stuttering (Falk et al., 2015) and dyslexia (Dellatolas et al., 2009; Meyler & Breznitz, 2005; Overly, 2003; Thomson & Goswami, 2008). Recent work has begun to relate these general timing deficits in dyslexic populations to the induction of meter specifically (Huss et al, 2011; Goswami et al, 2013), with Richards and Goswami (2019) specifically showing that dyslexic populations are less sensitive to misalignments between meter and syntax.

Dyslexic populations have also been shown to have abnormal phase alignment of entrained oscillations that accompany their sensorimotor synchronisation deficits (Colling et al., 2017). A

recent study expanded on this comparison by using more complex rhythms, both of which implied a beat at 2Hz but differed in the extent to which that beat was physically manifested in the signal (yielding rhythmically regular and irregular conditions). While both groups seemed to entrain similarly to regular rhythmic sequences, there was a clear difference for irregular rhythms. These differences in neuronal entrainment also correlated with behavioural differences in rhythmic sensorimotor synchronisation (Fiveash et al, 2020; figure 30).

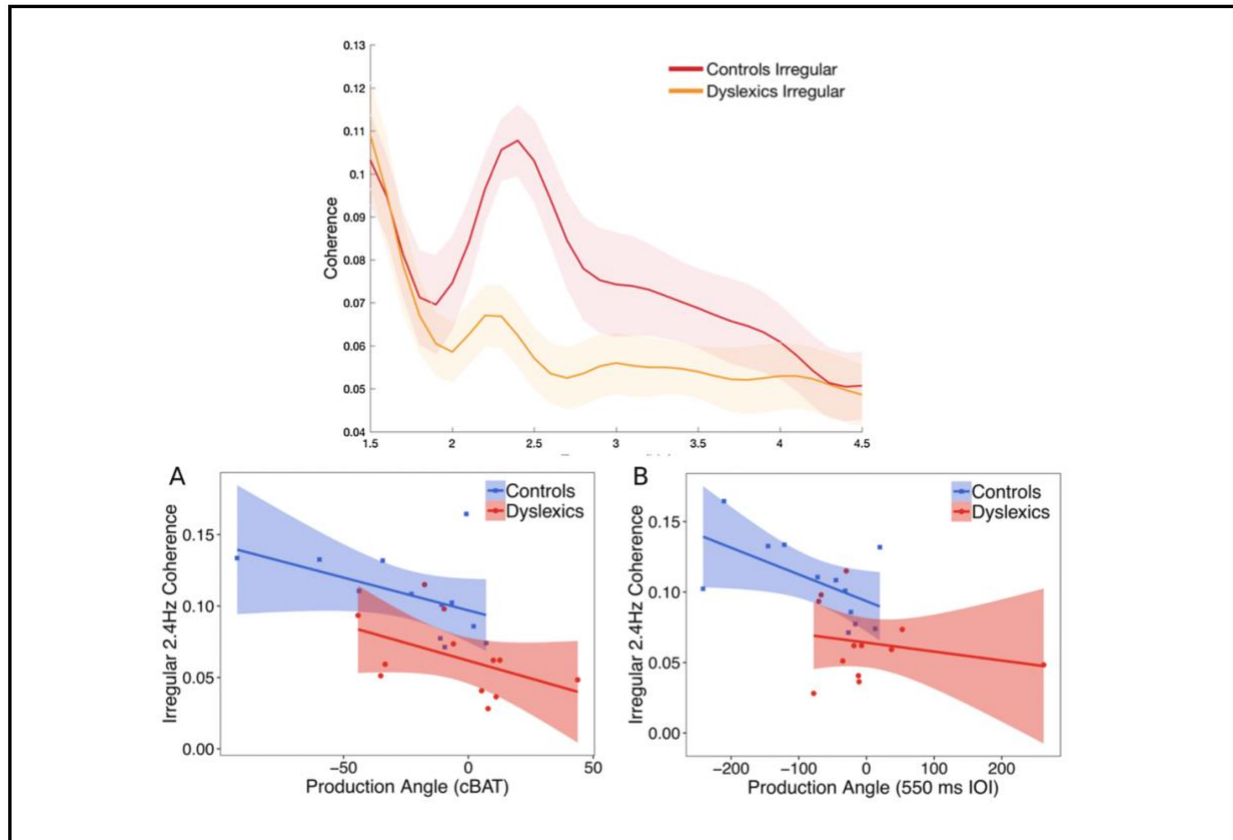


Figure 30: Differences between dyslexics and controls with regard to neural synchronisation to irregular rhythms (top), and correlations between these synchronisation measures and behavioural measurements of sensorimotor synchronisation (bottom; Fiveash et al, 2020)

Demonstrating the relevance of this to syntactic processing, Canette and colleagues (2020a) also showed that dyslexic populations not only exhibited differences in sensorimotor synchronisation but they also exhibited a later latency of the P600 ERP component in response to syntactic deviants, suggesting atypical processing.

Finally, from an evolutionary and developmental perspective, these findings are consistent with the idea that hierarchical rhythmic structure in music and language evolved out of hierarchical motor control (Lashley, 1951; Asano & Boeckx, 2015), and that specifically the syllable structure of speech evolved out of mandibular motor oscillations of the jaw in our primate ancestors (Fitch, 2019; MacNeilage, 1998). Indeed, recent modelling and experimental work suggests a tight interrelation between hierarchical perceptual structure and hierarchical motor control

(Tilsen, 2009, 2016, 2019), and in a way that is highly consistent with the oscillatory neural dynamics of motor control more generally (Churchland et al., 2012).

5.5. Summary

This chapter reviewed the cognitive importance of tuning the inside to the outside, as understood from the perspective of dynamic attending theory. It was shown that this process can be measured as the entrainment of neural oscillations (inside) to physical signals (outside). This entrainment occurs readily in both music and language, and specific mechanisms have been proposed for how this subserves or supports various cognitive functions. Of particular relevance to this thesis, neural delta oscillations were shown to track both syntactic phrase structure in speech (e.g. Ding et al., 2016) as well as metrical structure in music (e.g. Nozaradan et al., 2011). With regard to the latter, it was shown that the sensorimotor system is strongly implicated in supporting neural entrainment. This point helps to make sense of the effect of meter-syntax alignment on sensorimotor synchronisation observed in chapter 4. This is consistent with the reviewed literature on certain speech and language disorders, showing comorbidity in syntactic processing and sensorimotor coordination.

6. Study 2: Neural syncopation

Building on the behavioural foundation laid in chapter 4, and the theoretical review made in chapter 5, this study now probes the underlying neural dynamics of meter-syntax alignment. The primary aim is to resolve whether the primary driver of neural delta oscillations is syntax or meter. A version of the meter-syntax congruency paradigm is conducted while recording electroencephalography (EEG; $n = 29$). Three hypotheses are tested. The first, that delta tracks meter rather than syntax. The second, that a neutral 'no meter' condition would be detrimental to behavioural performance, and that it would not result in delta tracking. And finally, that desynchronisation of high-beta oscillations in frontal language areas (a correlate of verbal working memory encoding) would be stronger in congruent alignments over incongruent or neutral conditions. The results support all three hypotheses.

6.1. Introduction

6.1.1. Does delta track syntax or meter?

Chapter 4 showed that meter-syntax alignment affected both comprehension and sensorimotor coordination during the processing of complex sentences. Grounding these results in the brain, chapter 5 reviewed the role of neural oscillations and entrainment in various cognitive processes generally as well as those implicated in language and music processing specifically. It was shown that delta oscillations track both syntax in language and meter in music. But to the extent that both domains have syntactic and metrical structure, and that these structures are independent from each other, which does delta track: meter or syntax? This shared neural correlate will now be investigated in the context of the general experimental paradigm developed in Study 1, which manipulates meter while holding syntax constant.

Despite these studies showing delta tracking for both musical meter and linguistic syntax, there has been little attempt to reconcile these findings. In lieu of such an effort, delta has primarily been explained in terms of syntactic structure in language research, whilst largely ignoring meter. This is consistent with the pervasive Chomskyan bias for syntax over phonology (as described in chapter 2).

What do these syntactic interpretations of delta look like? A study from Lars Meyer and colleagues (2017) showed that delta oscillations reflected syntactic parsing decisions in syntactically ambiguous sentences (see also Gross et al., 2013). When these sentences were uttered with a neutral prosody the phase of delta-band activity was shown to be a strong predictor of the syntactic parse that participant would make. Participants sometimes even made a syntactic interpretation that contradicted the acoustically realised prosodic phrasing, and here too, delta seemed to be the most reliable correlate of the interpretation. Interpreting this, they argue that "delta-band oscillations provide an internal linguistic searchlight for the formation of syntactic phrases during auditory language comprehension."

A follow up study to this from Meyer and Gumbert (Meyer & Gumbert, 2018) claimed that delta phase functions to align neuronal excitability with syntactic informativeness. They suggest that

this alignment explains why delta biases syntactic parsing, and why delta tracks syntax more generally. Their theoretical idea is based upon two separate notions. The first is that oscillations reflect a modulation of the excitability of neuronal populations, and that this can enhance the processing of stimuli that are aligned to the most excited phase of the oscillation (this is a well understood physiological interpretation of neural oscillations; see chapter 5). The second aspect of their theory references previous work that characterises the information-theoretic informativeness of each incrementally processed word in a sentence (Levy, 2008). Specifically, they make use of the metric of 'syntactic surprisal', which is a computationally estimated measure of the unexpectedness of the syntactic category of an incoming word. Combining these two ideas, Meyer and Gumbert propose that delta serves to align neuronal excitability with syntactic surprisal to optimise processing.

Meyer and Gumbert go on to provide evidence for this hypothesis in an experiment in which participants detected grammatical violations in various positions within a sentence while EEG was recorded. Specifically, they showed correlations between delta phase and both syntactic surprisal in the sentence and response times and concluded that delta was indeed tracking syntax and facilitating its processing.

However, there are a few problems with this interpretation. Firstly, syntactic surprisal is also strongly correlated with position in phrase structure more generally. That is, words tend to be more predictable at the end position of phrase structure simply due the accumulation of information from previous words in that phrase. Thus, to the extent delta is tending to align with syntactic surprisal, it is certainly not clear that it is driven by syntactic informativeness over some other factor that might drive delta to align with phrase edges. Secondly, the electrode that they chose to analyse for the phase alignment analyses also showed the bizarre result that it only shifted by roughly half π radians over the course of the whole sentence. On the reasonable assumption (made by the authors) that the sentence corresponds with a delta cycle, its phase should systematically shift by the full 2π radians. But it did not. It is unclear how to interpret this. Minimally, it problematizes the interpretation that it is tracking a true delta oscillation and instead it may have been capturing some other pattern of neural activity that artifactually caused their result.⁵

Another major theoretical proposal is made by Martin and Doumas (2017), wherein delta is argued to reflect a neurocomputational mechanism for semantic compositionality. That is, they posit that delta tracking of syntax is a necessary consequence of combining meanings in a structured way. This claim is made with reference to a neuro-computational model originally conceived for modelling non-linguistic processes called DORA (Discovery Of Relations by Analogy). This model uses oscillation-like mechanisms to flexibly compose structured symbol-like representations (Doumas et al., 2008; Hummel & Holyoak, 1997, 2003; Martin & Doumas, 2019). Applying DORA to language generated computational dynamics that mirrored the electrophysiological delta dynamics observed by Ding and colleagues.

⁵ They did provide control analysis showing that this was not simply caused by common ERPs, however, this still does not explain this puzzling result.

Subsequent to this formal existence proof, electrophysiological studies have been conducted and argued to support this interpretation. Brennan and Martin (2019) recorded EEG while having participants listen to naturalistic speech stimuli. They showed a number of relationships between compositional linguistic structure and measures of phase synchronisation (inter-trial phase coherence) and cross-frequency coupling in delta, theta, and gamma oscillatory bands. They interpreted these varied patterns in oscillatory phase as supporting the idea that this is associated with (or are even causally driving) underlying compositional computations.

Another study by Kaufeld and colleagues (2020) recorded EEG while participants listened to speech that was either semantically and syntactically coherent (normal sentences), syntactically coherent but had no semantics (jabberwocky sentences⁶), or had coherent semantics but not syntactically productive (word lists). The results showed the highest degree of speech-brain tracking for the coherent sentences (with syntactically supported compositional meaning). This tracking was quantified in terms of phased-based mutual information between neural and speech signals (see: Keitel et al., 2018; Gross et al., 2013). They also had conditions in which the same speech stimuli were played in reverse, providing a control for prosodic confounds. They also found significant tracking for these sentences as well, although not to the same extent as the forward sentences. They interpret this as showing that while delta tracks acoustic prosodic cues, this may be enhanced by compositional computations.

However, there are problems with using reversed-speech as an acoustic control. Although spectrally identical to forward speech, this does not mean it is cognitively or perceptually identical. Speech amplitude envelopes are asymmetric, so reversing them is likely to result in unnatural speech envelopes, which are likely to be processed differently (see for importance of envelope characteristics: Schutz & Gillard, 2020). Indeed, the transformation of acoustic rhythms to neural rhythms is known to be nonlinear in nature (Nozaradan et al., 2016).

These and related interpretations of delta tracking are summarised in a recent theoretical review paper by Meyer, Sun, and Martin (2019). Their main argument is to distinguish *entrainment proper* from *intrinsic synchronicity*. Entrainment proper, or ‘exogenous cortical rhythm’, describes the phase-locking of neural oscillations to speech acoustics. In other words, a direct relationship between acoustic and neural rhythms modelled as entrained oscillators. They contrast this with intrinsic synchronicity, or ‘endogenous cortical rhythm’, which describe internal oscillatory activity arising from the generation of syntactic, semantic, and discourse representations, distinct from speech acoustics. This latter form of oscillatory generation includes that posited by Martin & Dumas (2017). They conclude by arguing that entrainment proper is insufficient to account for adaptive language processing given that speech signals are often only quasi-rhythmic. By having intrinsic generation of oscillations based on inferences about higher-order abstract linguistic structure, then this is thought to stabilise and make more robust the alignment between neural and linguistic rhythms (figure 31).

⁶ Jabberwocky sentences are defined by nonsense content words, supported by syntactic frames (word order, function words, inflectional morphology). Thus are thought to tap syntactic aspects of processing distinct from semantics (although there is debate around this, especially from constructional grammar perspectives).

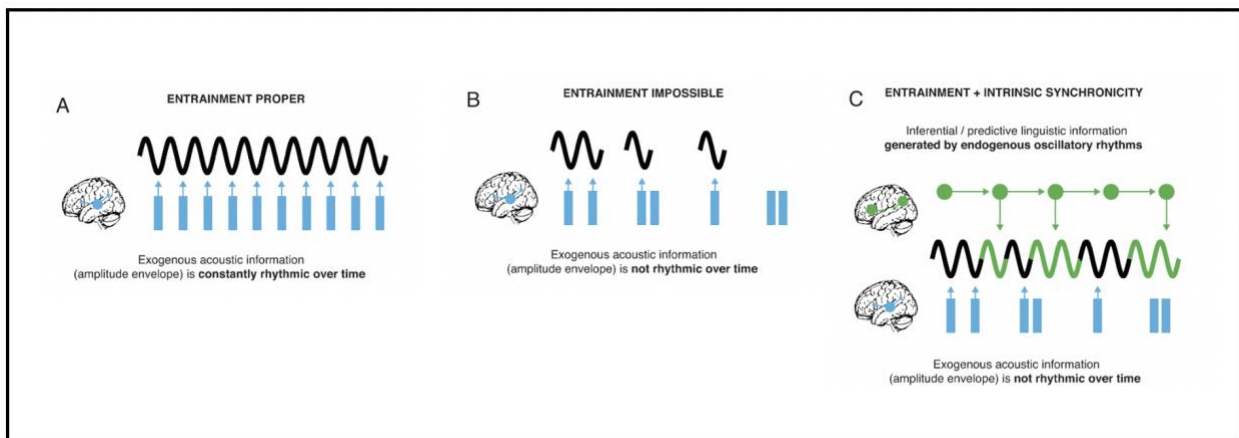


Figure 31: adapted from Meyer, Sun, & Martin (2019). Schematising their proposed difference between endogenous vs exogenous oscillations.

However, there are some issues with this theoretical framing. In response to their paper, Anne-Lise Giraud (2020) notes that the measured neural oscillations typically implicated in speech processing are generally not oscillators in the strict mathematically defined sense. Instead, Giraud notes that different families of neural oscillations (delta, theta, alpha, beta, gamma) have heterogeneous biophysical origins, and play different cognitive roles, but most importantly, cannot be uniformly characterised in terms of how they can become entrained to external signals. For example, as some of her own experimental and modelling work shows, theta oscillations can reliably track quasi-rhythmicity without having to posit additional internal generation or inference of structure (Giraud & Poeppel, 2012; Hyafil et al., 2015). Her model achieves this robustness through flexible bottom-up phase-resetting mechanisms (see also Rimmele et al., 2018). Giraud also provides evidence for how delta oscillations readily emerge as a consequence of certain neural architectures, without having to posit high-level linguistic explanations.

In another response to their paper, Kandylaki & Kotz (2020) note, along the lines I argue here, that prosody and rhythm perception may be underappreciated in this theoretical picture. Indeed, they note that it is intrinsically challenging to prise apart abstract syntactic aspects of comprehension, like argument structure building, from considerations of rhythmic prosody (Rothermich et al., 2012; Schmidt-Kassow & Kotz, 2009). This is important because the language system, and how oscillations are deployed within it, are likely making use of pre-existing capacities such as the processing of rhythm, shared with music (Kotz et al., 2018), and more generally how dynamic attending perspectives may apply (Kotz & Schwartz, 2010).

In the context of this ongoing discussion, the primary aim of the following study is to determine whether delta tracks meter or syntax. In other words, is there a unique contribution from internally neural activity that is specific to the generation of compositional meaning from syntactically structured language? Or is delta simply tracking meter? Answering this question will help bridge between the literatures on neural entrainment to meter in music (e.g. Nozaradan et al., 2011) and the recent literature on low-frequency oscillations in language processing. The

experimental paradigm developed and behaviourally grounded in chapter 4 allows this to be investigated directly by manipulating meter while keeping syntactic phrase structure constant.

6.1.2. Is a lack of meter worse or neutral?

If the priming of meter-syntax alignment affects comprehension, what happens when the prime is metrically neutral? In other words, instead of preceding each sentence with an isochronous sequence of tones, with amplitude accents on every other (binary) or every third (ternary) tone, what happens when this series of tones is unaccented throughout? The intuitive prediction may be that comprehension performance would be similar, or a little worse than the congruent meter-syntax alignment, but no worse than the incongruent alignments. This would be consistent with previous work on prosodic phrasing and how grouping cues can either reinforce, conflict, or be neutral to syntax-aligned groupings (Kjelgaard & Speer, 1999).

A less intuitive alternative is favoured here, based upon the oscillation-based model of short-term memory proposed by Hartley, Hurlstone, and Hitch (2016). The basic assumption of their model is that entrained neural oscillators act as a context signal for representing serial-order in short-term memory. All else being equal, an unaccented sequence of syllables with identical duration (i.e. neutral meter) is predicted to yield a particularly poor context signal (figure 32a). This is because the stimuli only support the entrainment of a syllable-rate oscillator, which phase-lock similarly across all syllables, and thus provide more-or-less the same 'context signal' for differentiating their serial position (the brightness of the colours in figure 32 represents how distinct the context signal is for each position). However, if there are rhythmic cues in the signal (figure 32b), or with top-down metrical interpretation, then additional oscillators could entrain and provide a more distinct context signal across the whole sequence (figure 32b), and thereby a more robust memory representation.

From this perspective, I propose that the different meter-syntax alignments affect comprehension because they shift which words in a phrase end up with the least distinct context signal (e.g. note the grey patches in the centre of each group in figure 32b). Since some words are systematically more informative than others (Levy, 2008), this could plausibly affect comprehension. A neutral meter condition, then, would be predicted to have the worst comprehension performance because it has the least distinct context signal under the assumptions of the model. This proposal has similarities to that of Meyer and Gumbert (2018). The difference is that I propose that delta is explicitly driven by metrical attending (rather than a more general 'syntactic search-light'), and that its alignment specifically functions to optimise serial-order representation in memory rather than a more general boost to perceptual discrimination or cognitive processing.

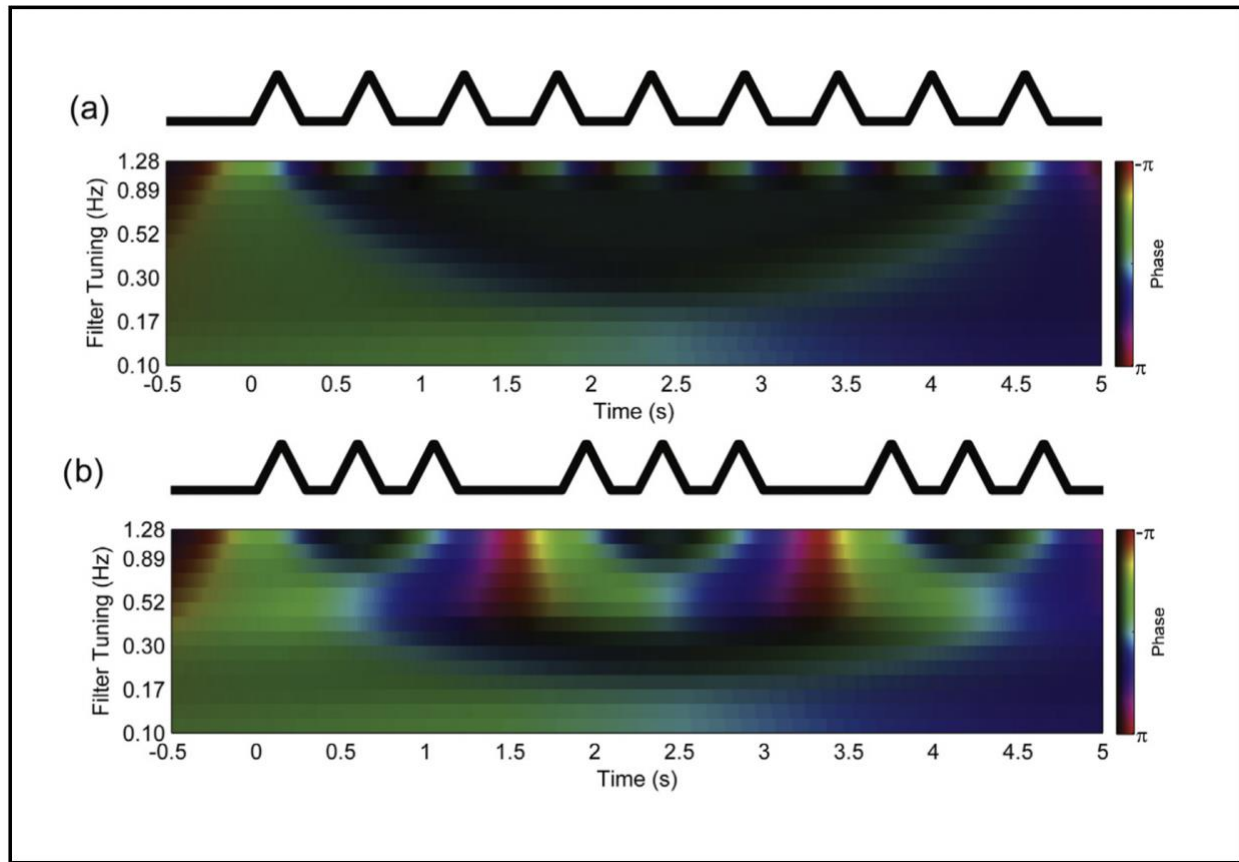


Figure 32: taken from Hartley, Hurlstone, & Hitch, (2016). The stronger the colour (i.e. not black) to more robust the context signal. a) shows the simulated context signal for a purely isochronous sequence, compared to b) when the rhythm allows low-frequency oscillators to entrain.

6.1.3. High-frequency oscillatory correlates of memory

Finally, the interpretation of the results in chapter 4 was that meter-syntax alignment influenced the encoding of the sentence in short-term memory. To better support this interpretation, the following study also seeks to find neural evidence to complement the sparse behavioural data (i.e. whether the participant answered the comprehension probe accurately or not, and how quickly).

A recent study by Bonhage and colleagues (2017) explored the electrophysiological correlates of working memory during language processing. They recorded EEG while participants read word lists, which afford little syntactic affordances for chunking, or syntactically organised sentences, which can be chunked according to their phrase structure. The participants were then probed for their memory of these materials. Their results showed increased encoding into memory was associated with both increased delta power (in line with the theoretical interpretation of delta in this thesis), and a reduction of power in the upper-beta frequency band (25-31Hz).

The latter observation of increased desynchronization (resulting in reduced power) in the upper-beta band relates clearly to more general neuro-cognitive theory of working memory. Recent research has provided compelling evidence for a model of working memory whose functioning is reflected in transient bursting in functionally dissociable gamma and high-beta frequency bands (Lundqvist et al., 2016; Miller et al., 2018). This model predicts that encoding in working memory correlates with desynchronisation of default-state beta-band activity in prefrontal areas, accompanied by increased gamma bursting. When this oscillatory bursting is trial averaged, it appears as sustained non-phased locked activity over the period of encoding. Indeed, this seems to correspond closely with the observed non-phaselocked high-beta desynchronisation observed by Bonhage and colleagues (2017) in the context of encoding linguistic information into working memory.

Low-beta and alpha frequencies are also associated with encoding in working memory (Klimesch, 2012; Roux & Uhlhaas, 2014). However, a recent study by Lundqvist and colleagues (Lundqvist et al., 2020) provides some evidence that helps make sense of how these different bands can all be involved. Specifically, they show that frequency appears to increase ascending the cortical hierarchy from primary sensory areas to high-level integration areas in the prefrontal cortex, such that alpha desynchronisation in sensory areas may play a similar top-down control and inhibition function to that of beta oscillations in frontal areas. Furthermore, they suggest that an increase in frequency for more frontal areas may have the functional purpose of allowing better temporal separation of individual bursts and therefore better separation of multiple representations being integrated into a coherent memory representation. Given that syntactic comprehension requires the integration of multiple representations into hierarchical structures, it makes sense that prefrontal unification areas (Hagoort, 2004, 2019) may rely more upon higher frequency beta than corresponding working memory sites in the parietal cortex also associated with language processing relying upon alpha (Meyer, Obleser, & Friederici, 2013).

Thus, the question of whether high-beta activity in frontal language areas is differentially modulated by meter-syntax congruency is also asked in this study.

6.2. Experiment 3

6.2.1. Experimental design

6.2.1.1. Syntactic complexity and meter-syntax alignment

The same experimental design logic behind the syntactic complexity and meter-syntax alignment manipulations from Study 1 is applied here. Specifically, the subject-extracted sentences are set to a binary meter and the object-extracted sentences to a ternary meter. There were two reasons for this decision. One is to avoid the rhythmic confound discussed in experiment 2 arising from setting the subject-extracted sentence to a ternary meter. The other reason is that having both binary and ternary meters will help to show two distinct patterns of delta entrainment. This helps to show robustness in the results and helps rule out the possibility that a particular delta frequency was arising for some other non-meter cause.

To implement a neutral ‘no meter’ condition, the target sentence is preceded by *unaccented* auditory tones, as compared to the other conditions in which the sentence is preceded by accented tones that prime a specific metrical interpretation (see figure 33).

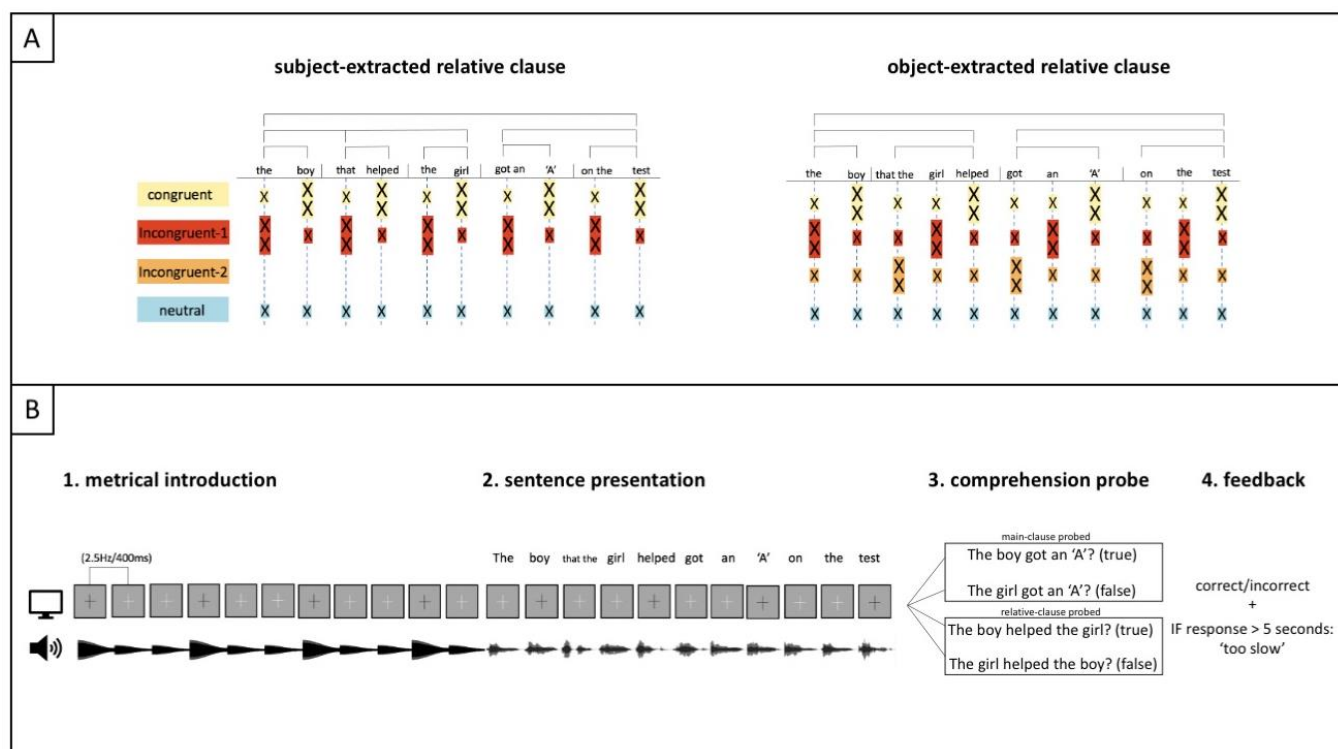


Figure 33: Top: meter-syntax alignment conditions for experiment 3. Bottom: Trial schematic.

6.2.1.2. Linguistic materials

Partly to compensate for the smaller sample size and for the extra meter-syntax congruency manipulation, the sentence materials were further expanded out to a total of 112 English sentences, with all the same structural and probe characteristics as in Study 1 (see appendix A for full sentence list).

6.2.1.3. Auditory materials

Both the auditory beat stimuli used to prime the metrical interpretations, and the artificially synthesised and preprocessed speech stimuli were constructed in an identical fashion to experiment 2.

6.2.1.4. Experimental procedure

Participants were seated at a chair in front of a computer and supplied with EEG compatible insert earphones (ER-3A, Etymotic Research) adjusted to a fixed comfortable listening volume. The experiment was run by a program written in python (largely using the PsychoPy python library).

After an introduction to the task led by the experimenter, participants completed a series of practice trials with a 'neutral meter' before being encouraged to ask any clarifying questions. This practise session lasted approximately 5-10 minutes. Participants then completed a block of 'neutral' meter-syntax alignment trials before the other trials such as to not bias their metrical interpretation. This, however, has the downside of not allowing order to be counterbalanced with regard to the 'neutral' alignment condition.

After the neutral trials, participants then completed four additional practise trials with specific metrical alignments. This allowed them to get used to the new requirement of maintaining a metrical interpretation of the sentence. They then completed the rest of the trials as a single experimental block with a short break halfway through. To ensure the participants did not lose track of the particular metrical alignment, since they were not tapping a drum-pad like experiment 2, the fixation cross during sentence presentation faintly flashed with a black outline in time with the strong beats. This was calibrated to be subtle enough to not be distracting.

The trial procedure is otherwise identical to that used in experiment 2, except that there was no finger tapping (figure 33b). During all trials, participants were also instructed to remain still and relax so as to not induce motor artefacts in the EEG data (and were encouraged to stretch in between trials).

6.2.1.5. EEG Recording

Continuous EEG data was recorded from 64 electrodes arranged according to the standard 10–10 electrode placement system (Oostenveld & Praamstra, 2001) using a BrainVision ActiChamp system, digitized at a 1000-Hz sample rate and referenced online to Cz.

6.2.1.6. Participants

29 native English-speaking undergraduate psychology students (17 female, 12 male) from the University of Sydney area took part in this study in exchange for course credit. They were naive to the purpose of the experiment and had normal hearing, and normal or corrected vision, and no prior history of speech and language disorders. They were between 18 and 54 years of age ($M = 24.9$, $SD = 8.1$). The mean comprehension accuracy of three of these participants was lower than 60% (50% is guessing) so their data was excluded from analysis, leaving 26 participants. This smaller sample size (compared to $n=40$ in Experiments 1 & 2) largely reflects the practicalities of running considerably more time-intensive EEG studies. As such, the study has less statistical power to replicate the previously observed behavioural effects. However, this sample size is larger than most of the EEG studies mentioned in the introduction (~10-20 participants). Thus, it is expected to be sufficient to detect the primary neural effects of interest.

6.2.2. Predictions

Behavioural predictions I sought to replicate the core behavioural result shown in Study 1. Namely, that the incongruent meter-syntax alignments would interfere with the ability to respond accurately to comprehension probes. I additionally predicted that the neutral meter would also negatively affect comprehension accuracy and response times. However, due to the sample

size, it is unlikely there would be enough statistical power to detect the smaller effects on response times (and again, this was not the primary focus).

Delta tracking of meter Delta-oscillations are predicted to entrain to metrical structure. For all conditions, a spectral peak at 2.5Hz (i.e. the syllable rate of the stimuli) is trivially predicted. The crucial observation is then whether there is significant spectral energy at higher-level meter related frequencies. In the subject-extracted sentence conditions (which have a binary metrical prime), a delta peak at 1.25Hz is predicted. For the object-extracted sentences (with ternary metrical prime) peaks at 0.83Hz and 1.66Hz are predicted (1.66Hz is the harmonic of this ternary meter frequency as is often observed during the perception of ternary meter, see Nozaradan et al., 2011).

To determine whether delta tracks meter or syntax, two observations will be critical. Firstly, if both meter and syntax independently contribute delta activity then there should be differences in neural power between the different alignment conditions. This is because simultaneous oscillations at the same frequency but different phases should partially cancel each other out. Indeed, this should be clearest for the comparison in the subject-extracted sentences where such hypothetical oscillations in congruent and incongruent conditions should be perfectly counter-phase to each other. If only meter contributes to delta tracking, there should be no significant differences between conditions in terms of neural power at meter-related frequencies. I predict the latter.

In addition to looking at differences in power, a clear indicator of whether delta tracks meter or syntax will be to also analyse the phase of delta. If delta only tracks syntax then the phase of the delta oscillations should not vary across conditions (and indeed to persist in the 'no meter' condition). However, if delta tracks only meter, then the phase of the delta oscillation should be clearly staggered evenly across the different conditions.

Modulation of non-phaselocked upper-beta power Finally, as a reflection of the effect of meter-syntax alignment on the encoding of the sentence in short-term memory, the congruent conditions are predicted to show greater desynchronisation (lower relative power) of high-beta oscillations (25-32Hz) than incongruent conditions and neutral conditions (consistent with Bonhage et al., 2017).

6.3. Behavioural results

Comprehension data were analysed similarly to the first two experiments: using mixed-effects logistic regression with fixed-effects for congruency (congruent, incongruent, neutral), syntactic complexity (subject-RC, object-RC), probed clause (main-clause, relative-clause), and probe framing (positive or negative). A term for trial count was not included because the neutral trials were always the first 32 trials of the experiment, thus confounding condition with trial count. Random intercepts for participants and items were also included in the model. Response times (RTs) were analysed using linear mixed-effects regression with the same structure.

The comprehension accuracy results (table 6) show a significant effect of the incongruent-1 alignment and a significant effect of the neutral alignment. However, the effect of the incongruent-2 alignment on comprehension did not replicate.

Table 6: Estimate of fixed effects for accuracy model

	β	Std. Error	Z value	P value
(Intercept)	1.205	0.177	6.813	<0.001 ***
Congruency (incongruent-1)	-0.264	0.125	-2.111	0.035 *
Congruency (incongruent-2)	-0.074	0.154	-0.481	0.630
Congruency (neutral)	-0.338	0.124	-2.720	0.007 **
Sentence extraction (subject)	0.699	0.108	6.491	<0.001 ***
Probed clause (relative-clause)	-0.592	0.093	-6.348	<0.001 ***
Probe framing (positive)	0.747	0.094	7.918	<0.001 ***

There was also a significant effect on response times (table 7) for the incongruent-1 condition and the neutral 'no meter' condition, but not the incongruent-2 condition.

Table 7: Estimate of fixed effects for response times model (log-transformed units)

	β	Std. Error	t value	P value
(Intercept)	0.665	0.055	11.997	<0.001 ***
Congruency (incongruent-1)	0.042	0.017	2.490	0.013 *
Congruency (incongruent-2)	0.021	0.021	0.996	0.319
Congruency (neutral)	0.082	0.016	4.907	<0.001 ***
RC extraction (subject)	-0.087	0.017	-5.066	<0.001 ***
Probed clause (relative-clause)	0.061	0.013	4.864	<0.001 ***
Probe framing (positive)	-0.082	0.013	-6.526	<0.001 ***

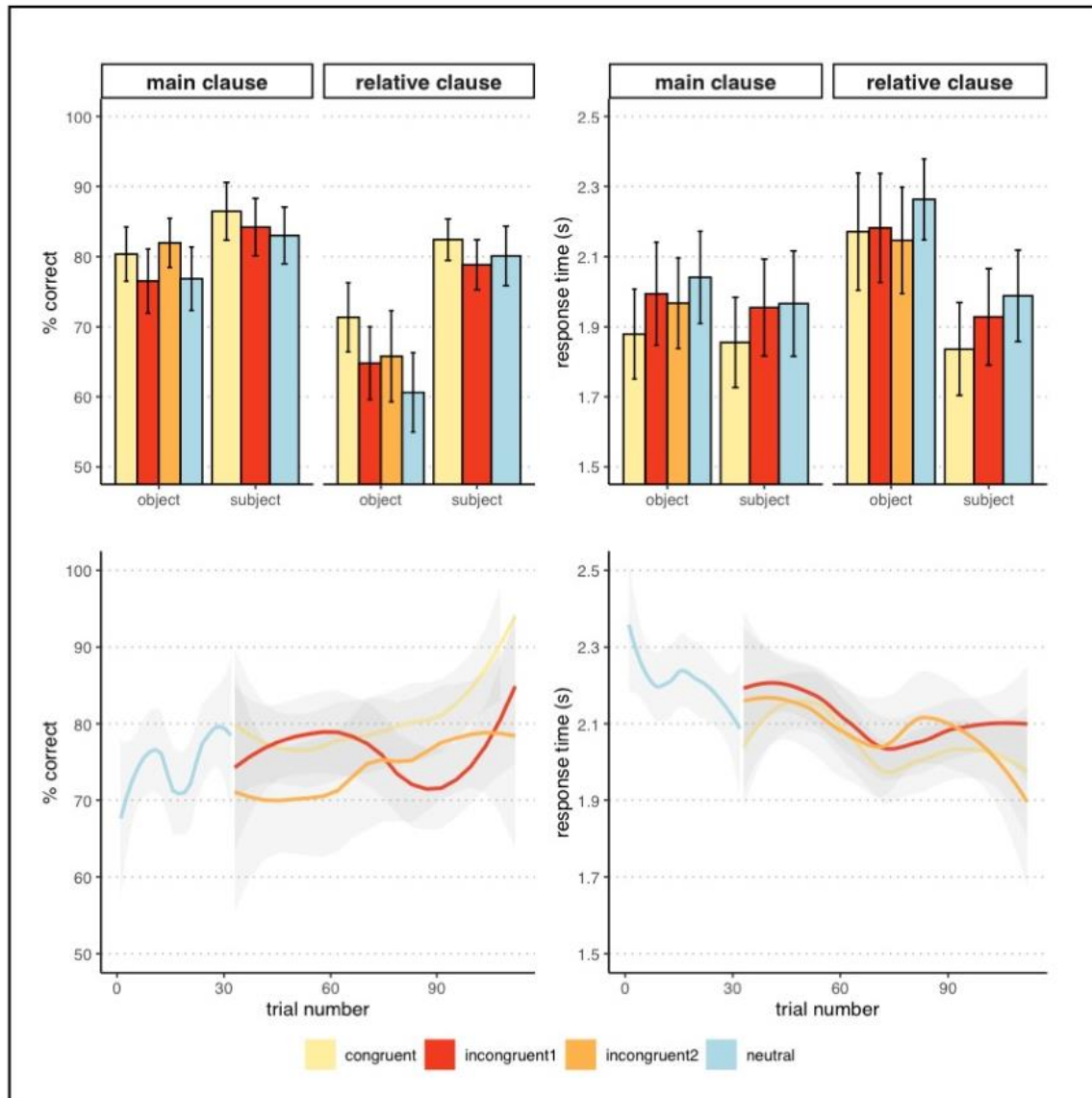


Figure 34: Experiment 3. Left: comprehension accuracy. Right: response times. Bottom: how these results distribute over the trials in the experiment (averaging over conditions).

6.4. Neural results

Preprocessing

Preprocessing was performed offline using a custom Matlab script, taking advantage of a number of EEGLab functions (Delorme & Makeig, 2004). Data were first filtered using a Hamming windowed FIR filter with 0.1Hz highpass and 50Hz lowpass. Then trial epochs were created ranging from 1 second prior to trial commencement to 12 seconds after. Data were then manually inspected for artefacts, rejecting trials that were overly noisy or contained other artefacts. Electrode interpolation was used sparingly for particularly noisy electrodes. After this inspection procedure, data were average rereferenced and then ICA (using the runica algorithm) was run to remove eye-blink artefacts.

GED source separation

A source-separation stage was then performed using generalised eigendecomposition (GED) implemented in a custom Matlab script. GED is an extension of other simpler forms of linear source separation such as Principal Components Analysis (PCA). This approach uses linear spatiotemporal filters to create a weighted sum of all the electrodes that optimises a ratio between certain minimisation and maximisation criteria (Cohen, 2017a; De Cheveigné & Parra, 2014). Roughly speaking, this approach can be said to isolate a source based upon an optimisation of a signal to noise ratio, where what counts as signal (maximisation; an $\mathbf{S}$ matrix) and what counts as noise (minimisation; an $\mathbf{R}$ matrix) is defined by the user in some way.

GED was used to isolate two components of interest for further analysis. The first was an *entrainment component*, which was intended to isolate activity related to low-frequency entrainment. The second was a *language processing component*, intended to capture neural activity selective to core linguistic operations. The primary advantage of analysing weighted-average components generated through GED is an increase in the sensitivity of the analysis to signal over noise (Cohen, 2017a, 2017b). Additionally, by computing these components orthogonally to the eventual statistical contrast to be analysed, as here, dangers of circular inference can be avoided allowing for more robust inferences as compared to other approaches such as arbitrarily selected ‘electrodes of interest’ (Kriegeskorte et al., 2009). Source separation is also applied here at the group-average level, affording the least possible chance of overfitting to noise.

6.4.1. Metrical entrainment source

The approach to computing the *entrainment source* was to use the ‘Rhythmic Entrainment Source Separation’ variant of GED (RESS; Cohen & Gulbinaite, 2017). Data were first epoched around just the speech component of each trial (ignoring the initial beat introduction). To define the ‘signal’ part of the optimisation algorithm, the speech data (all conditions, all participants) were narrow-band filtered with a peak of 2.5Hz (the syllable presentation rate) and a narrow full-width at half maximum (FWHM) of .2Hz. The channel-by-channel covariance of this narrow-band filtered signal constituted the $\mathbf{S}$ matrix. The ‘noise’ part used the exact same EEG data narrow-band filtered at a frequency 0.4Hz either side of the peak frequency (2.1Hz and 2.9Hz) with FWHMs of .4Hz. Covariance matrices were computed for each of these filtered datasets, and the $\mathbf{R}$ matrix was defined as their average.

Eigendecomposition was then applied to the ratio of $\mathbf{S}$ and $\mathbf{R}$ matrices and the component with the highest eigenvalue was selected. This yielded a filter forward model with a prominent fronto-central scalp topography (figure 36) typical of beat based neuronal entrainment (e.g. Nozaradan et al, 2011 among others). This forward model was then applied back to the raw data of all participants for further analysis.

6.4.1.1. Spectral analysis

To determine whether there was entrainment at meter-related frequencies, a static spectral analysis was performed. First, trial data were epoched around just the speech section of each

trial, omitting the metrical prime section. For each participant and condition, trials were then averaged to improve signal to noise, attenuating non-stationarities.

Typically for estimating such low frequency oscillations very long trials are used, on the order of 30 seconds in duration (e.g. Nozaradan et al., 2011). However, due to task constraints, each speech section of the trial in this experiment was only 4 seconds long. As such, the ability for the Fourier transform to reliably estimate spectral power is reduced since noise in the data would be less likely to be averaged out. Furthermore, edge artefacts are a potential problem with estimating low frequency activity with short trials. To attenuate these artefacts, a Hamming window was applied to each of the trial averaged timeseries. This also has the advantage of attenuating any potential ERP response to the change from auditory beats to speech. The Fourier transform is then applied to this windowed data. Zero-padding was used to achieve a spectral resolution of 0.01Hz. And Fourier coefficients were derived in units of power.

The resulting spectra of the binary trials are plotted in figure 38a and the ternary trials in figure 38b. To make comparison across all conditions easier, figure 38e shows bar plots for all conditions at just the four frequencies of interest. To account for any possible spectral leakage the values shown for each frequency and condition were averaged across 5 Fourier coefficients centred on the frequency of interest.

1.25Hz activity (binary meter)

Both congruent and incongruent subject-extracted sentences were preceded with a binary metrical prime (strong beat every other beat). If this caused the sentence to be processed in a binary meter, then there should be a spectral peak at 1.25Hz. This is indeed what was found as is visually clear in figure 38a. This power increase is significantly higher than in the ternary conditions, where there is no expected power at 1.25Hz ($M = 0.108$, 95% CI $[-0.031\ 0.184]$; $t(25) = 2.908$, $p = 0.007$; paired-samples two-sided t test; $d = 0.800$). There was no significant difference between congruent and incongruent conditions ($M = -0.007$, 95% CI $[-0.090\ 0.074]$; $t(25) = -0.197$, $p = 0.845$), consistent with my prediction. There was also a significant difference between the binary conditions and the neutral binary-condition ($M = 0.079$, 95% CI $[0.001\ 0.158]$; $t(25) = 2.083$, $p = 0.047$; $d = 0.572$), as well as a significant difference between the neutral conditions and the ternary conditions ($M = 0.028$, 95% CI $[0.003\ 0.054]$; $t(25) = 2.302$, $p = 0.030$; $d = 0.653$).

0.83Hz and 1.66Hz activity (ternary meter)

Congruent, incongruent-1, and incongruent-2 conditions for the object-extracted sentences were preceded with a ternary metrical prime (strong beat every three beats). If this meter is perceived during the sentence, a peak at 0.83Hz is expected. This can clearly be seen in figure 38a and 38e. Power in ternary conditions was indeed significantly higher at 0.83Hz than in the binary conditions, in which no peak is expected at 0.83Hz ($M = 0.064$, 95% CI $[0.016\ 0.112]$; $t(25) = 2.760$, $p = 0.011$; $d = 0.726$). Power in the ternary conditions was also significantly higher than the neutral ternary-condition ($M = 0.054$, 95% CI $[0.001\ 0.107]$; $t(25) = 2.103$, $p = 0.045$; $d = 0.598$). There was no significant difference between neural power between the different ternary conditions ($F(2, 75) = 0.279$, $p = 0.797$), nor was there a statistically significant difference

between the neutral condition and the binary meter conditions ($M = 0.010$, 95% CI [-0.026 0.046]; $t(25) = 0.569$, $p = 0.574$).

It is also expected that there by a spectral peak at 1.66Hz for ternary meter (i.e. the harmonic of the 0.83Hz periodicity). Consistent with the other results, neural power at 1.66Hz was significantly higher in the ternary conditions than in either the binary conditions ($M = 0.053$, 95% CI [0.021 0.086]; $t(25) = 3.364$, $p = 0.002$; $d = 0.846$) or the ternary-neutral condition ($M = 0.064$, 95% CI [0.031 0.096]; $t(25) = 4.004$, $p = <0.001$; $d = 1.036$). Neural power again did not differ at 1.66Hz between the different ternary meter congruencies ($F(2, 75) = 0.072$, $p = 0.931$),

2.5Hz (syllable rate; exploratory analysis)

Another prominent pattern in these data, that I did not predict, was that neural power at the syllable rate (2.5Hz) would be higher in the conditions with metrical primes over those with neutral primes, as can be seen clearly in figure 35. This difference is statistically significant for both the ternary conditions over the ternary-neutral conditions ($M = 0.115$, 95% CI [0.057 0.174]; $t(25) = 4.048$, $p = <0.001$; $d = 0.726$), as well as the binary conditions over the binary-neutral condition ($M = 0.225$, 95% CI [0.135 0.315]; $t(25) = 5.157$, $p = <0.001$; $d = 1.067$). There was also significantly higher power in the binary conditions at 2.5Hz than there was in the ternary conditions ($M = 0.106$, 95% CI [0.050 0.162]; $t(25) = 3.888$, $p = <0.001$; $d = 0.436$).

Phase alignment of delta oscillations

Does the phase of delta track meter or syntax? The answer to this question is clearly seen in figures 35c and 35d, which show the neural response to the sentences narrow-band filtered around the metrical frequency of interest (1.25Hz for binary trials, 0.83Hz for ternary trials). If delta tracked syntactic structure, then activity should be more-or-less synchronised across metrical alignment conditions. However, as can be clearly seen, delta activity systematically varies in phase between metrical alignment conditions, such that there are two evenly spaced apart oscillations for binary conditions (figure 35c) and three for the ternary conditions (figure 35d). There is no clear pattern for the neutral conditions. This is precisely as predicted if delta tracked the perceived meter rather than syntax.

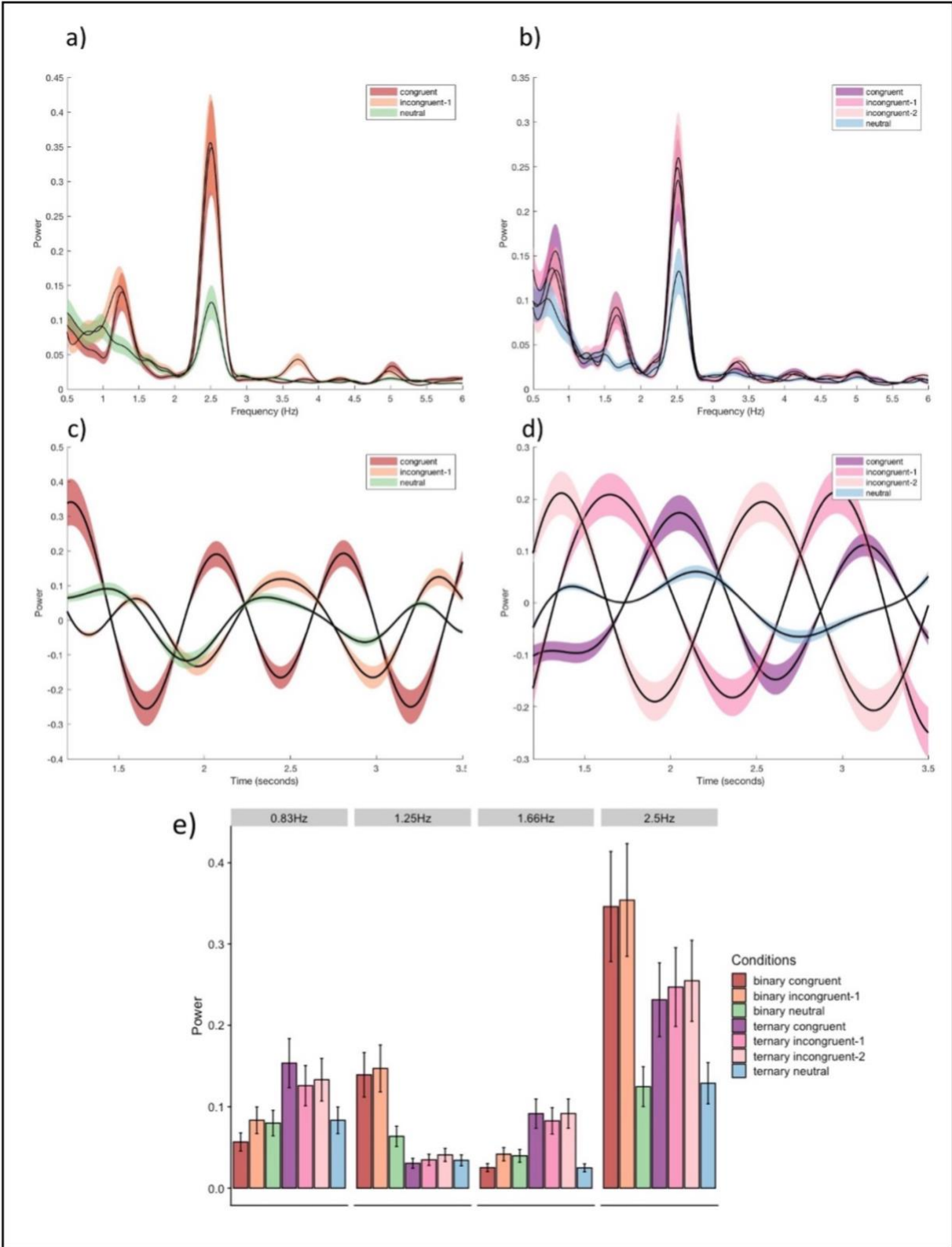


Figure 35: Spectral results from Experiment 4. a) & b) show neural spectra for subject- (binary) and object-extracted (ternary) sentences respectively. c) and d) show these same data as a timeseries narrow-band filtered at the meter-related frequency. Coloured shading in a, b, c, and d represents the standard error of the mean. e) shows the spectra from all conditions on a single plot.

6.4.1.2. Time-frequency analysis

To further support the interpretation that these neural data reflect metrical attending, a time-frequency analysis was conducted to supplement the static spectral analysis. This was implemented by convolving the signal with a family of complex Morlet wavelets, resulting in an estimate of oscillatory power at each time step and at frequencies ranging from 1 to 50 Hz in linear 1 Hz steps. The time-frequency resolution of the wavelets was frequency dependent such that the full-width at half maximums of the wavelets scaled logarithmically from 0.2Hz to 0.4Hz over the 50 frequency steps. The resulting time-frequency map saved timepoints from 350ms before time-locking event to 600ms after (avoiding edge artefacts), and then downsampled the resulting map to a resolution of 100Hz. All further time-frequency analyses in this chapter are conducted with these same parameters.

The data from each trial of the experiment was broken up into epochs ranging from -500ms to 800ms centred on the onset of each word as a time-locking event. For each participant, time-frequency maps were then computed for each of these individual epochs before then averaging over trials for each meter-syntax alignment condition, and each beat level (i.e. separate maps were computed for words aligning with strong and weak beats in each of those conditions). The entire metrical beat introduction section of the trial was then used as a baseline (a similar baselining strategy was used in Gordon, Magne, & Large, 2011) against which a decibel-normalisation procedure was applied to the resulting trial-averaged time-frequency maps.

A comparison of nonphaselocked timefrequency activity for words occurring on strong beats to those of weak beats was made to further test the presence of metrical attending. This comparison was statistically evaluated using a non-parametric permutation-testing procedure, using a significance threshold of $p = 0.05$ and 1000 permutations per comparison. To correct for multiple comparisons, a further conservative pixel-based cluster thresholding approach was used. The plotted results show a map of z-values that differentiate phase-locked oscillatory power, and the black-outlines show the clusters of statistically significant z-values that survived the cluster thresholding procedure.

In line with previous work on timefrequency correlates of meter (e.g. Fujioka, Ross, & Trainor, 2015 among others), the resulting difference map (figure 36) shows significantly higher beta power leading up to the beat, followed by a rapid desynchronisation after the beat.

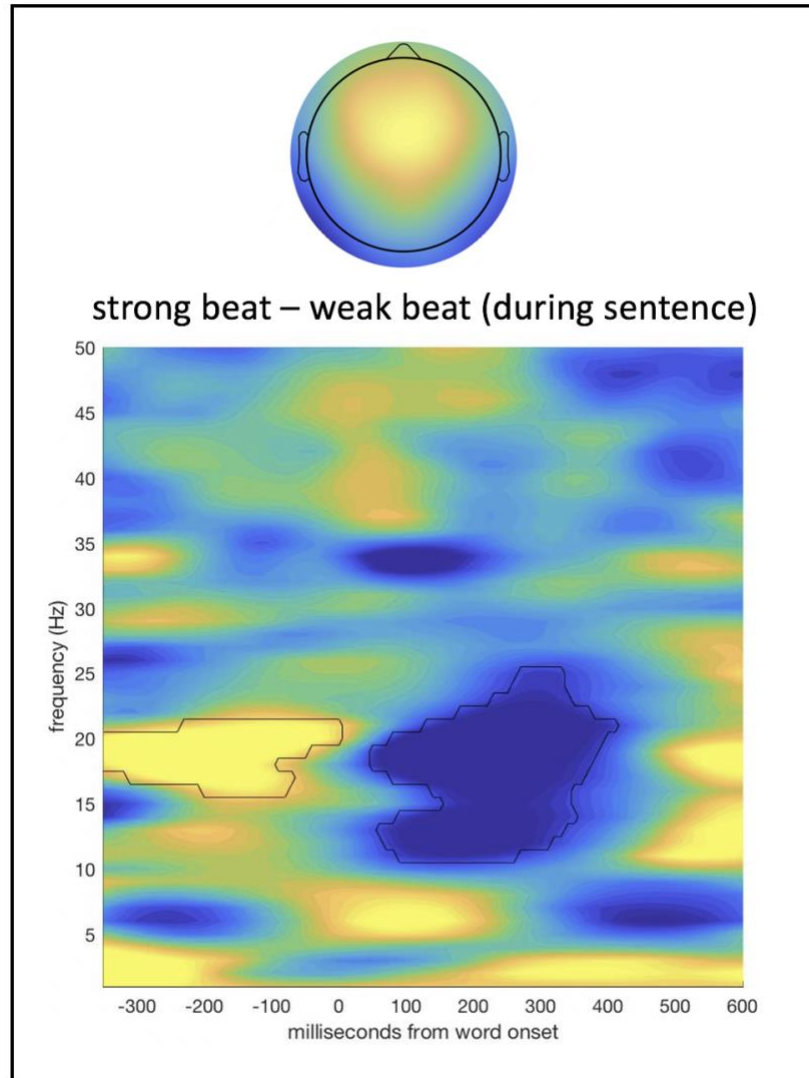


Figure 36: Time-frequency difference map of non-phaselocked neural activity contrasting metrically strong words against metrically weak words (areas surrounded by black outline are statistically significant).

6.4.2. Language network source

To test the effect of meter-syntax alignment on more language-specific processing, a ‘language source’ was isolated using GED. Firstly, the EEG data were narrow-band filtered across the upper-beta/lower-gamma gamma band (peak: 35Hz, FWHM: 10Hz). This decision was based on previous research finding this frequency range being strongly involved in core linguistic processing (Bastiaansen & Hagoort, 2006; Lewis et al., 2015), and indeed the correlates of working memory encoding are also featured in this frequency range (Lundqvist et al., 2016). After narrow-band filtering, the **S** matrix was defined as the the activity corresponding to the first two words of each trial (e.g. “the boy”) and the **R** matrix was defined as the first ‘weak-strong’ beat pair of each metrical introduction (e.g. 2nd and 3rd tone in binary conditions and 3rd and 4th tone in ternary conditions). Initially, source separation was tried with the whole speech section as the **S** matrix and the whole beat section as the **R** matrix, yielding similar but less

stable results. This is likely a result of extra variance and nonstationarities inherent in data over longer time periods.

Applying the GED procedure to these **S** and **R** covariance matrices yielded the following filter-forward model, with a plausible scalp distribution seeming to correspond to left-lateralized superior temporal lobe and inferior frontal areas. This forward model was then applied to the (unfiltered) data of all participants for further analysis.

To assess the effect of meter-syntax alignment on this language processing source, two difference maps were computed. The first shows the difference between non-phaselocked time-frequency power between congruent and incongruent conditions (figure 37a). The second shows the difference between congruent and neutral conditions (figure 37b). In line with my predictions, there was significantly lower high-beta power in the congruent conditions for both comparisons.

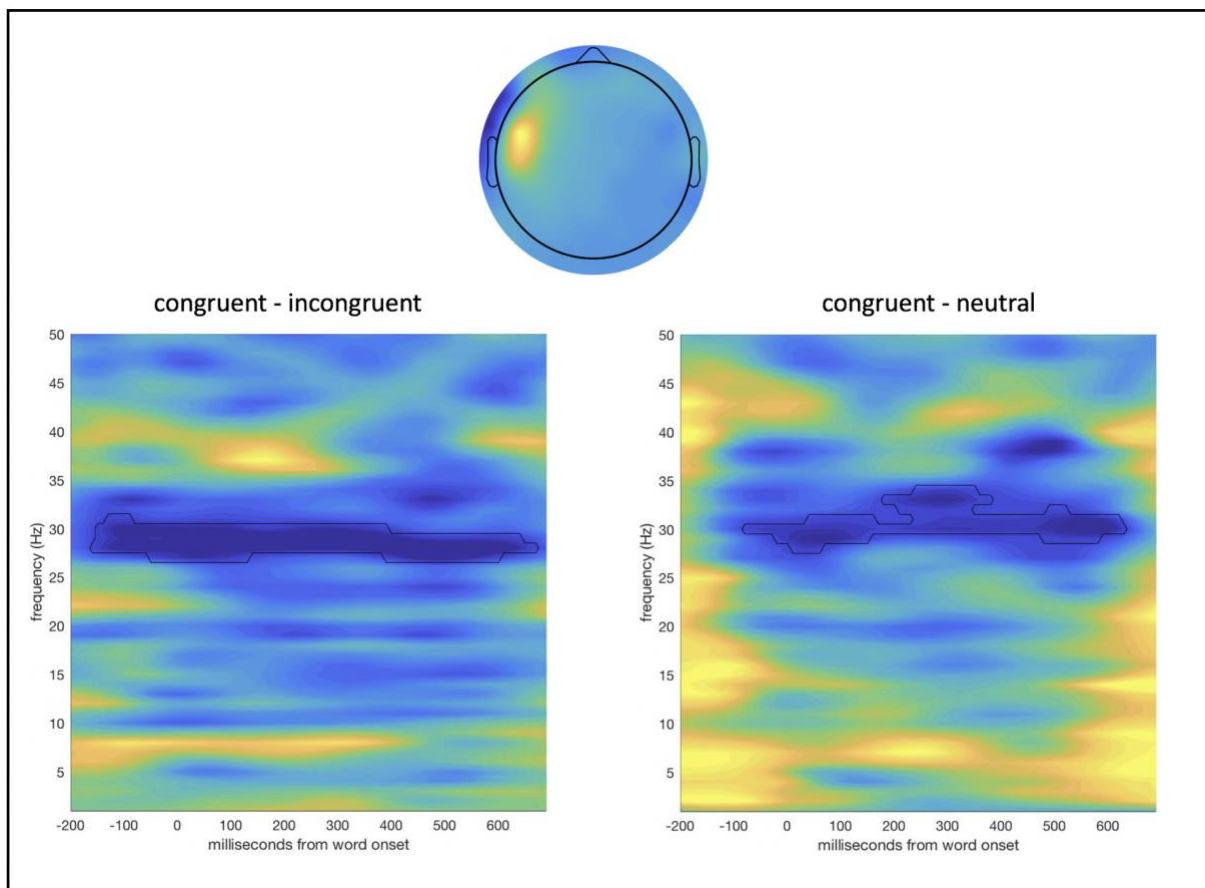


Figure 37: Top: a forward model of language comprehension source. Bottom: difference maps for time-frequency activity (areas surrounded by black outline are statistically significant).

6.5. Discussion

The results of this study replicated the core behavioural result from study 1 while also probing its underlying neural dynamics. As predicted, delta oscillations tracked perceived metrical

structure in the acoustically neutral speech rather than following syntactic phrase structure, going against recent theories linking delta to more higher-order linguistic generators. Unlike study 1, this study also included a 'neutral' meter condition where sentences were primed by an unaccented beat, this also had clear behavioural and neural effects consistent with predictions derived from a recent model of short-term memory. Finally, analysis of a 'language component' showed alignment related differences in high-beta oscillations, supporting the interpretation that congruent meter-syntax alignment supported more robust encoding in short-term/working memory.

The main finding in this study was that delta oscillations tracked meter, not syntax. As discussed in the introduction, and in chapter 2, the field has tended to devalue prosody and rhythm in favour of the most abstract aspects of language (also see Kandylaki & Kotz, 2020). A number of recent studies have suggested that delta oscillations reflect an abstract syntactic or compositional function (e.g. Martin & Doumas, 2017; Meyer et al., 2017; Meyer & Gumbert, 2018). I have suggested here that this is incorrect and an example of this tendency to ignore prosodic structure. The results of this study support the alternative interpretation that delta tracks metrical structure (Nozaradan et al., 2011) or, more generally, processes of dynamic attending (Large & Jones, 1999; Kotz & Schwartze, 2010) and there was little to no evidence to support these other theories.

Specifically, the phase of delta oscillations reliably tracked the perceived metrical structure, which shifted between the various alignment conditions as opposed the syntactic structure which remained constant. This phase tracking with meter contradicts a strong interpretation of the compositional meaning model (Martin & Doumas, 2017; Brennan & Martin, 2019), which predicts delta phase aligns to compositional computations. A weaker claim of this theory might predict that compositional computations contribute a smaller degree of additional delta activity that would combine with other sources in the measured signal (e.g. Kaufeld et al., 2020; Meyer, Sun, & Martin, 2019). If this were true, delta power should vary between the different meter-syntax alignment conditions since non-phase aligned delta sources should at least partially cancel out. However, the data in the present study find no evidence for this either, since the different alignments did not significantly differ in delta power at the phrasal/metrical rate.

If some form of this compositional delta theory were correct, then one would also expect to see delta tracking for the conditions in which there was a 'neutral' metrical prime (an unaccented series of isochronous tones prior to the speech). There was some suggestion of this for the binary-neutral condition and nothing for the ternary-neutral condition. However, one should be careful about interpreting the weak binary-neutral result. While there is slightly elevated activity at this frequency when taking this frequency in isolation, looking at it in the context of the whole spectra, there is a notable absence of a peak relative to surrounding Fourier coefficients (figure 35a), meaning that this relatively raised activity is likely artefactual.

The observed detrimental effect of neutral meter on comprehension relative to even the incongruent conditions may seem counter-intuitive. However, as discussed in section 6.1.2, this result is consistent with the oscillation-based model of serial-order short-term memory proposed

by Hartley, Hurlstone, and Hitch (2016). Consistent with the assumptions of this interpretation, the data only showed neural entrainment at the syllable rate for the neutral conditions. Furthermore, there was significantly less neural power at this frequency in the neutral meter condition than there was in the other metrical conditions. This may have therefore further contributed to there being a weak context signal with which to support memory processes.

How can a lack of delta tracking for the 'neutral meter' condition be reconciled with previous findings showing delta despite not having any explicit metrical prime (e.g. Ding et al., 2016; Teng et al., 2020)? Unlike these previous studies, the present experiment only presented one sentence per trial as compared to ten consecutive sentences per trial (without break) in the study from Ding and colleagues. Isochronous and structural repetitive stimuli like these are well accepted as strongly inducing metrical percepts in music (e.g. Lerdahl & Jackendoff, 1983). For example, a study by White (2017) showed that when all else was equal, tonal syntax dictated the perception of meter in musical stimuli. It is therefore likely that these studies were incidentally inducing metrical percepts, despite the lack of overt prosodic cues.

Anecdotally supporting this, I heard David Poeppel demo the English stimuli from the Ding study at a talk in Köln in February 2019, and my immediate subjective impression was that I was hearing the speech in a metrical context. The findings of the present study therefore not only have important implications for neuro-oscillatory theories of language processing but also methodological implications concerning not confusing the acoustic cues of prosody with their perception, as seems to have at least implicitly been the case in some prior studies (Ding et al., 2016; Meyer et al., 2017).

The results also showed that congruent alignment resulted in greater desynchronisation of upper-beta oscillations as compared to incongruent or neutral conditions. This was observed in the language source, which seemed to reflect activity in left-lateralized temporal and frontal language areas. This pattern is consistent with other prior work showing upper-beta desynchronization for working memory in language (Bonhage et al., 2017), as well as more general models of working memory (Lundqvist et al., 2016, 2018; Miller, Lundquist, & Bastos, 2018). While further research is definitely needed, this appears to be consistent with the interpretation that metrical alignment influences memory processes.

A limitation the study design used was that the trials for 'neutral meter' condition were all done in one block prior to the rest of the trials (which were randomised for order). This was done deliberately so as to not bias their metrical interpretation. However, this leaves open a potential practise effect confound whereby performance may have been better later in the experimental session simply due to practise with the task. This cannot be ruled out entirely, and further research should be done to replicate this effect without the confound. Nonetheless, there were practise trials prior to the neutral meter block, which should help attenuate this potential confound. The result is also consistent with other previous research showing that speech stresses improve the memory performance of otherwise rhythmically identical stimuli (Ryan, 1969; Boucher, 2006).

6.6. Conclusion

The results of this study further support the proposed effect of metrical alignment on sentence processing. It was also shown that delta tracks meter rather than syntax, going against some recent theoretical suggestions that have underestimated the involvement of abstract prosodic structure in sentence processing. It was also shown that a lack of metrical tracking altogether is more costly to comprehension than various forms of misalignment, which is consistent with the oscillation-based memory model from Hartley, Hurlstone, and Hitch (2016).

7. Study 3: Algebraic syncopation

It has been claimed so far that the effect of meter-syntax alignment is mediated by how metrical alignment affects short-term memory. The primary aim of this final study is to more directly investigate whether this interpretation is correct. Specifically, on closer analysis, the study design used in the prior experiments does not rule out an alternative hypothesis whereby meter supports individual item memory rather than serial-order memory. The final experiment ($n = 69$) is designed such as to disentangle these possibilities by using a non-linguistic grouping task that removes the rich semantic constraints inherent to language which confound this interpretation of studies 1 and 2. The results show evidence in favour of both hypotheses: that metrical alignment affects both individual item memory as well as serial-order robustness.

7.1. Introduction

The main interpretation of the results so far is that the alignment of meter to phrase structure influences the robustness of serial order short-term memory. This is partly motivated by recent computational models, which posit the use of a ‘context signal’ to represent serial order, and that this context signal be represented by the state of entrained neural oscillations (Hartley, Hurlstone, & Hitch, 2016). Given that meter is associated with the entrainment of endogenous oscillators (Large & Jones, 1999; Nozaradan et al., 2011; chapter 6), it seems reasonable to assume that they could be used as part of this context signal. Indeed, meter affects serial order memory in both language (Boucher, 2006) and music (Mathias, Palmer, & Pfordresher, 2015).

However, there remain reasonable objections to this being the only factor at play. Firstly, as discussed in study 1 (chapter 4), I predicted an interaction between meter-syntax alignment and syntactic complexity, such that alignment would have a stronger effect on sentences with higher syntactic/memory demands (Gibson, 1998; 2000). This prediction was partly based on the assumption that non-local dependencies tax short-term memory resources. However, this was not supported by the data.

Secondly, while serial order memory is important for syntactically extended sentences (King & Just, 1991), so is individual item memory. Serial order and individual items are thought to be dissociable in short-term/working memory (Burgess & Hitch, 2006; Henson et al., 2003; Ng & Maybery, 2002). Item memory may independently contribute to task performance in the experiments so far. For example, in the subject-extracted congruent condition from experiment 1, content words aligned with strong beats (“boy”, “helped”, “girl”, “A”, “test”), whereas in the incongruent condition there are largely function words on the strong beat (“the”, “that”, “the”, “got an”, “on the”). Content words are more informative to sentence meaning, so a more robust encoding of them could result in better comprehension, all else being equal (including robustness of serial-order). Thus, the congruency effect may simply be the result of disrupting the encoding of the most informative items in short-term memory, separate from their serial order representation. Or indeed it could be some combination of these factors.

It is difficult to distinguish these alternatives in the experimental paradigms presented so far because the language system has evolved mechanisms to cope with noisy memory representations. In particular, mishearing a particular word or misremembering the ordering of a sequence of them can often be compensated for by semantic heuristics (Mollica et al., 2020; Ferreira et al., 2002; Christianson, 2016). For example, in the sentence “the police officer held out her ___”, people will readily interpret the sentence in terms of the final word as “badge” regardless of whether they heard it clearly or not. Thus, if participants are more robustly encoding meaningful content words in the congruent alignments, this might help them infer sentence meaning more so than if they were just robustly encoding function-words.

Interpreted within the framework of Dynamic Attending Theory (Large & Jones, 1999; chapter 5), there has indeed been evidence that alignment to musical beats can support individual item memory. For example, a recent study by Hickey and colleagues presented visual images of objects to participants while they listened to metrically structured music. Implicit memory for these objects was higher for those aligned with neurally entrained strong beats (Hickey et al., 2020).

In linguistic contexts, meter has also been shown to affect lexical access. That is, the alignment of meter affects the degree to which the neural representation of a word’s meaning is activated and represented in short-term memory during sentence processing. This has been shown both behaviourally and related to neural correlates of sentence comprehension (Gordon et al., 2011; Kember et al., 2019; Magne et al., 2007). Recent experiments also show that rhythmic stimulation during a maintenance phase of a working-memory task can improve task performance (Plancher et al., 2018), potentially by allowing greater attentional energy for refreshing items in memory.

It may be the case, then, that meter-syntax alignment supports comprehension by more robustly encoding syntactically informative words into working memory (indeed, this is similar to the claim made by Meyer & Gumbert, 2018 discussed in the last chapter).

7.1.1. Metrical modulations of serial order memory

Prevailing models of short-term memory posit a dissociation between item and serial order representation, and specifically relate the representation of serial order to a timing/context signal (Henson, 1998; Ng & Maybery, 2002; Henson et al., 2003; Burgess & Hitch, 2006). Some of these theories implement this context signal as an oscillator whose state at each item presentation functions as an associative cue for serial recall (Brown, Preece, & Hulme, 2000).

Hartley, Hurlstone, & Hitch (2016) extend this to the entrainment of multiple neural oscillators at different time scales and propose that these are synchronised to external signals through bottom-up phase-locking with the amplitude envelope, as has been well-described empirically for speech perception (Luo & Poeppel, 2007; chapter 5). This model provides an account for why rhythmic manipulations of item presentation affect the robustness of serial-order recall. They show this by comparing model simulations against the results of a number of serial-order memory experiments that manipulate the timing of item presentation (replicating and extending:

Ryan, 1969). The results confirm a good fit of model and human recall and highlight systematic effects of timing on memory.

The plots below (figure 38a) show human memory for ungrouped (isochronous) presentation against a presentation in rhythmic groups of threes (implemented by inserting a short pause after every third item). Not only does rhythmic grouping result in superior memory, but strikingly, when rhythmically grouped, transposition errors tend to track with group position rather than just local proximity. In other words, transposition errors are more likely between groups rather than within groups as shown by the peaks in figure 38b for the grouped conditions.

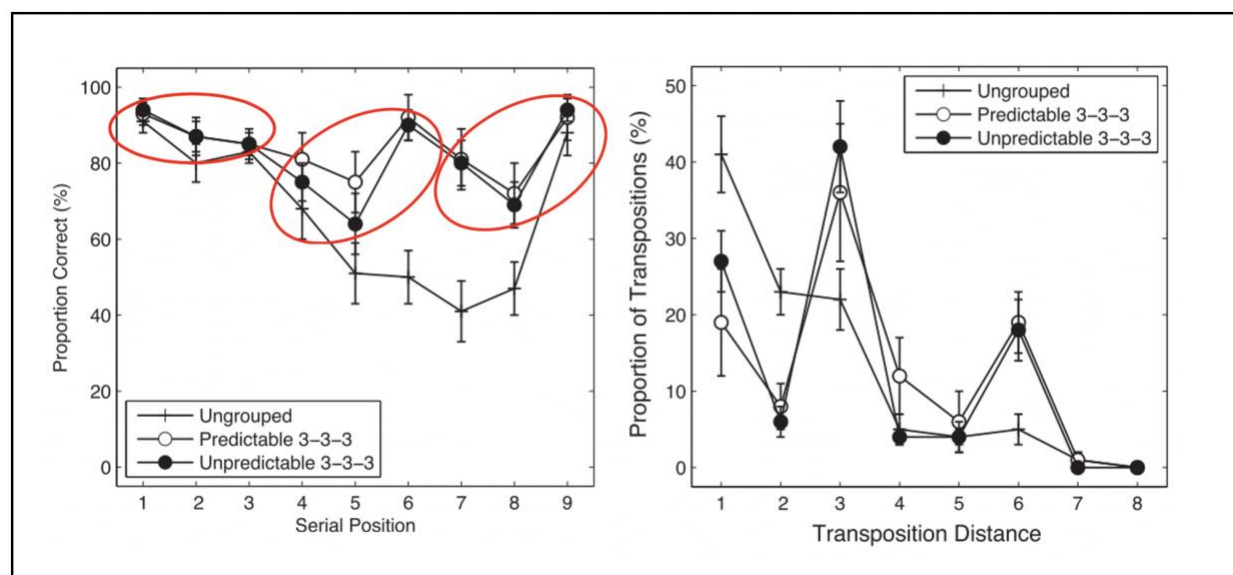


Figure 38: Adapted from Hartley, Hurlstone, & Hitch (2016). Left: recall performance for different positions in sequence. Red circles highlight the grouping in threes. Right: frequency of different transposition distances. It is notable that the most likely interposition error (besides swapping adjacent items) is swapping items that are 3 items away (i.e. have same position in the next group).

Another characteristic pattern is that serial position errors follow a ‘scalloping’ pattern, where serial memory is least robust in the middle of rhythmic groups and most robust at the ends. This pattern comes out in both their behavioural data and the model simulations. A related finding is that response-times to probed items that begin rhythmic groups are reliably slower than at the end of groups (Hurlstone, 2019). These patterns may help to explain why shifting the alignment of meter relative to syntactic phrase structure affects comprehension. In other words, the nuclear-stress rule may reflect an optimal alignment of the oscillatory context signal such as to make the most important information within phrase groups (with regard to serial-order robustness) the most robust.

Another related possibility is that the incongruent meter-syntax alignment interrupts the ability to attend higher metrical levels to the sentence. For example, in the binary meter condition every second word is primed to have a metrical accent. However, as is commonly accepted in music theory, there is a tendency to perceptually attribute an even higher metrical level if it is consistent with the structure of the stimuli (Lerdahl & Jackendoff, 1983). Therefore, one might

perceptually attribute the following higher-level accent pattern, even though it was not explicitly cued: “the **boy** that helped the **girl** got an ‘A’ on the **test**” (relative accent strength indicated by bolding). Indeed, speech production data indicate that people naturally phrase their speech to signal up to five levels of metrical structure if the language supports it, as often especially clear in children’s literature with regular meter reinforced by rhyming (Breen, 2018).

My impression is that this higher-level meter naturally emerges in the congruent alignment but not in either of the incongruent conditions. If the congruent meter-syntax alignment is better able to support the additional entrainment of higher metrical levels, then this would predict even more robust representation of serial order (a more distinct context signal), with less likely transposition errors within larger phrase groups.

More generally, it is argued that rhythmic grouping for enhanced serial-order representation is a domain-general property of verbal/auditory short-term memory, manifesting similar patterns of results in both music and language (Gorin et al., 2016, 2018b, 2018a; Mathias, Palmer, & Pfordresher, 2015). And there appears to be growing recognition of the involvement of low-frequency neural delta oscillations in memory-based processes (Boucher et al., 2019; Ghitza, 2017; Rimmele et al., 2020; chapter 6), and for an association between rhythmic timing abilities and serial-order recall more generally (Gilbert et al., 2017; Henson et al., 2003; Saito, 2001).

7.1.2. Summary

The current and final study aims to address these issues. That is, to provide evidence to adjudicate more reliably whether the meter-syntax alignment manipulation affects item or serial-order memory (or neither or both).

7.2. Experiment 4

7.2.1. Method

Experiment 4 takes inspiration from the set of experiments on algebra processing first discussed in chapter 3 (Landy & Goldstone, 2007). These studies showed that formally irrelevant perceptual features (such as spatial proximity) affect how people processed hierarchical structure in algebra. This analogous to the argument being made here: that the formally irrelevant prosodic aspect of the signal also plays a fundamental role in language comprehension.

In their original experiments, Landy & Goldstone (2007) presented algebraic expressions after which followed a probe expression that scrambled the order of the operands. Participants judged whether the scrambled version was algebraically equivalent to the original. Equivalency depends upon the hierarchical grouping determined by the mathematical operators and the order in which they are applied. This order of operations canonically follows the order-of-precedence convention (e.g. sometimes referred to as the acronym ‘PIDMAS’), which, for example, dictates that multiplication is performed before addition. The grouping implied by

operator precedence affects equivalence judgements since serial order does not matter within a group. For example:

$$A * B + C * D \leftrightarrow D * C + B * A \quad (\text{equivalent})$$

$$A * B + C * D \leftrightarrow D * B + C * A \quad (\text{not equivalent})$$

In their experiments, Landy and Goldstone manipulated the participant's ability to make equivalency judgments by varying the visual proximity such that it was either consistent or inconsistent with the correct grouping.

$$A * B + C * D \quad (\text{consistent})$$

$$A * B + C * D \quad (\text{inconsistent})$$

Consistency had a large effect on speeded equivalency judgements. Crucially, however, this effect was only shown for what they called 'sensitive trials'. Sensitive trials were those in which the equivalence of the probe (what they called 'validity') differed depending on whether multiplication precedes addition (as is the standard convention) or whether addition preceded multiplication. In other words, if participants used spatial proximity to parse the expression, they would make a lot of mistakes when proximity is inconsistent in the sensitive trials. This is precisely what they found (figure 39).

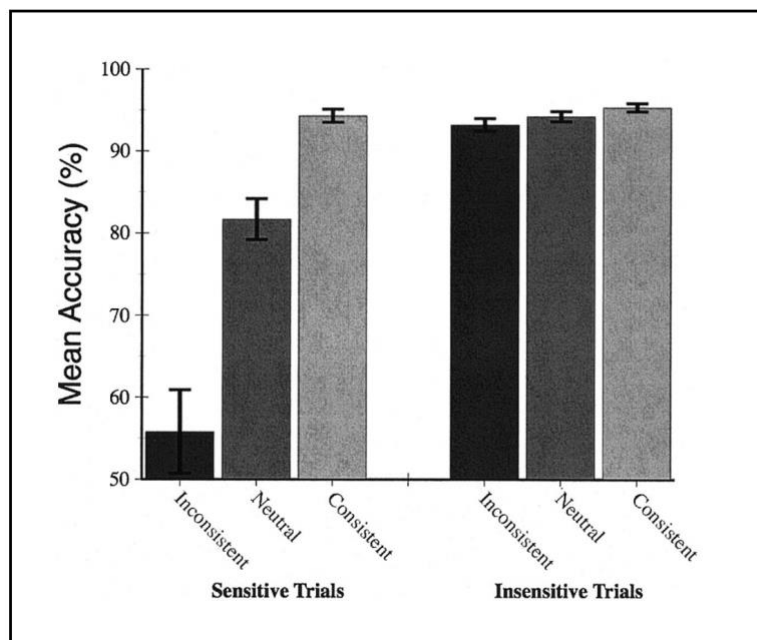


Figure 39: task accuracy results from Landy & Goldstone (2007).

Although obviously different from natural language comprehension, task performance in making these algebraic equivalency judgements also relies upon serial order to parse hierarchical

syntactic dependencies. But unlike language, operands and operators lack the kind of rich semantic content words in language have. This means that semantics cannot be used to repair noisy encoding. Therefore, any effect of meter-syntax alignment in this task should more directly reflect short-term memory without confounds arising from compensatory semantic heuristics.

7.2.1.1. Syntactic complexity

The previous experiments manipulated syntactic complexity by varying the extraction of a relative clause. Syntactic complexity can also be operationalised for these algebra expressions in a comparable way. For example, an expression like “A + B + C + D” is maximally simple, as it lacks hierarchical organisation, and as such, serial order is irrelevant to its truth condition. An expression like “A * B + C * D” is more complex because it involves the coordination of two higher-level constituents, only within which order does not matter. Finally, an expression like “A + B * C + D” is most complex. Although it only has two hierarchical levels like the previous example, there is an additional distance-based integration and memory cost (Gibson, 1998, 2000; see discussion in chapter 4) between the A and the D operands (figure 40).

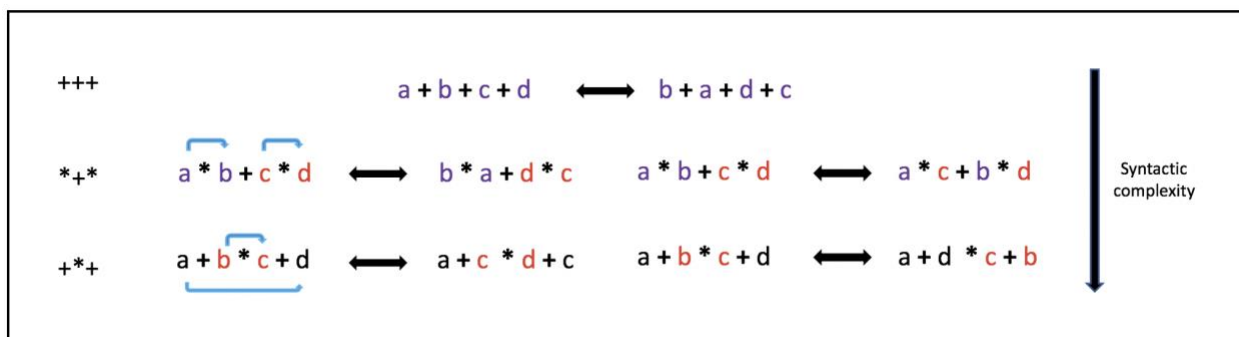


Figure 40: syntactic complexity of the algebra structures (in terms of dependency locality)

7.2.1.2. Metrical alignment

In the following experiment, algebra expressions are presented one item at a time as a visual rapid serial presentation (RSVP) synchronised to an auditory beat (similar to experiment 1). This contrasts with Landy and Goldstone’s design, which presented the whole expression simultaneously on a screen (2007). And instead of manipulating spatial proximity, I manipulate meter-syntax alignment like in Studies 1 and 2. To distinguish how the alignment of meter might affect serial order and item memory differently, meter was also manipulated at two distinct hierarchical levels (figure 41)

Level 1 meter aligns its strong beat to either the operands (congruent) or to the operators (incongruent). Roughly speaking, the operands are an analogue of content words in language and the operators an analogue of function words. Level 2 meter takes the congruent Level 1 meter (ensuring all operands coincide with strong beats) and adds an additional hierarchical level, resulting in a further strong beat on every other operand. This higher-level metrical level aligns its strong beat to either the rightmost operand in the group (congruent) or the leftmost operand in the group (incongruent).

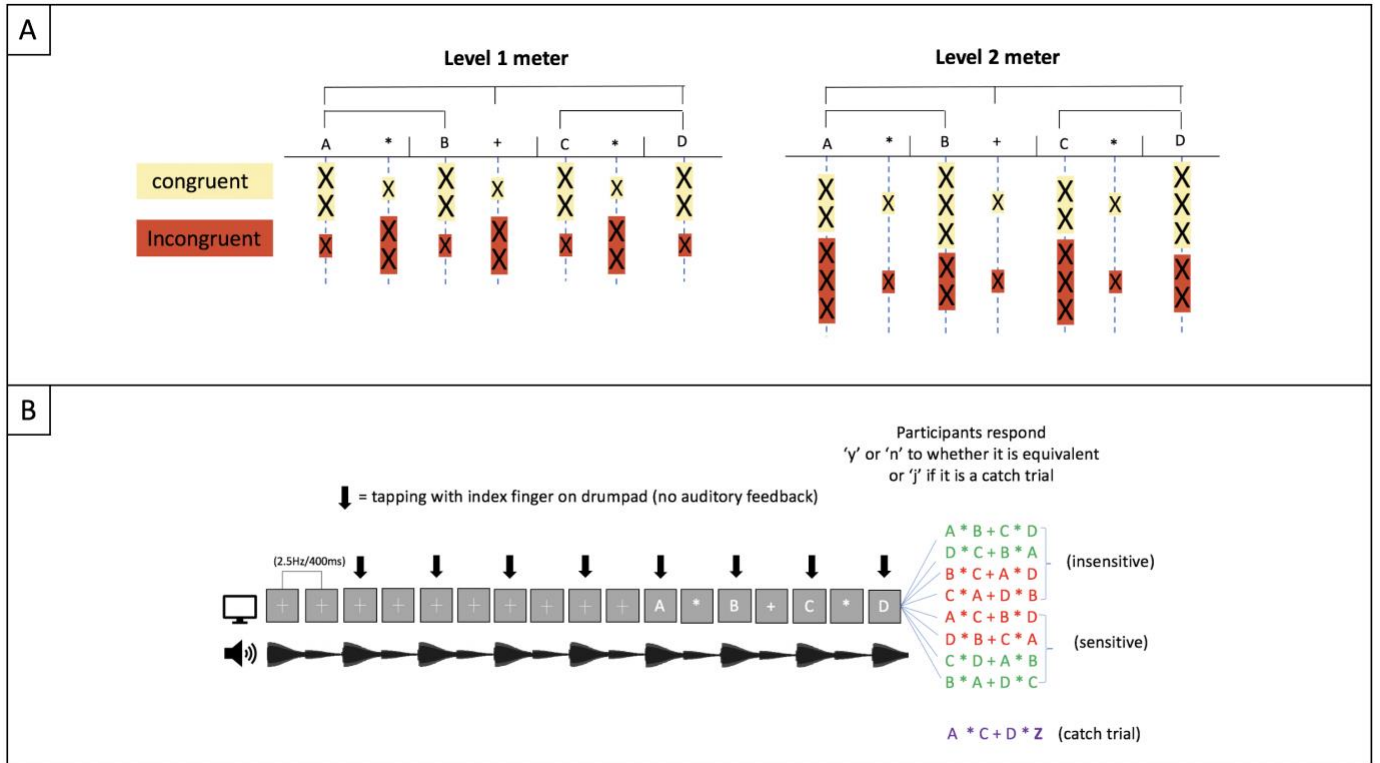


Figure 41: Top: Metrical alignment for Experiment 4 (only the $++$ structure shown to exemplify). Bottom: Trial schematic for experiment 4.

7.2.1.3. Auditory materials

The auditory beats that implement these different meter-syntax alignments was generated, as before, in a custom Python script and consisted of a 333Hz pure tone in which a 2.5Hz beat was induced by amplitude-modulating the signal with an asymmetric Hanning window with 80% depth and a 19:1 ratio of rise-to-fall time.

The Level 1 metrical accent was then induced by manipulating amplitude, with a 50% volume increase every other beat. The Level 2 metrical accent was induced by additionally adding a tone an octave lower once every four beats (applying this to the congruent level 1 alignment beat).

7.2.1.4. Sensitivity and probes

I adopt the same design as Landy and Goldstone (2007) for controlling the permutations of the probes in each trial. That is, the probe could be one of eight permutations of the original operands. This selected set of 8 permutations (out of a possible 24 permutations) allows Validity (whether equivalent to original) and Sensitivity to be balanced across trials, as represented in table 1 (for $++$ structure) and table 2 (for $++$ structure) below. The $++$ trials are not shown in a table as all permutations are valid and none are sensitive, however, the same permutations are applied.

Table 8 (these condition tables are identical to those in Landy & Goldstone, 2007)

Permutation	Stimulus	Probe	Valid	Valid if + precedes *?	Sensitivity
a b c d	$a + b * c + d$	$= a + b * c + d$	True	True	Insensitive
d c b a	$a + b * c + d$	$= d + c * b + a$	True	True	Insensitive
b c a d	$a + b * c + d$	$= b + c * a + d$	False	False	Insensitive
c a d b	$a + b * c + d$	$= c + a * d + b$	False	False	Insensitive
a c b d	$a + b * c + d$	$= a + c * b + d$	True	False	Sensitive
d b c a	$a + b * c + d$	$= d + b * c + a$	True	False	Sensitive
c d a b	$a + b * c + d$	$= c + d * a + b$	False	True	Sensitive
b a d c	$a + b * c + d$	$= b + a * d + c$	False	True	Sensitive

Table 9

Permutation	Stimulus	Probe	Valid	Valid if + precedes *?	Sensitivity
a b c d	$a * b + c * d$	$= a * b + c * d$	True	True	Insensitive
d c b a	$a * b + c * d$	$= d * c + b * a$	True	True	Insensitive
b c a d	$a * b + c * d$	$= b * c + a * d$	False	False	Insensitive
c a d b	$a * b + c * d$	$= c * a + d * b$	False	False	Insensitive
a c b d	$a * b + c * d$	$= a * c + b * d$	False	True	Sensitive
d b c a	$a * b + c * d$	$= d * b + c * a$	False	True	Sensitive
c d a b	$a * b + c * d$	$= c * d + a * b$	True	False	Sensitive
b a d c	$a * b + c * d$	$= b * a + d * c$	True	False	Sensitive

To ensure that participants were attending to all possible groups in each trial, an additional ‘catch trial’ procedure was used. In such catch trials, a random operand in the permuted probe is substituted for a different letter that was not present in the original. Participants are required to identify when such catch trials take place as these occur randomly throughout the experiment. This procedure was especially necessary to ensure that the $++$ structure was being processed as intended. That is, without catch trials, participants could adopt a strategy whereby they only need to attend to the middle group of $++$ trials and could safely ignore the outer group. This ensures that the processing assumption inherent to the syntactic complexity definitions in figure 40 are met.

7.2.1.5. Algebra materials

For each participant, a set of 192 trial equations were constructed, balanced equally between the three equation structures (+++, *+*, ++*) and meter-syntax congruency conditions. For these resulting trials, the probe equation for each was constructed by permuting the original operands according to one of the eight permutations as described in tables 1 and 2, ensuring each experimental condition had an equal number of each of the eight probe permutations. The specific operands were instantiated with random letters (drawing without replacement), with the letters i, l, and o being omitted from the possible letters due to their similarity to other symbols.

A further 60 'catch' trials were then constructed (balanced roughly between the different equation structures, and probe permutations). Each probe in the catch trial was a permutation of the original where a random operand is substituted with a different letter that was not present in the original equation. The main trials and the catch trials are then randomly mixed together, constituting a final number of 252 trials.

7.2.1.6. Experimental procedure

Participants were seated at a chair in front of a computer and supplied with headphones adjusted to a comfortable listening volume. They were also provided with a MIDI drum pad (Korg Nanopad; same as experiment 2). The experiment was run by a custom program written in python (largely using the PsychoPy python library). Participants were guided through a computer-based introduction by the experimenter, before the participants then completed a series of practise trials, including familiarisation with the tapping component, before then being encouraged to ask any clarifying questions before the main experiment. They then completed the main experiment as a single block, where they were forced to take a break of at least 15 seconds once every 64 trials.

Each trial is commenced by the participant pressing the spacebar. Once the trial begins, participants are instructed to hold their gaze on a fixation cross centre-screen while the auditory beat introductory period begins (playing two full bars of the meter). Participants are instructed to begin tapping in time with this beat as soon as they can and to continue tapping at the same rate while the trial equation appears on the screen.

After the full equation has been presented, there is a 400ms delay before the probe equation appears on the screen (in full, rather than sequentially presented). Participants are prompted to respond as quickly as possible, indicating whether the probe is equivalent to the original (pressing the "y" key) or not equivalent (pressing the "n" key). If they believe the trial is a catch trial (i.e. one of the letters has been replaced), they press the "j" key. If participants take longer than 5 seconds to respond, they are prompted to speed up on the next trial. Corrective feedback is given after each trial and participants are encouraged to balance trial performance (speed and accuracy) with attention on tapping in time as accurately as they can.

7.2.1.7. Participants

After three rounds of initial piloting and refinements, the main experiment recruited 69 undergraduate students between 17 and 40 years of age ($M = 20.7$, $SD = 4.4$; 45 female, 16 male, 8 undisclosed) from the University of Sydney, whose time fulfilled a partial course credit requirement. They had normal hearing, and normal or corrected vision, and no prior history of speech and language disorders. Given the complexity of the design and our hypotheses, a larger sample size was used, collecting data until the end of the semester.

7.2.1.8. Predictions

If metrical alignment only affects item activation, as described in section 7.1.1, then a congruency effect is only expected for level 1 meter and not for level 2 meter. This is because level 1 meter contrasts accenting the items that must be recalled (congruent) versus accenting the symbols that dictate how they should be grouped (incongruent). This differs from the level 2 metrical alignment, which contrasts accenting the items at either the right edge (congruent) or left edge (incongruent) of the group. Neither the left- or right-edge operand is informationally privileged with regard to the task of determining algebraic equivalency and neither can in principle be predicted from the other. Thus, under the 'item activation' explanation, the congruency manipulation in the level 2 alignment should not affect task performance. Additionally, catch trial performance should reflect the degree to which meter affects single item encoding (since catch trial performance is entirely dependent on item memory rather than serial order).

If meter-syntax alignment affects serial-order, then level 2 meter-syntax alignments should affect task performance and not Level 1 alignment. That is, only level 2 meter corresponds to syntactic groups. Thus, to the extent the alignment of meter at this level can bias grouping or affect the robustness of serial-order within groups, this is expected to influence task performance.

A third possibility is that both item-specific and serial-order factors shape task performance. Given the prior literature, this seems most likely. However, given the relatively simplicity of the task, and the small effect sizes previously reported for modulations of item memory by metrical beats (e.g. Hickey et al., 2020), it is likely that any observed effect here will be small and perhaps only manifest in response times. Thus, an effect on response times is predicted for level 1 meter. A more robust effect of serial-order memory is thus predicted for level 2 metrical alignment on comprehension accuracy and response times (in line with the experiments so far in this thesis). This effect is predicted to be stronger in the sensitive probe conditions, as sensitive trials are more likely to result in error if the serial-order representation of the sequence is compromised.

Thus, meter-syntax congruency is predicted to affect task performance in both level 1 and level 2 alignments. Additionally, syntactic complexity and probe-sensitivity are predicted to show significant main effects on task accuracy and response-times. Finally, as with experiment 2, it is predicted that meter-syntax alignment will also affect sensorimotor synchronisation.

7.2.2. Results

Comprehension data were analysed using mixed-effects logistic regression. Separate models were used for the Level 1 and Level 2 meter conditions. Each model had fixed-effects for congruency (congruent, incongruent), structure (+++, ***, +**), metrical level (level 1, level 2), sensitivity (sensitive, insensitive), and trial. A random intercept was included for participants but a further intercept for equation did not significantly improve model fit ($p = 0.999$; likely because the majority of items were unique due to randomisation). Model fit was significantly improved by adding a structure by sensitivity interaction term ($p = 0.013$), but not by adding a further structure by congruency term ($p = 0.111$), as such the former model is adopted.

Accuracy Starting with the Level 1 meter results (table 10), the effect of congruency was not significant, with the rest of the terms in the model showing clear significant effects. In contrast, the Level 2 meter results (table 11) showed a significant effect of congruency. For both metrical alignments, there were also significant main effects of sensitivity and an interaction between sensitivity and structure.

Table 10: Estimate of fixed effects for accuracy model for Meter Level 1

	β	Std. Error	Z value	p value
(Intercept)	2.400	0.138	17.427	<.001 ***
Congruency (incongruent)	-0.129	0.067	-1.912	0.056 .
Structure (***)	-0.954	0.113	-8.451	<.001 ***
Structure (+++)	-1.317	0.108	-12.170	<.001 ***
Sensitivity (sensitive)	-0.598	0.100	-5.965	<.001 ***
Sensitivity (sensitive) x Structure (***)	0.469	0.151	3.112	0.001 **
Trial	0.002	<.001	4.884	<.001 ***

Table 11: Estimate of fixed effects for accuracy model for Meter Level 2

	β	Std. Error	Z value	p value
(Intercept)	2.396	0.139	17.207	<.001 ***

Congruency (incongruent)	-0.180	0.068	-2.639	0.008 **
Structure (***)	-0.800	0.115	-6.973	<.001 ***
Structure (+++)	-1.232	0.109	-11.340	<.001 ***
Sensitivity (sensitive)	-0.717	0.102	-7.056	<.001 ***
Sensitivity (sensitive) x Structure (***)	0.470	0.154	3.055	0.002 ***
Trial	0.002	<.001	4.892	<.001 ***

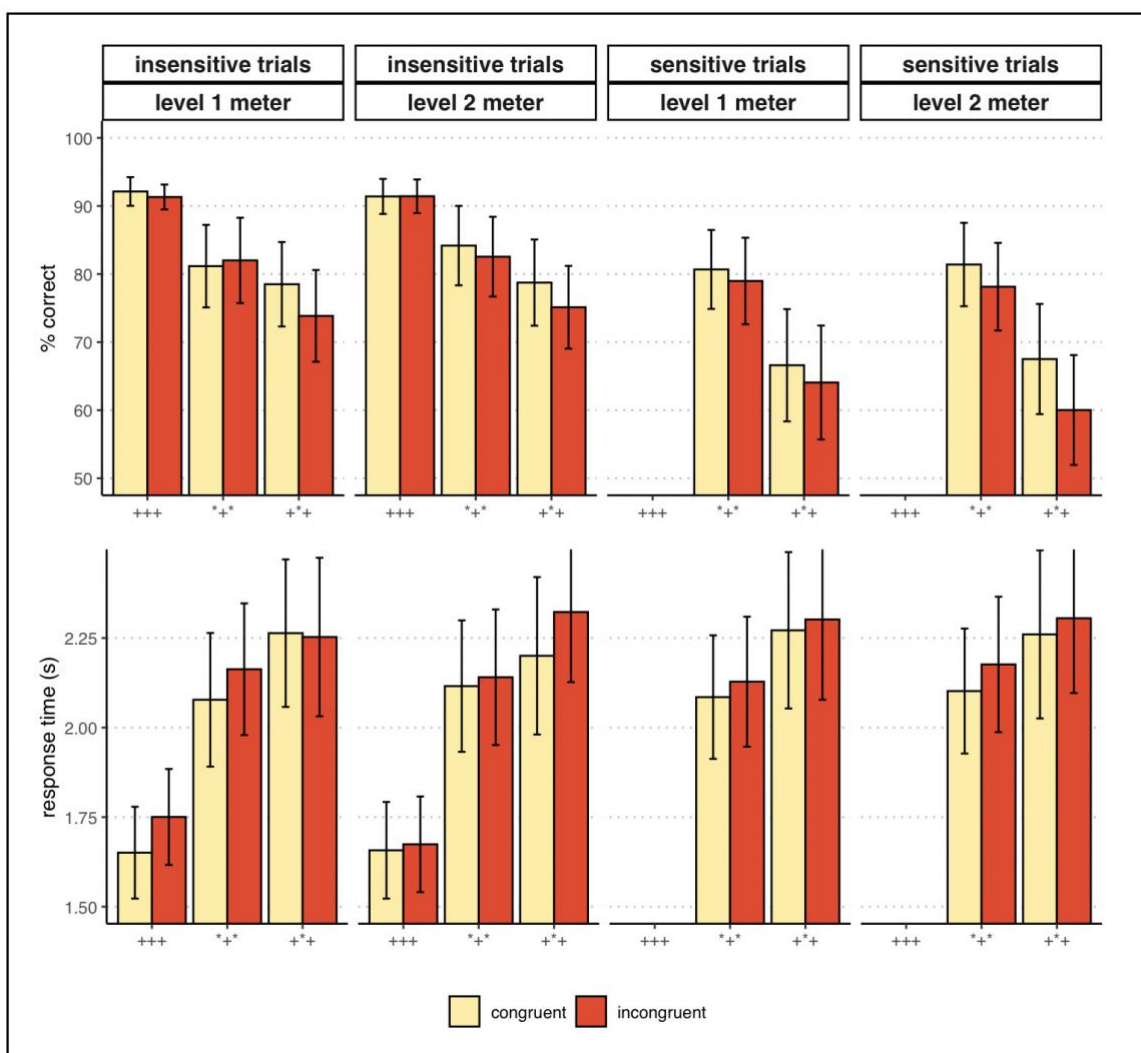


Figure 42: Experiment 4 results for accuracy (top) and response times (bottom). Errors bars represent SEM.

Response times

Model selection Response times (RTs) were analysed using linear mixed-effects regression. Neither of the interaction terms significantly improved model fit ($p = 0.669$; $p = 0.150$), nor did sensitivity ($p = 0.776$) so these were discarded. As before, the response time is defined by the

interval between when the probe is displayed on the screen to when the participant indicates their response by keypress.

Results Both level 1 (table 12) and 2 (table 13) meter metrical alignments showed significant main effects of congruency and for the other predictors.

Table 12: Estimate of fixed effects for response times model for Level 1 Meter (log-transformed units)

	β	Std. Error	t value	p value
(Intercept)	1.924	0.069	27.613	<.001 ***
Congruency (incongruent)	0.060	0.020	2.982	0.003 **
Structure (***)	0.402	0.025	16.303	<.001 ***
Structure (+++)	0.549	0.025	22.267	<.001 ***
Trial	-0.002	<.001	-14.089	<.001 ***

Table 13: Estimate of fixed effects for response times model for Level 2 Meter (log-transformed units)

	β	Std. Error	t value	p value
(Intercept)	1.878	0.069	27.349	<.001 ***
Congruency (incongruent)	0.046	0.020	2.301	0.021 *
Structure (***)	0.459	0.025	18.666	<.001 ***
Structure (+++)	0.587	0.025	23.833	<.001 ***
Trial	-0.002	<0.001	-13.490	<.001 ***

Catch trials Additionally, the performance on the ‘catch’ trials (figure 43) was analysed using a two-sided t-test. This was not statistically significant for level 1 meter ($t = 1.525$, $df = 2045.4$, $p = 0.127$) or level 2 meter ($t = 0.325$, $df = 2075.8$, $p\text{-value} = 0.745$).

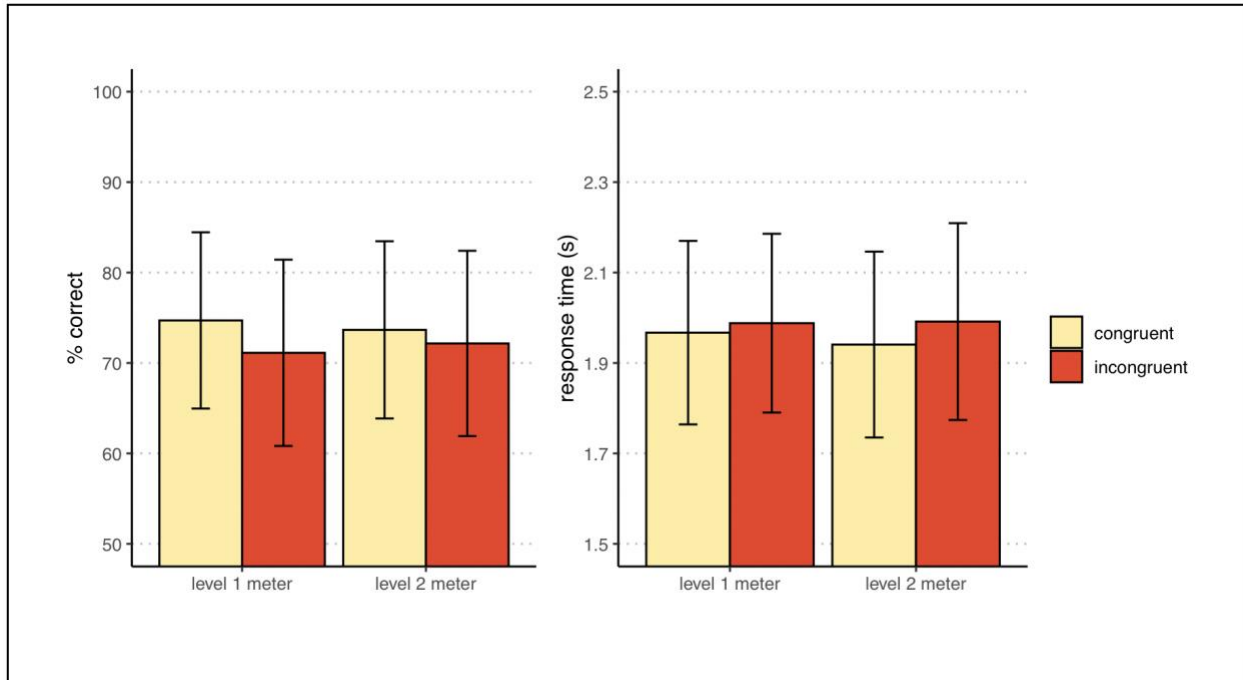


Figure 43: catch trials results from Experiment 4.

Tapping data Analysis of the tapping data (trial averaged; figure 44, 46) also revealed a significant effect of congruency on tapping asynchronies such that tapping during the Level 1 Meter trials tended to be earlier in the incongruent conditions ($t = 6.1307$, $df = 8118.8$, $p < 0.001$). This effect was not significant for the level 2 meter conditions ($t = 0.042$, $df = 8309$, $p = 0.967$).

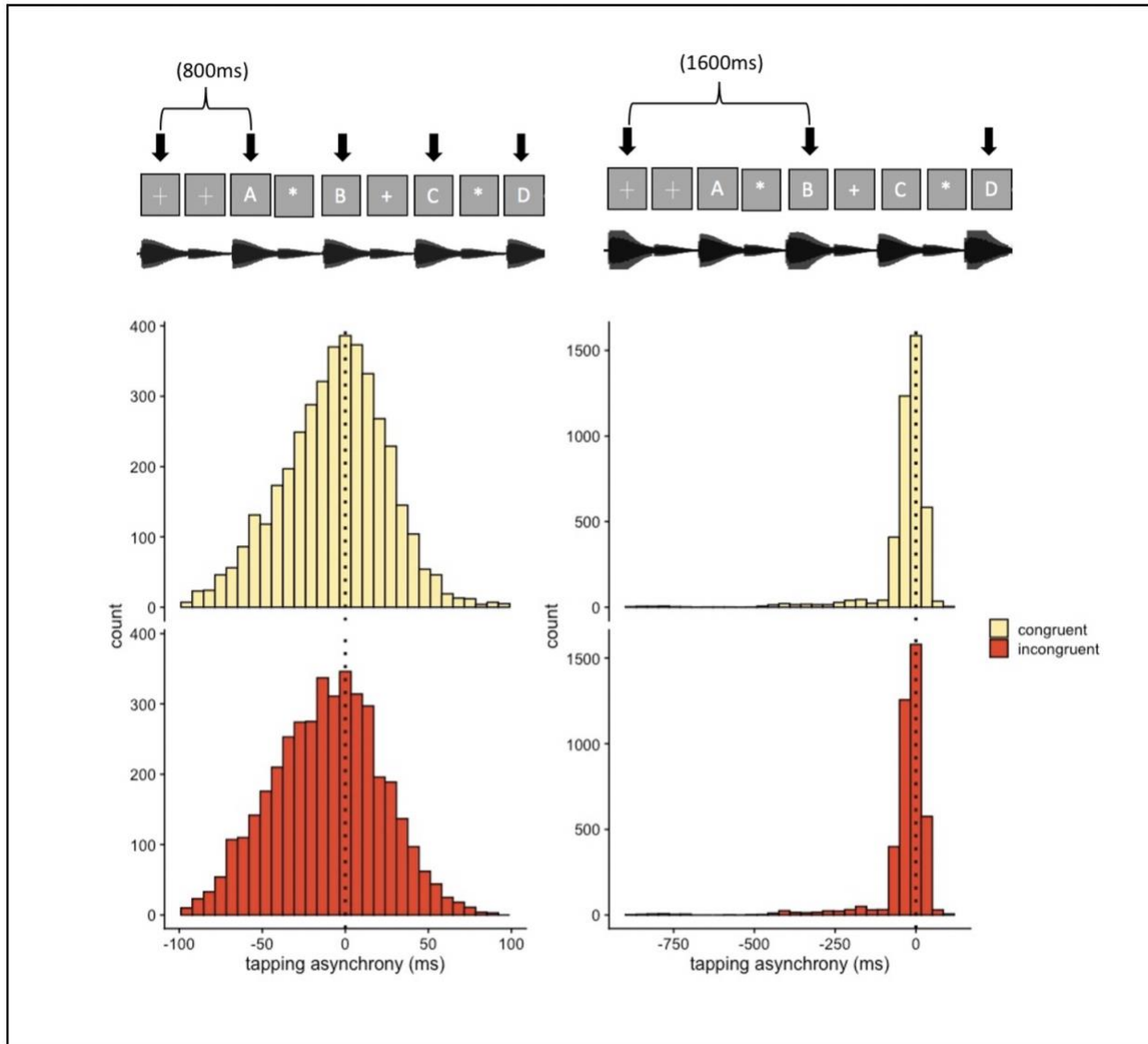


Figure 44: Mean asynchronies (trial averaged) for Experiment 4. The examples above each graph show the congruent alignments for both level 1 meter (left) and level 2 meter (right).

Unlike in experiment 2, however, there was no effect on the stability of finger tapping (mean standard deviation of asynchronies; figure 45, figure 46) for either level 1 meter ($t = 0.580$, $df = 7681.7$, $p = 0.562$) or level 2 meter ($t = 0.799$, $df = 7948.3$, $p = 0.424$).

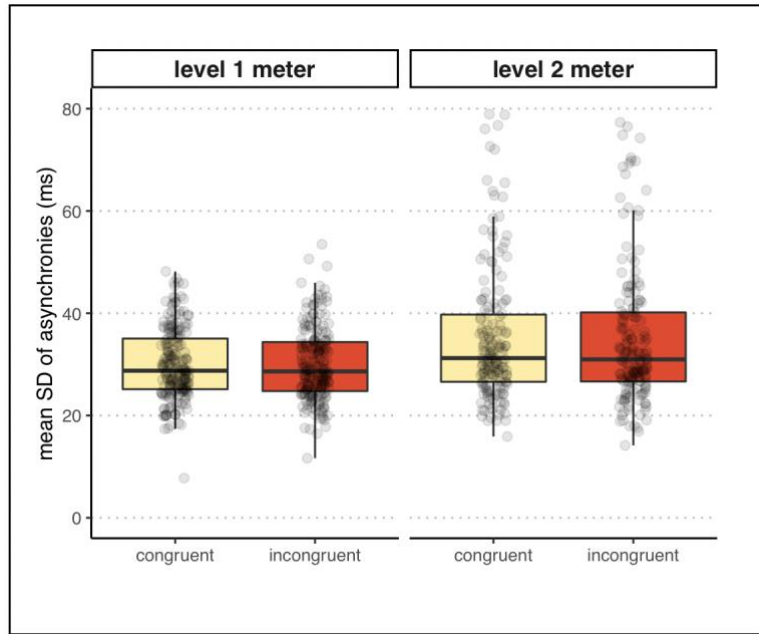


Figure 45: stability of sensorimotor synchronisation (mean SD of asynchronies) in Experiment 4.

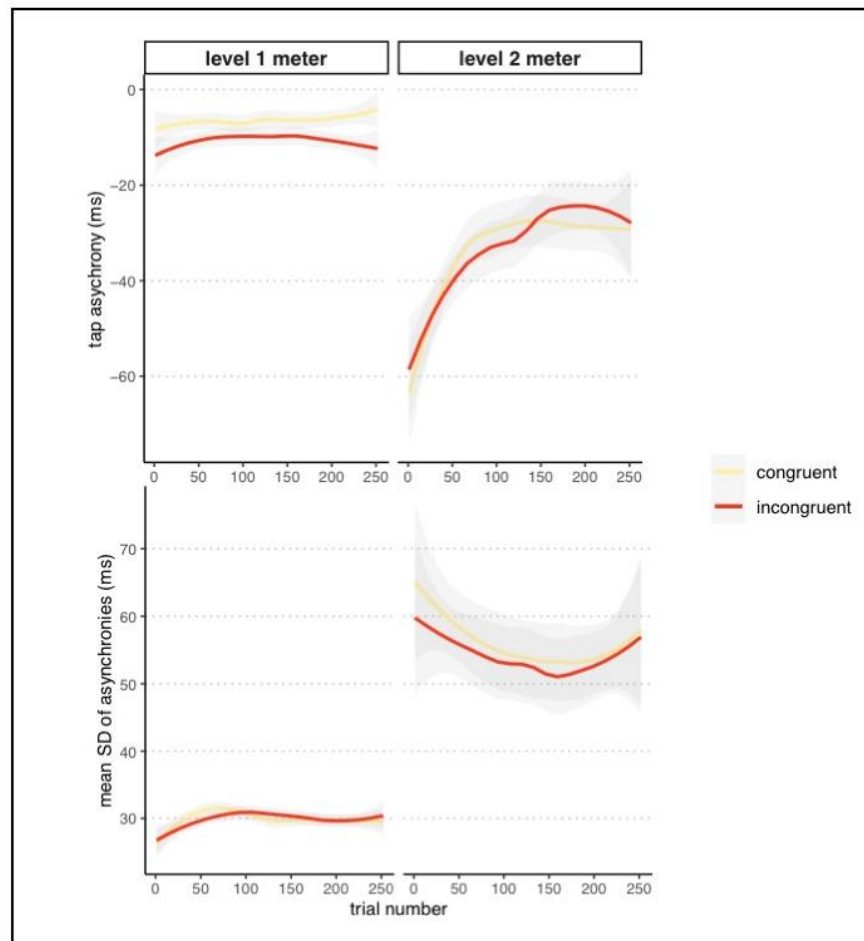


Figure 46: tap asynchronies (top) and stability (bottom) as a function of trials in the experiment.

Median split Due to the high accuracy and larger sample size, I conducted an additional exploratory median-split analysis (on task accuracy). This indeed turned out to be informative, showing that the congruency effect was appreciably stronger in the lower performing participants. This is perhaps clearest in figure 47 in which shows how the effect distributes over the course of the experiment (aggregating over all three equation structures). In particular, for the lower performing participants in the sensitive trials (where the congruency effect is predicted to be strongest), there is a distinct separation of congruent and incongruent performance for both accuracy and response times. Although, curiously, this interacts with trial position in an opposite pattern: larger effect for accuracy later in the experiment and earlier for the response times.

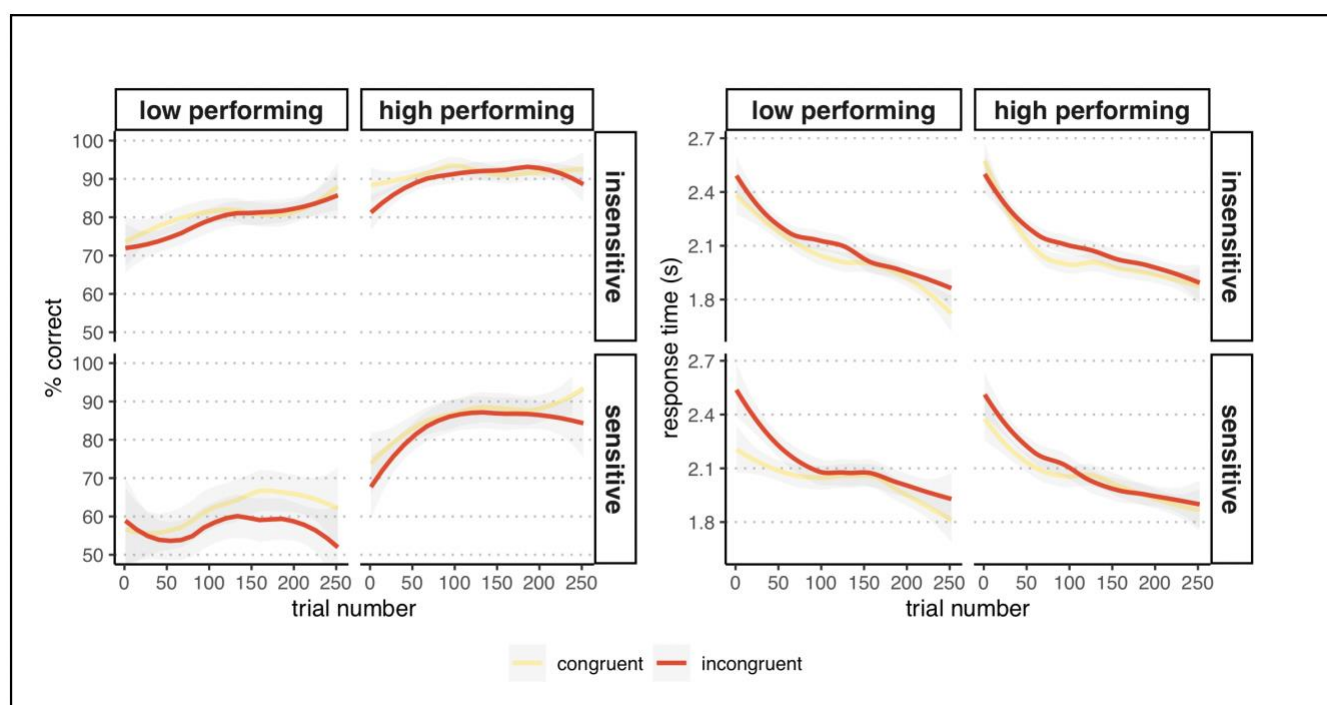


Figure 47: Task accuracy (left) and response times (right) as a function of trials in the experiment, each of which is further broken down by a median-split (high and low performing) and the sensitivity of the comprehension probe.

7.2.3. Discussion

This final experiment replicated the meter-syntax congruency in a non-linguistic domain (algebraic grouping), although with some interesting nuances. By applying it to a non-linguistic task it extended these findings by clarify the mediating influence of meter on serial-order short-term memory. Unlike these prior experiments, it also manipulated meter at two hierarchically distinct levels. This allowed the possible effects of meter on item memory and serial-order memory to be disentangled and the results supported a contribution from both aspects of memory to the effect on task performance. Although the effect of serial order appears to have a stronger influence.

It is informative to first consider the observed effect on the hierarchically flat '+++' algebraic structure. Task performance for this structure does not depend on serial-order memory (all

permutations of this structure are equivalent); one only needs item memory to not fail the 'catch trials'. Indeed, there was no difference in task accuracy for metrical congruency for either metrical type. But there was an effect on response times (for the level-1 meter condition). This makes sense under the assumption that level-1 metrical alignment affects individual item encoding and level-2 affecting serial-order encoding. Serial order is irrelevant for the '+++' structure, thus consistent with no effect on task accuracy. Differences in item encoding then make sense for slower response times since it might take longer to make the judgement of whether the given trial is a catch trial (whose determination relies only on individual item memory).

Consistent with this interpretation of the '+++' results, task accuracy for the hierarchically organised structures ('*+*', '+*+') was sensitive to meter-syntax congruency. This effect was primarily attested in the 'sensitive' conditions in which the probes were more demanding of a robust syntactic encoding of the stimulus, and there was little to no effect for the 'insensitive trials'. It was also strongest for the structure with non-local dependencies ('*++'). This suggests that whatever is causing the congruency effect, it is particular to process of encoding the structural dependency relationships among items, rather than just being a generic task or memory effect. Serial-order memory is a reasonable candidate for something that could mediate this, as this is particularly key for processing syntactic non-local dependency structures.

However, one caveat of this finding is that the size of this congruency effect is somewhat smaller than in studies 1 and 2. I offer two possible explanation for this. Firstly, the task in this experiment is different from that of the syntactic comprehension task in prior experiments. Determining the grouping of these algebraic terms is a consciously explicit process different from the more automated process of determining syntactically licenced linguistic meaning in sentence processing.

Secondly, and perhaps more importantly, the syntactic structure of the materials in this experiment are less complex than those of the language materials used previously. The most syntactically complex structure used (*++) presents the simplest possible manifestation of a non-local dependency. This is less complex than the simplest syntactic construction in the language experiments (the subject-extracted relative clause structure). Part of the need for this reduced complexity relates to the first point: that language processing is an overlearned task in which proficient speakers become experts over years of experience, and therefore many aspects of language comprehension are automated and operate beneath conscious awareness.

The smaller effect size observed in this experiment is then consistent with the reduced syntactic complexity of the stimuli and therefore the reduced demands upon short-term memory. From this perspective, the observed differences in the median-split analysis may then be a result of individual differences in short-term/working memory, such that those with lower capacities were more likely to show a congruency effect. However, unfortunately, I did not independently measure memory capacity in these participants, so this interpretation remains only speculative and should be investigated in future research.

While the response-time differences for the ‘+++’ structures support the contribution of level-1 metrical alignment to individual item encoding, the accuracy results just discussed for the ‘*+*’ and ‘+*+’ structures for the level-2 metrical alignment support the existence of serial-order memory effects. That is, it is hard to explain the effect of level-2 alignment in any other way than affecting serial order or implied grouping. It simply does not make sense that it could have been affected by item encoding. This is because each operand in the syntactic groups is equally informative to task performance, and level-2 metrical alignment just shifts which one of these operands coincides with the metrical accent. Both congruent and incongruent conditions have an equal number of strong and weak beat operands. Thus, metrical modulations of item memory are insufficient to explain the observed results.

The results also showed a significant effect of meter-syntax alignment on sensorimotor synchronisation. However, unlike with experiment 2, the results here were more limited. Specifically, the alignment of level-1 meter affected the pattern of asynchronies such that participants tended to over-anticipate the beat, tapping earlier in incongruent conditions than congruent conditions. This effect, however, was limited only to level-1 meter, with no effect for level-2 meter. There was also no meter-syntax alignment effect on the stability of this tapping (mean standard deviations of asynchronies), which was what was observed in experiment 2, but not in the present study.

There are at least two possible reasons for this discrepancy. Firstly, in experiment 2, participants tapped in time with auditory speech, whereas in this experiment, they tapped in time with amplitude modulated sine-tones (which were synchronised to visual presentation of the equations). These tones presented a clear and perfectly consistent amplitude envelope to synchronise with, whereas the amplitude envelope of speech is far more complex and variable (even despite various forms of normalisation applied in experiments 2 and 3). It is possible, therefore, that this complexity interacted with the metrical alignment in experiment 2 and resulted in less stable tapping.

A second potential factor is that in experiment 2 the speech that participants tapped along to also was the carrier of the task relevant content (language). In the present experiment, the content was shown on a screen visually such that it was synchronised with the auditory tones they were tapped along to. Further research will be required to understand how these factors interact with regard to the effect of meter-syntax alignment on sensorimotor synchronisation.

One interesting possibility for future research would be to use this task for participants who speak typologically distinct languages regarding meter-syntax alignment. That is, this algebraic task can be understood more-or-less cross-linguistically. This would allow the linguistic experience of participants to be compared in the same task. For example, comparing native English speakers against native speakers of languages like Japanese or Turkish, that are expected to have opposite nuclear-stress alignments to that in English (see: Iversen et al., 2008; Yoshida et al., 2010). While it would likely be the case that their level-1 metrical alignment results are similar, their level-2 alignment results would be predicted to show opposite patterns of interference from metrical alignment.

7.3. Conclusion

Building on Studies 1 and 2, this final study provided further constraints on the interpretation of the meter-syntax congruency effect. It was shown that the effect generalised to a working memory task involving non-linguistic stimuli. Due to the nature of these stimuli (lacking semantic content), a cleaner interpretation of the effect of meter on short-term memory was shown. Specifically, it appears that it affects both individual item encoding as well as serial-order memory when meter aligns to the level of larger syntactic sequences. This seems consistent with recent claims that short-term memory systems are largely shared between music and language and that this exhibits a close relation to rhythm.

8. General discussion

This thesis investigated the effect of meter-syntax alignment on sentence comprehension, sensorimotor synchronisation, and neural entrainment. It found sentence comprehension to be optimal when metrical accent aligns to the right node of syntactic phrases, consistent with the canonical nuclear-stress rule. Shifting the phrasal accent to other positions (incongruent alignment) disrupted both comprehension and sensorimotor coordination. Probing the underlying neural dynamics of this effect, it was shown that neural delta oscillations tracked the perceived meter rather than the syntactic structure. This is consistent with neural resonance theories of metrical attending and inconsistent with higher-order syntactic generators of delta that have been proposed recently. When only the syllable-rate oscillator entrains to the sentence (i.e. no meter), comprehension is worse than when meter is misaligned. This makes sense under the assumptions of the BUMP model of serial-order short-term memory (Hartley, Hurlstone, & Hitch, 2016). The final experiment aimed to more directly distinguish whether metrical alignment affects short-term memory for individual items or the serial-order in which they appear. It primarily found support for the latter—suggesting that modulations of serial-order memory are responsible for the meter-syntax congruency effect. This final chapter attempts to make sense of these results in a broader context, spelling out their theoretical and practical implications.

8.1. Theoretical implications

Chapter 2 posed the question of what language is and what language comprehension entails. Two answers to these questions were presented and contrasted. A key point of contention between them was the more technical question of whether language is best understood in terms of an idealised computational procedure for generating grammatical structures, à la Chomsky, or in terms of how we socially communicate given our various cognitive and environmental constraints and affordances.

The Chomskyan view downplays phonology and the sensorimotor system in language processing. This implies that metrical alignment at most plays an ancillary role relating to externalisation and communication, which are viewed as irrelevant to the ultimate design of language. Metrical alignment is therefore seen as an ad hoc accommodation to the realities of linguistic performance. Competence ('pure language') is seen as residing in a fully formed and encapsulated syntactic system which only interacts with performance systems through narrowly defined interfaces. These interactions are dictated by the action of syntax, and phonological and semantic systems simply follow its orders, passively translating syntactic structures into sound and meaning.

The contrasting view advocated for here rejects this syntactic hegemony, and relatedly, rejects the dissociation of language design from factors of social communication and cognitive constraints. Articulated in the Parallel Architecture (Jackendoff, 1997, 2002; Jackendoff &

Audring, 2020), an equal coalition of linked syntactic, phonological, and semantic systems constitute linguistic competence, and does so in a way that recognises a less stark divide between theoretical competence and practical performance. This architecture leaves room for metrical alignment to have a more important role. I will now briefly speculate on a few ways that this may be so and how the results of this thesis can inform this.

8.1.1 Perceptual representations and conscious awareness

If phonology is just for externalisation, as claimed by Chomsky, then it should only affect comprehension during communication (when language is externalised and internalised). It should not play a role in non-communicative contexts such as self-directed ‘internal speech’. This is because syntactic objects come first in this architecture and then phonetic objects are only afterward ‘spelled out’. This assertion is challenged by recent studies on auditory perceptual simulation of language (when people imagine distinct voices in their head when reading or otherwise; Zhou & Christianson, 2016a; 2016b; Zhou, Garnsey, & Christianson, 2019). These studies show that explicit auditory perceptual imagery affects processing depth and robustness: the more vivid the auditory imagery, the more robust the syntactic comprehension.

The authors speculate that auditory perceptual simulation achieves this by more richly activating syntax-aligned prosodic representations. These prosodic representations then scaffold syntactic comprehension making it more robust. The results of this thesis are consistent with this view: that the alignment of meter to syntax is not so much for communicating syntax through prosody (although it likely does this as well), as instead serving a more basic cognitive function in regulating short-term/working memory. This kind of story can only really make sense in something like the Parallel Architecture, which allows syntactic, phonological, and semantic representations to interact more-or-less freely without mandating that one should be derived from another in any particular order.

If this more cognitive view of prosody is correct, then it helps to explain why prosodic phrasing is so pervasive, even in non-ambiguous speech (Speer et al., 2011), or when silently reading (Breen, 2014; Zhou & Christianson, 2016). Indeed, syntactic reanalysis is thought to *necessitate* (implicit) prosodic reanalysis (Bader, 1998), further highlighting this tight relation. Frazier, Carlson, and Clifton relatedly conclude that “the prosodic representation of the sentence is the *essential skeleton* that holds different syllables together and indexes an item across representation types (phonological, syntactic, semantic), thereby permitting an utterance to be retained in memory while it is processed” (my emphasis; Frazier, Carlson, & Clifton, 2006).

From a complementary perspective, as discussed in chapter 2, the phonetic form of language may be thought of as a ‘conscious handle’ for manipulating linguistic information (Jackendoff, 1997, p. 192; 1987). As reviewed by Stanislas Dehaene (Dehaene, 2014), the ability to hold and manipulate information in the conscious workspace has a considerable amplifying effect on processing depth and interaction with memory systems. Consistent with this, word recognition and semantic priming can take place subliminally, but deeper combinatorial syntactic integration

requires the amplifying powers of conscious awareness (Rabagliati et al., 2018; van Gaal et al., 2014).

Putting these pieces together, clothing language in perceptually grounded phonetic forms may be necessary to systematically take advantage of the amplifying powers of the conscious workspace. Prosodic structure (groups and grids) may then be a specific way of structuring this perceptual representation that further provides cognitive constraints and affordances that support deeper syntactic comprehension (above and beyond the utility of prosodic phrasing for facilitating communication). This thesis specifically provided evidence for how meter coordinates our memory representation of linguistic sequences (among other functions).

8.1.2 Meter-syntax alignment as embodied linguistic skill

The perspective just argued highlights the importance of perceptuomotor processes and representations in the cognition of language. I will now further speculate on how this might relate to language learning and development from an embodied skill acquisition perspective. Embodied cognition describes how seemingly abstract cognitive tasks like *symbolic reasoning* can be implemented or supported by perceptual-motor processes and representations (Goldstone & Barsalou, 1998; Goldstone et al., 2015; Landy et al., 2014; Wilson, 2002; Clark, 2015). These embodied factors are implicated in the development of expertise in many domains (Kellman & Garrigan, 2009; Kellman & Massey, 2013; Landy, 2018; Marghetis et al., 2016). Goldstone, Landy, and Son (2010) describe this in terms of developing a ‘Rigged-Up Perception-Action Systems’ (RUPAS): converting effortful cognitive processes into learned and automatic perception and action routines.

Mathematics is typically thought of as an abstract higher-order domain. But consistent with the ‘rigging-up’ view, mathematical expertise repurposes systems for visual object attention to parse the hierarchical structure of algebraic expressions (e.g. like those in chapter 7; Landy & Goldstone, 2007; Marghetis et al., 2016; also see Kellman & Massey, 2013). Indeed, this is also reflected in how the brain recycles perceptuo-motor circuits for mathematical processing (Maruheyama et al., 2012; Dehaene & Cohen, 2007). Chess is another domain classically thought to be the pinnacle of human higher-order abstract reasoning. But here too, expertise rigs-up the visual system to do cognitive heavy-lifting (Chase & Simon, 1973; De Groot, 1965; for similar neural result see also: Bilalić, 2016). Taken together, this highlights that while novices effortfully think through solutions, the progression to expertise allows one to simply *perceive* them. De Groot remarks:

We know that increasing experience and knowledge in a specific field (chess, for instance) has the effect that things (properties, etc.) which, at earlier stages, had to be abstracted, or even inferred are apt to be immediately perceived at later stages. To a rather large extent, abstraction is replaced by perception... (pp. 33–34)

How does this relate to language? Firstly, it is worth noting that the general motivation behind the Rigged-Up Perception-Action System theory parallels those already proposed for linguistic theories such as ‘Good Enough’ processing (Ferreira et al., 2002; Ferreira & Patson, 2007;

Christianson, 2016) and other theories that explain aspects of language processing in terms of cognitive efficiency and parsimony (Christiansen & Chater, 2008, 2015). These approaches in turn align with more general tendencies in biological systems; what Daniel Dennett calls ‘Good Tricks’ (Dennett, 1995); what Herbert Simon described in the domain of decision-making as ‘satisficing’ (Simon, 1956). In other words, cognitive systems efficiently solve problems with whatever available resources, whether they be amodal computational resources or more grounded sensorimotor ones (Clark, 2015).

The embodied skill acquisition framework also fits with recent calls to reconceptualise language acquisition and development as a form of skill acquisition (Christiansen & Chater, 2018). Traditionally, language acquisition is framed as a kind of inference problem where the child has an innate structural device specified by Universal Grammar but must infer the parameters of that device from their linguistic experiences. In other words, language learners are seen as ‘mini linguists’. The alternative language-as-skill framework, advocated for by Christiansen and Chater argues that learning to understand and produce language is practical, not theoretical, and that it is in principle no different from other forms of skill acquisition like learning a musical instrument or learning how to ride a bike.

Putting all of this together, prosody-syntax correspondence can be reconceptualised in terms of how perceptuo-motor processes become adapted to fit the needs of linguistic skill. This view contrasts with viewing prosody-syntax correspondence as an innate part of linguistic competence (and perhaps genetically encoded as part of Universal Grammar), or just a trivial accommodation for externalisation (e.g. Adger, 2003). It also goes against the idea that its underlying neural substrate as delta oscillations is an obligatory response arising from compositional/syntactic computations (e.g. Martin & Doumas, 2017; Meyer, Sun, & Martin, 2019). Instead, in the view I am advocating for here, prosody-syntax correspondence is the result of a ‘rigging-up’ of the perception-action system to make syntactic comprehension more effective and robust by aligning the linguistic task to the oscillatory constraints and affordances of metrical rhythm.

This perspective means that metrical alignment is to some extent arbitrary. Different metrical alignments do not catastrophically interfere with comprehension, as is clear in how language can be metrically manipulated in song yet still understood (e.g. Temperley, 1999, 2019). The nuclear-stress rule may simply reflect an optimal alignment for processing syntactic phrases. I emphasise, however, that this is not only optimal for externalisation (as Chomsky might accept) but optimal for the internal cognitive processing of syntactic constructions in general.

It is also important to emphasise that this alignment is not a trivial given, but something dynamically negotiated. Lower levels of the metrical hierarchy are tethered tightly to the properties of the acoustic signal. Indeed, neural theta rhythms continue to track the acoustic amplitude modulations of syllables regardless of conscious awareness (Gui et al., 2020; Makov et al., 2017). By contrast, metrical periodicities at the delta oscillation level are subject to conscious top-down attention, or top-down signalling from other less conscious aspects of processing (e.g. our acquired metrical reflexes that we do not need to think about). This

flexibility allows meter to malleably adapt to abstract structure in signals like music and language without having to be accompanied by a clear physical correlate.

Becoming an expert language user, in this view, entails training perception and action routines to automatically align delta-level meter in an optimal way for syntactic comprehension. This is analogous to how a chess master trains their perceptual system to automatically parse abstract relational patterns from a chess board, which can then be more deliberately reasoned about (De Groot, 1965). In both cases, freeing up cognitive resources for other aspects of processing that cannot be so easily automated.

Evidence can be found for this in how linguistic experience with typologically distinct languages affects general auditory processing. The nuclear-stress rule aligns metrical prominence to the rightmost phrasal node in languages like English, and to the leftmost phrasal node in languages like Japanese. This difference is argued to arise from differences in syntactic structure between these two languages (Cinque, 1993). However, this cultural difference not only affects their linguistic preferences but also more general auditory processing, as evidenced by studies showing that English and Japanese speakers tend to perceive different metrical alignments from acoustically identical rhythms in a way that is consistent with their realisation of the nuclear-stress rule (Iversen et al., 2008; Yoshida et al., 2010). This indicates that meter-syntax alignment is not just specific to language, but rather an adaption of the auditory system to the demands of language processing, consistent with the ‘rigging-up’ view.

People also rhythmically gesture in time with speech in ways that reinforce phrasal prosody (so-called “beat gesture”; McNeil, 1992), and this is often done automatically without awareness. Such gestures are known to influence neural entrainment to speech rhythms (Biau et al., 2015). And more generally, beat gesture is known to work synergistically with acoustically signalled prosodic prominence in speech (Kushch et al., 2018; Morett & Fraundorf, 2019). Beat gesture may therefore also reflect a rigging-up of automatic action-routines to support and supplement perceptual processes in comprehension. This is likely to be mediated by the neural mechanisms involved in *active sensing* discussed in chapter 5, and more generally how the motor system interfaces and coordinates with the perceptual system. This might also help to explain why metrical misalignment affected sensorimotor synchronisation, as shown in experiments 2 and 4.

8.1.3 An evolutionary perspective

This embodied perspective also fits into questions of language evolution. In *The Descent of Man* (Darwin, 1871), Charles Darwin suggested that some early analogue of song was once the primary communicative medium of our hominin ancestors, from which our modern language capacity gradually grew: the so-called musical protolanguage hypothesis. Many modern instantiations of this and related ideas continue to be proposed and debated today, in addition to other related precursor systems and evolutionary trajectories (Arbib, 2012; Brown, 2000, 2017; Fitch, 2010, 2017; Mehr et al., 2020; Mehr & Krasnow, 2017; Mithen, 2006; Tomlinson, 2016).

Of particular relevance to the present discussion is the idea that a precursor stage resembled something like speech prosody (Brown, 2017; Fitch, 2010) embedded within a broader system

of generative phonology, whose combinatorial (Lipkind et al., 2013) and hierarchical (Sainburg et al., 2019) properties seem to be shared, at least to some extent, with other complex vocal learners (Filippi et al., 2019).

Within our own primate lineage, combinatorial *syllable structure* is thought to have evolved out of nonvocal facial displays such as rhythmic lip-smacking (Ghazanfar et al., 2013; MacNeilage, 1998; Poeppel & Assaneo, 2020). This is then proposed to have been combined with a capacity for hierarchical rhythmic structure, likely mediated by expansion in the brain's dorsal auditory pathway (Rilling et al., 2008; Scott et al., 2009; van der Lely & Pinker, 2014), whose hierarchical aspects may have been exapted from hierarchical action/motor control and planning (Asano & Boeckx, 2015; Fitch, 2019). Together, this bare phonological hierarchy is thought to have paved the way for a more abstract syntactic phrasal hierarchy (Fitch, 2019; see also Brown, 2017; Jackendoff, 2002). In other words, generic circuits involved in action-planning may have been recycled and repurposed for language, not only at the developmental timescale, where they are rigged-up through practise to make processing more effective (Dehaene & Cohen, 2007), but also at the evolutionary timescale (Anderson, 2010), perhaps giving rise to the cultural development of syntactic constructions in the first place and also implicating the evolution of music (Hilton et al., 2021).

This phonology first, syntax last (or later) hypothesis (as advocated in: Fitch, 2019; 2010; Jackendoff, 1997; 2002; Culicover & Jackendoff, 2005; Everett, 2017) contrasts with the assertions of mainstream Chomskyan theory (Chomsky, 1995; Chomsky & Berwick, 2016; Tallerman, 2013), however, it seems to increasingly prove to be a more coherent story to tell of the available data. Especially, it seems to make better sense of the deep parallels between music and language concerning shared rhythmic structure (Lerdahl & Jackendoff, 1983; Patel, 2008; Lerdahl, 2013; Heffner & Slevc, 2015; Hawkins, 2014; Beier & Ferreira, 2018).

8.2. Practical implications

What does understanding meter-syntax alignment through the lens of embodied skill acquisition get us? I argue here that it brings new perspectives on how to think about language development (both typical and atypical) and language learning more generally, as well as music-to-language transfer effects.

A classic concept in language acquisition research is 'prosodic bootstrapping'. As discussed in chapter 3, this describes how infants use acoustic prosodic cues to bootstrap various aspects of the language acquisition process, ranging from determining word/clause boundaries (Hawthorne & Gerken, 2014; Hirsh-Pasek et al., 1987; Jusczyk et al., 1992; Männel & Friederici, 2009) to determining the ordering of words within syntactic constituents (Bernard & Gervain, 2012; Gervain et al., 2008; Gervain & Werker, 2013; Toro et al., 2016). Indeed, while there has been substantial theoretical and empirical interest in how infants use probabilistic information to learn language structure (Marcus et al., 1999; Morgan & Newport, 1981; Sanders et al., 2002; Thompson & Newport, 2007), prosodic cues have an even stronger effect that can override

learned statistical contingencies (Johnson & Jusczyk, 2001). Clearly prosody is important in early language acquisition.

Prosodic bootstrapping is typically viewed as underlying a concrete-to-abstract shift in language development. That is, children use prosody as a temporary scaffold to acquire the more abstract and formal rules of language and then once they have acquired this more abstract knowledge, they use these formal rules instead to guide their adult competency: This is implicit in the notion that this stage is about ‘setting parameters’ of a pre-existing grammatical competence. Indeed, similar concrete-to-abstract shifts have been described in a number of domains such as mathematics and science, and the idea has roots in Piaget’s influential work on child development (Piaget, 1952).

However, a number of recent studies have challenged this view, some proposing the opposite progression: an increasingly dependence on concrete representations (Simons & Keil, 1995; Varma & Schwartz, 2011). For instance, Braithwaite and colleagues show that in acquiring arithmetic competence, children come to increasingly rely upon embodied concrete strategies for making sense of algebraic syntax (Braithwaite et al., 2016; see also chapter 7). Engaging with this educationally, Ottmar and Landy show that the instructional sequence of ‘concreteness fading’ to be the most effective. That is, starting off by explicitly scaffolding perceptual and embodied strategies and progressively fading these supports away.

An obvious parallel obtains here between concreteness fading in math learning and prosodic bootstrapping in language. However, the key distinction in the more recent theory is that rather than just temporarily using concrete representations to learn abstract rules, it is thought that perceptual scaffolding is used to ‘rig-up’ the perception and action systems to do this processing more automatically and flexibly without having to rely upon explicit cues. Indeed, as discussed in chapter 3, linguistically competent adults readily perceive nuclear-stress in syntactic phrases regardless of whether acoustic stress is present (Cole et al., 2019), and the same is true for the perception of prosodic grouping (Buxó-Lugo & Watson, 2016). By contrast, infants require explicit acoustic cues to make such distinctions (Hawthorne & Gerken, 2014; Männel & Friederici, 2009).

Becoming an expert language user may therefore be a bit like learning to hallucinate prosodic structures where syntactic structures make them likely, even in the absence of confirming acoustic cues. Such changes to higher-level perception are common to many domains of expertise (Landy, 2018). Language, then, may be no different.

This view highlights the importance of active perceptual processing rather than the more traditional emphasis on abstract rules. The practical significance of this is then that language acquisition (and second-language acquisition more broadly) may benefit from instructional techniques developed to target perceptual learning, including those made possible by computer interfaces (see discussion in: Kellman & Garrigan, 2009; Kellman & Massey, 2013).

This may be especially helpful for those with atypical language development. For example, developmental dyslexia is thought to relate to atypical neural entrainment to speech and metrical rhythms (Goswami, 2011; Colling et al., 2017; Fiveash et al., 2020; Huss et al., 2011), and part of this may stem from low-level perceptual deficits in tracking amplitude envelope rise time characteristics in speech signals (reviewed: Goswami, 2015). Although the discussion of meter-syntax alignment in the previous section implied a more active top-down perceptual alignment process, if children have difficulties in basic bottom-up induction of speech stress, they may fail to take advantage of early prosodic bootstrapping that typically developing children use to ‘rig-up’ their perception-action systems for syntactic comprehension. Indeed, Richards and Goswami (2019) show that children with developmental language disorders are specifically impaired in detecting violations to meter-syntax alignment.

If rhythmic structures are as important as I claim to the cognition of language, then there are also implications for how musical abilities and engagement might have transfer effects to language abilities. There has been broad interest in musical transfer effects for some time, partly arising from media-hype around the debunked ‘Mozart effect’ —which purports that listening to classical music before an academic test improves performance and spatial reasoning abilities. Rather than something specific to music, the (small) effect is driven by generic arousal and mood enhancements (Thompson, Schellenberg, & Husain, 2001; Husain, Thompson, & Schellenberg, 2002). More recently, other claims have been made around whether musical engagement might enhance IQ (Schellenberg, 2004), mathematical abilities (critical perspective: Haimson et al., 2011), and specific aspects of executive function (namely ‘updating’ as measured with n-back tasks; Slevc et al., 2016). However, a recent multilevel meta-analysis questions some of these more general effects (Sala & Gobet, 2019).

A more promising connection, however, seems to hold for music-to-language transfer effects. Part of what makes this connection more plausible is the fact that music and language have a number of well-established cognitive and neural parallels, ranging from a reliance on fine-grained auditory processing to higher-level parallels relating to hierarchical rhythmic structure and long-range structural integration. If music and language have overlapping neural implementations, then strengthening these shared networks should be of mutual benefit. As noted by Ani Patel, music also has a number of other properties that make it especially likely to drive improvements in language (and less likely than the other way around; although language experience can also shape perception, e.g. Kyle, Sun, & Tierney, 2020). Namely, music generally requires a higher degree of *precision* in pitch and rhythm, music tends to be more *emotionally* engaging, *repetitive*, and conducive of focused *attention* (Patel, 2011, 2012).

With these conditions in place, it appears that both pitch and rhythmic aspects of music can enhance prosodic processing in language. For pitch, musical experience is associated with enhanced low-level pitch processing as reflected in subcortical circuitry previously assumed to be impervious to experience dependent plasticity (Kraus & Chandrasekaran, 2010; Wong et al., 2007). Highlighting the practical significance of this, music lessons are shown to enhance the ability to decode emotional prosody in speech (which heavily relies upon pitch cues; Thompson, Schellenberg, Hussain, 2004). Indeed, randomised controlled trials involving musical training

show enhanced processing of fine-scale speech information after as little as one year of training (Kraus et al., 2014).

More specifically relevant to this thesis, however, is the association between rhythm abilities and both phonological (Hausen et al., 2013; Tierney et al., 2017; Tierney & Kraus, 2014; Woodruff-Carr et al., 2014; 2017; Ozernov-Palchik & Patel, 2018) and grammatical abilities in language (Gordon et al., 2015a; 2015b; Bonacina et al., 2018). The effect on grammatical abilities is also consistent with more general associations between musical experience and enhanced syntactic processing in language (Jentschke & Koelsch, 2009; Brod & Opitz, 2012; Patel & Morgan, 2016).

From traditional perspectives this observation is perplexing—why should rhythm affect something as abstract as syntax? However, this result makes a great deal of sense under the framework advocated for here where rhythmic structure functions as a cognitive support. Musical training may therefore benefit syntactic comprehension in two ways. It may help to refine bottom-up perceptual processing (Kraus & Chandrasekaran, 2010; Wong et al., 2007; Tierney & Kraus, 2014; Tierney et al., 2015; Woodruff Carr et al., 2014), thus helping people to more finely discriminate acoustically signalled prosodic information (and therefore benefit from acoustically signalled prosody-syntax alignment). More speculatively, musical training may also help to refine the ability to modulate perception from the top-down such as to align hierarchical rhythmic structures with other more abstract tonal or linguistic hierarchies. Patel and Morgan make some speculations along these lines, although with less of a focus on hierarchical rhythmic structures and more on general long-range syntactic integration (Patel & Morgan, 2016).

A question of practical importance is then to ask how best to support and implement such music-to-language transfer effects. Musical activity is highly varied, and it would seem to be problematic to assume that all musical engagement would have the same effects. Anecdotally speaking from my own experience of many years in elite musical training institutions, it is easier than one may think to get a formal degree in musical performance and still neglect basic aspects of rhythm perception (more so in certain classical music traditions). An important concept in future research, therefore, will be to explicitly describe and monitor ‘treatment fidelity’ (Wiens & Gordon, 2018). In other words, it is of limited value to merely note that some group engaged in some generic musical activity or had music lessons. To yield maximally informative results, future research must determine what specific sorts of musical activities yield benefit, and to measure the extent to which engagement in these activities were followed and to what degree.

8.3 Conclusion

I conclude this thesis by returning to how my former guitar teacher Tim Kain taught me to be mindful of rhythmic structure. I struggled with rhythm for a long time as a musician—despite having played music regularly since I was about 13 years old. But Tim patiently and repeatedly

redirected my attention to rhythmic structure and its alignment to other structures and helped me think about it in more direct and tangible ways by providing embodied metaphors such as mapping 'strong' and 'weak' metrical beats onto 'down' and 'up' gestures. This was a key insight that transformed my musicianship deeply.

The thread running through the course of this PhD journey has been the hunch that this insight might also shed light onto how we process language: What if, in becoming expert language users, we also must learn to control our mental rhythms in relation to our linguistic task? I have provided some suggestive evidence in support of this presumption and rambled about various theoretical framings thereof. Minimally, I have shown that meter-syntax alignment affects our ability to comprehend sentences and to coordinate our actions with syntactically organised rhythmic sequences (such as speech). I also showed the relation of this to the dynamics of how our brains resonate to metrical percepts. Further investigation remains for future research, and I eagerly look forward to finding out which bits (presumably, all of it) that I got wrong.

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Appendix A

Below are the sentences used in experiments 1,2, and 3. Experiment 1 only used the first 48 of these, experiment 2 the first 72, and experiment 3 used all 112 (it is indicated below where these cut-offs are).

Subject extracted sentences

The boy that helped the girl got_an A on_the test
The clerk that liked the boss had_a desk by_the window
The guest that kissed the host brought_a cake to_the party
The priest that thanked the nun left_the church in_a hurry
The thief that saw the guard had_a gun in_his holster
The crook that warned the thief fled_the town the_next morning
The knight that helped the king sent_a gift from_his castle
The cop that met the spy wrote_a book about_the case
The nurse that blamed the coach checked_the file of_the gymnast
The count that knew the queen owned_a castle by_the lake
The scout that punched the coach had_a fight with_a manager
The cat that fought the dog licked_its wounds in_the corner
The whale that bit the shark won_the fight in_the end
The maid that loved the chef quit_the job at_the house
The bum that scared the cop crossed_the street at_the light
The man that phoned the nurse left_his pills at_the office
The priest that paid the cook signed_the check at_the bank
The dean that heard the guard made_a call about_the matter
The friend that teased the bride told_a joke about_the past
The fox that chased the wolf hurt_its paws on_the way
The groom that charmed the aunt raised_a toast to_the parents
The nun that blessed the monk lit_a candle on_the table

The guy that thanked the judge left_the room with_a smile
The king that pleased the guest poured_the wine from_the jug
The girl that pushed the nerd broke_the vase with_the flowers
The owl that scared the bat made_a loop in_the air
The car that pulled the truck had_a scratch on_the door
The rod that bent the pipe had_a hole in_the middle
The mule that liked the duck left_the farm the_next day
The niece that kissed the aunt sang_a song for_the guests
The boat that chased the yacht made_a turn at_the boathouse
The desk that scratched the bed was_too old to_be moved
The cook that hugged the maid had_a teer on_her cheek
The boss that mocked the clerk had_a crush on_the intern
The fruit that squashed the cake made_a mess in_the bag
The dean that called the boy had_a voice full_of anger
The thug that saw the man had_a hat on_his head
The truck that bumped the car had_a man at_the wheel
The prince that helped the girl had_a head of_blonde hair
The bug that attacked the ant was_as big as_a coin
The shelf that bumped the piano had_some dust on_its top
The tradie that thanked the man left_his wallet in_the car
The chief that thanked the girl was_too late for_the bus
The sailor that pushed the banker hurt_his foot in_the brawl
The bloke that helped the mum gave_a gift to_say thanks
The dad that knew the chef had_a son with_a limp
The dancer that liked the guy was_a chef in_the city
The kid that followed the dad had_some mud on_his shoes # **Experiment 1 stops here**
The monk that liked the wife put_a stone in_the basket
The vet that helped the Dad left_the house in_a hurry
The lion that saw the goat drank_some water from_the stream
The creep that noticed the clown had_a scratch on_his chin
The judge that thanked the poet put_a note on_the desk
The cop that thanked the snitch hurt_her hand in_the fight
The fraud that helped the duke had_a sip from_his glass
The bro that heard the brat broke_the vase on_the shelf
The men that blamed the crew had_some lunch on_the job
The jock that phoned the kid wore_a watch that_was new
The wolf that chased the bear had_some blood on_its fur
The elf that fought the dwarf left_the room full_of anger
The punk that liked the boss had_a son in_the school
The mayor that followed the crowd was_not happy with_the result
The niece that liked the date lit_the candle on_the table
The guard that mocked the chef had_a scar on_her ankle
The puppy that nudged the goat dug_a hole in_the grass
The guy that helped the bard had_a date on_that night

The gool that loved the goth told_a joke to_their friends
The spouse that thanked the teen was_too late for_the meeting
The geek that blamed the crook closed_the door with_the switch
The chap that saw the gent had_a rose in_his jacket
The lady that helped the baker had_a head of_brown hair
The beast that heard the witch had_a plan that_was evil # **Experiment 2 stops here**
The mug that chipped the plate was_too old for_the shop
The coin that scratched the knife had_a price that_was high
The coach that called the kid had_a voice that_was strange
The clown that knew the geek had_a party on_that day
The poet that blamed the maid wrote_a letter to_the mayor
The girl that pushed the boy had_a grudge that_was strong
The wife that knew the cop poured_a glass of_cold water
The bride that loved the man made_a note of_what happened
The duke that warned the Queen fought_the man in_the castle
The host that pleased the bloke had_a friend that_was near
The liar that knew the scout left_the house in_a fury
The vet that saw the liar wore_a hat on_her head
The bard that mocked the guard told_a lie to_the boss
The jock that thanked the kid had_a smile on_his face
The goat that bit the mule was_not happy with_the food
The man that punched the thug had_a grimace on_his face
The elf that fought the orc made_a cry that_was loud
The priest that called the judge made_a promise on_that night
The witch that liked the punk poured_a glass of_red wine
The goth that knew the poet left_the church in_a hurry
The duke that loved the bard left_a note on_the stool
The cop that saw the creep dropped_a book by_the lake
The Queen that punched the Prince knew_the truth of_the matter
The vase that cracked the jar was_then sold at_the market
The team that helped the crew had_some lunch in_the sun
The host that liked the team was_then told of_the result
The Dad that saw the uncle waved_his hand to_say hi
The aunt that knew the man made_a cake to_say thanks
The Dad that loved the niece hid_the ball in_the vase
The niece that helped the bloke had_a head of_blonde hair
The son that mocked the guy crossed_the street near_the tree
The child that pushed the friend told_a lie to_the man
The chef that saw the child had_a smile on_her face
The chair that knocked the desk had_a price that_was high
The beast that scared the knight hit_the stone with_a fist
The wife that phoned the boss knew_the story was_too long
The host that warned the friend had_a date on_that night
The gent that loved the lady told_a story about_the trip

The mate that blamed the tradie had_a trip the_next day
The man that pushed the mate was_too drunk to_fight back

Object extracted sentences

The boy that_the girl helped got an A on the test
The clerk that_the boss liked had a desk by the window
The guest that_the host kissed brought a cake to the party
The priest that_the nun thanked left the church in a hurry
The thief that_the guard saw had a gun in his holster
The crook that_the thief warned fled the town the next morning
The knight that_the king helped sent a gift from his castle
The cop that_the spy met wrote a book about the case
The nurse that_the coach blamed checked the file of the gymnast
The count that_the queen knew owned a castle by the lake
The scout that_the coach punched had a fight with a manager
The cat that_the dog fought licked its wounds in the corner
The whale that_the shark bit won the fight in the end
The maid that_the chef loved quit the job at the house
The bum that_the cop scared crossed the street at the light
The man that_the nurse phoned left his pills at the office
The priest that_the cook paid signed the check at the bank
The dean that_the guard heard made a call about the matter
The friend that_the bride teased told a joke about the past
The fox that_the wolf chased hurt its paws on the way
The groom that_the aunt charmed raised a toast to the parents
The nun that_the monk blessed lit a candle on the table
The guy that_the judge thanked left the room with a smile
The king that_the guest pleased poured the wine from the jug
The girl that_the nerd thanked broke the vase with the flowers
The owl that_the bat scared made a loop in the air
The car that_the truck pulled had a scratch on the door
The rod that_the pipe bent had a hole in the middle
The mule that_the duck liked left the farm the next day
The niece that_the aunt kissed sang a song for the guests
The boat that_the yacht chased made a turn at the boathouse
The desk that_the bed scratched was too old to be moved
The cook that_the maid hugged had a tear on her cheek
The boss that_the clerk mocked had a crush on the intern
The fruit that_the cake squashed made a mess in the bag
The dean that_the boy called had a voice full of anger
The thug that_the man saw had a hat on his head
The truck that_the car bumped had a man at the wheel
The prince that_the girl helped had a head of blonde hair
The bug that_the ant attacked was as big as a coin

The shelf that_the piano bumped had some dust on its top
The tradie that_the man thanked left his wallet in the car
The chief that_the girl thanked was too late for the bus
The sailor that_the banker pushed hurt his foot in the brawl
The bloke that_the mum helped gave a gift to say thanks
The dad that_the chef knew had a son with a limp
The dancer that_the guy liked was a chef in the city
The kid that_the dad followed had some mud on his shoes # **Experiment 1 stops here**
The monk that_the wife liked put a stone in the basket
The vet that_the Dad helped left the house in a hurry
The lion that_the goat saw drank some water from the stream
The creep that_the clown noticed had a scratch on his chin
The judge that_the poet thanked put a note on the desk
The cop that_the snitch thanked hurt her hand in the fight
The fraud that_the duke helped had a sip from his glass
The bro that_the brat heard broke the vase on the shelf
The men that_the crew blamed had some lunch on the job
The jock that_the kid phoned wore a watch that was new
The wolf that_the bear chased had some blood on its fur
The elf that_the dwarf fought left the room full of anger
The punk that_the boss liked had a son in the school
The mayor that_the crowd followed was not happy with the result
The niece that_the date liked lit the candle on the table
The guard that_the chef mocked had a scar on her ankle
The puppy that_the goat nudged dug a hole in the grass
The guy that_the bard helped had a date on that night
The gool that_the goth loved told a joke to their friends
The spouse that_the teen thanked was too late for the meeting
The geek that_the crook blamed closed the door with the switch
The chap that_the gent saw had a rose in his jacket
The lady that_the baker helped had a head of brown hair
The beast that_the witch heard had a plan that was evil # **Experiment 2 stops here**
The mug that_the plate chipped was too old for the shop
The coin that_the knife scratched had a price that was high
The coach that_the kid called had a voice that was strange
The clown that_the geek knew had a party on that day
The poet that_the maid blamed wrote a letter to the mayor
The girl that_the boy pushed had a grudge that was strong
The wife that_the cop knew poured a glass of cold water
The bride that_the man loved made a note of what happened
The duke that_the Queen warned fought the man in the castle
The host that_the bloke pleased had a friend that was near
The liar that_the scout knew left the house in a fury
The vet that_the liar saw wore a hat on her head

The bard that_the guard mocked told a lie to the boss
The jock that_the kid thanked had a smile on his face
The goat that_the mule bit was not happy with the food
The man that_the thug punched had a grimace on his face
The elf that_the orc fought made a cry that was loud
The priest that_the judge called made a promise on that night
The witch that_the punk liked poured a glass of red wine
The goth that_the poet knew left the church in a hurry
The duke that_the bard loved left a note on the stool
The cop that_the creep saw dropped a book by the lake
The Queen that_the Prince punched knew the truth of the matter
The vase that_the jar cracked was then sold at the market
The team that_the crew helped had some lunch in the sun
The host that_the team liked was then told of the result
The Dad that_the uncle saw waved his hand to say hi
The aunt that_the man knew made a cake to say thanks
The Dad that_the niece loved hid the ball in the vase
The niece that_the bloke helped had a head of blonde hair
The son that_the guy mocked crossed the street near the tree
The child that_the friend pushed told a lie to the man
The chef that_the child saw had a smile on her face
The chair that_the desk knocked had a price that was high
The beast that_the knight scared hit the stone with a fist
The wife that_the boss phoned knew the story was too long
The host that_the friend warned had a date on that night
The gent that_the lady loved told a story about the trip
The mate that_the tradie blamed had a trip the next day
The man that_the mate pushed was too drunk to fight back